ONLINE CONTROL FOR STORAGE SYSTEMS AND PROJECTION-FREE CONVEX OPTIMIZATION VIA FIRST-ORDER METHODS

by

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Dedication

Dedicated to my parents, Mohammad and Soraya, and my sister, Kiana, whose unwavering love and support have sustained me throughout my Ph.D. journey.

Acknowledgments

I am deeply grateful to my advisor, Professor Michael J. Neely, for his guidance, patience, and encouragement throughout my Ph.D. His thoughtful feedback and careful approach to research have helped me grow as a researcher and have shaped this dissertation. I am thankful for his continued support and for the opportunity to learn from him.

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Abstract

This thesis develops first-order optimization and online learning methods with explicit performance guarantees for nonsmooth convex optimization, storage control, and transmission-rate adaptation. The optimization methods avoid costly domain projections. The storage methods account for the effect of current decisions on future resource availability. Decision parameterizations and auxiliary variables expose convex structure and make the algorithmic updates tractable. For rate adaptation from binary acknowledgments, confidence bounds guide transmission-rate selection and yield bounds on time-average expected queue length.

For nonsmooth convex optimization, methods combining linear optimization over the original domain with subgradient queries and projections on an auxiliary set achieve an $O(T^{-1/2})$ upper bound on the objective gap using at most T queries to each oracle. For problems with additional convex inequality constraints, the thesis applies the algorithm published jointly with Michael J. Neely in 2024 to the positive parts of the constraints, with a parameter choice and analysis for exact deterministic oracles. The constraint-violation norm decreases at the same rate when an optimal Lagrange multiplier for the functional constraints exists. The algorithm does not require the multiplier or a bound on its norm as input.

For energy-harvesting allocation across $n \geq 2$ subsystems, a controller updates total usage and allocation proportions while respecting energy availability. For a planned horizon T and sufficient battery capacity of order $\sqrt{T}$, independent of n, it achieves $O(\sqrt{T(1+\log n)})$ expected cumulative

regret against the best fixed nonnegative allocation in hindsight whose total usage does not exceed the mean energy arrival. Arrivals and total per-slot usage lie in $[0, 1]$. Costs are convex and revealed after allocation, with a fixed componentwise bound on the selected cost subgradients. The arrivals are i.i.d., and their next-slot conditional mean, given the full past history, equals their common mean.

For scalar storage control with a known retention coefficient $\alpha \in (0, 1)$, sufficient capacity to avoid saturation, bounded adversarial arrivals, and convex costs, simplex disturbance-action control (SDAC) policies specify allocations of past arrivals over time. An entropic SDAC controller learns these allocation fractions while respecting resource availability. It achieves $O(\sqrt{T \log(H+1)})$ regret against fixed SDAC policies with memory H and $O(\sqrt{T \log\log T})$ regret against fixed infinite-memory SDAC policies. Each comparator is evaluated on its own feasible trajectory under the same arrivals and costs. These bounds assume full cost feedback, with $L_x, L_u > 0$ bounding the state and action components of the selected cost subgradients. For each memory length $H \geq 1$ and all sufficiently large T, every causal feasible controller has worst-case expected regret at least $\frac{\alpha}{8}(L_x/(1-\alpha) + L_u)\sqrt{T}$ against the fixed memory-H SDAC class. For fixed H, α, L_x, L_u, the $\sqrt{T}$ dependence on the horizon T is optimal.

For an initially empty communication queue with independent i.i.d. arrival and channel-capacity sequences in $[0, 1]$, a transmitter chooses rates from a continuous interval and learns their mean offered service from binary acknowledgments. Given a known positive lower bound ε on the difference between the best achievable mean offered service and the mean arrival rate, UCB1 on a finite rate grid achieves time-average expected queue length $O((1 + \log(1/\varepsilon))/\varepsilon^2)$, uniformly over all horizons. As $\varepsilon \downarrow 0$, the converse result from the original joint paper on rate adaptation gives a worst-case lower bound of $\Omega(\varepsilon^{-2})$ for every causal policy in some admissible environment

and at some horizon. For this UCB1 controller, a matching $\Omega(\varepsilon^{-2}\log(1/\varepsilon))$ lower bound holds at some horizon even with deterministic arrivals and capacities.

Preface

Disclosure of AI Assistance

I used ChatGPT, including GPT-5.6 and GPT-6, in preparing this dissertation. The account below describes the scope of this assistance by chapter. Brief chapter-opening notes also identify AI-generated material and specific mathematical contributions.

Abstract and Chapters 1 and 8. AI provided substantial assistance in rewriting and reorganizing my initial drafts of the abstract, introduction, and conclusion.

Chapter 2. This chapter was generated primarily using AI. The assistance included rewriting and reorganizing my initial drafts, selecting relevant material from other chapters, and bringing it together into a coherent presentation of the mathematical background.

Chapter 3. I used ChatGPT, primarily GPT-6 with some use of GPT-5.6, to refine proofs and improve the chapter's organization and editorial presentation.

Chapter 4. GPT-6 Astra suggested replacing the constraints by their positive parts within the original update rules, developed the corresponding proofs, and helped with the chapter's exposition. ChatGPT also helped me rewrite and shorten the proof, improve the parameter choices, and reduce the constants.

Chapter 5. Most of the changes made to the published work for this chapter were produced with AI assistance at my request. These included reorganizing the exposition and proofs, making the conditional-mean assumption explicit, correcting and completing the sufficient battery-capacity calculation, tightening the term involving the initial potential in the regret bound, and making step-size tuning explicit.

Chapter 6. I used ChatGPT to refine and reorganize my proofs and improve their presentation, constants, and arguments. The lower-bound construction and proofs in Section 6.4 were developed primarily using ChatGPT (5.6 sol), as disclosed in the source manuscript.

Chapter 7. My use of AI in preparing the original joint paper was limited to editorial assistance. For the dissertation, AI helped improve constants, refine proof steps, and improve the presentation of the analysis for a known lower bound on service slack. The statement and proof of Theorem 7.9, which establishes the necessity of the logarithmic factor for the specific UCB1 algorithm used here, were generated by AI for this dissertation and are not part of the original paper.

Formatting and submission support. ChatGPT also assisted with the dissertation's front matter, including this Preface, as well as formatting, bibliography checks, and preparation for submission.

Chapter 1

Introduction

This thesis studies constrained optimization and online resource control under different oracle and feedback models. The restrictions arise from costly domain projections, resource availability that depends on past actions, and service rates that must be learned from binary feedback. In the optimization problems, linear optimization constructs points in the original domain while subgradient updates use an auxiliary domain. In energy allocation and scalar storage control, parameterizing allocations or policies permits first-order learning, while battery feedback and action clipping enforce resource availability. The guarantees bound objective gap, constraint violation, or regret. The scalar storage guarantee allows adversarial arrival and cost sequences. For transmission-rate adaptation, confidence bounds based on binary feedback yield a bound on time-average expected queue length.

1.1 Models and performance criteria

Table 1.1 compares the models, available information, performance criteria, and feasibility requirements.

Table 1.1: Principal distinctions between the research models. Formal assumptions are stated in each chapter.

Chapter	Model and information	Comparison and feasibility
3	Convex objective; linear oracle on $\mathcal{X}$, subgradients on $\mathcal{Y} \supseteq \mathcal{X}$, and easy projection onto $\mathcal{Y}$.	Fixed minimizer over $\mathcal{X}$; output in $\mathcal{X}$, with auxiliary queries permitted outside $\mathcal{X}$.
4	Additional convex inequality constraints; linear oracle on $\mathcal{X}$; objective and constraint subgradients and constraint values on $\mathcal{Y}$; easy projection onto $\mathcal{Y}$.	Upper bound on objective gap at $\bar{x} \in \mathcal{X}$; upper bound on constraint violation if an optimal Lagrange multiplier exists.
5	Finite battery capacity; i.i.d. energy arrivals from an unknown distribution, observed before allocation; the full convex allocation-cost function is revealed afterward.	Expected regret against a fixed allocation whose per-slot expenditure is at most the mean energy arrival; only the controller must respect the available battery energy on every slot.
6	Leaky storage with retention coefficient $0 < \alpha < 1$ and enough capacity to avoid saturation; arbitrary arrivals in $[0, 1]$ and entire jointly convex state–action cost functions revealed after acting.	Regret against fixed policies that allocate fractions of past arrivals over time; controller withdrawals never exceed the available resource.
7	Independent i.i.d. arrival and capacity sequences with unknown distributions; known lower bound ε on service slack. Rate in $[0, 1]$ chosen before observing capacity; binary acknowledgment even for an empty queue.	A bound on time-average expected queue length that is independent of T.

1.2 Contributions

In Chapters 3 and 4, T denotes the query budget for each oracle, including initialization. In the online problems, T denotes the number of rounds.

1.2.1 Projection-free nonsmooth optimization

Chapter 3 develops a method that separates the construction of feasible points from the evaluation of objective subgradients. Linear optimization generates points in the original convex domain, while projected subgradient updates operate on an auxiliary domain containing it. A multiplier couples the updates. The method returns a feasible output with objective gap $O(T^{-1/2})$. Each update requires one linear optimization call, one subgradient call, and one auxiliary projection. The resulting oracle complexity is $O(\epsilon^{-2})$ for an objective gap of at most ϵ.

1.2.2 Projection-free optimization with functional constraints

Chapter 4 formulates a joint bound on objective gap and constraint violation. The output belongs to the original domain, and its objective gap has an $O(T^{-1/2})$ upper bound. If an optimal Lagrange multiplier exists, the constraint violation norm also satisfies an $O(T^{-1/2})$ bound. The multiplier affects the guarantee but is not required to choose the algorithm parameters.

1.2.3 Online allocation with stochastic energy arrivals

Chapter 5 develops an energy-allocation controller by writing each allocation as the product of total energy usage and a vector of allocation proportions. This representation permits separate updates: a projected first-order update with battery feedback controls total usage, while an entropic mirror-descent update controls the proportions. The controller respects available energy on every slot and, under the conditional-mean assumption (5.7), achieves $O(\sqrt{T(1+\log n)})$ expected regret

with sufficient battery capacity $O(\sqrt{T})$. The benchmark is the best fixed nonnegative allocation whose total usage does not exceed the mean energy arrival. With the componentwise subgradient bound fixed, the dependence on the number of subsystems improves from $O(\sqrt{n})$ in Yu and Neely [1] to $O(\sqrt{\log n})$ for $n \geq 2$, while the sufficient capacity is independent of n.

1.2.4 Online control of scalar storage

Chapter 6 introduces simplex disturbance-action control (SDAC), which parameterizes storage policies by fractions of past arrivals assigned to future withdrawals. Entropic mirror descent updates the policy parameters, and action clipping limits each withdrawal to the resource currently available. For arbitrary arrivals in $[0, 1]$ and full cost feedback, with retention and cost parameters fixed, the controller achieves $O(\sqrt{T \log(H+1)})$ regret against the best fixed SDAC policy with memory H.

Choosing the memory length as a function of the horizon and bounding the cost of truncation gives $O(\sqrt{T \log\log T})$ regret against the best fixed infinite-memory SDAC policy (Corollary 6.9). This policy class includes every feasible scalar fixed-fraction policy under the same dynamics (Corollary 6.8), so the guarantee also applies to that comparison class.

For retention $\alpha \in (0, 1)$ and positive state and action cost Lipschitz bounds L_x, L_u, define $G_\alpha := \alpha(L_x/(1-\alpha) + L_u)$. For every integer $H \geq 1$ and horizon satisfying (6.50), and for every causal feasible controller, including randomized controllers, Theorem 6.10 supplies a deterministic arrival and cost sequence fixed before the interaction on which expected regret against the fixed memory-H SDAC class is at least $G_\alpha\sqrt{T}/8$. The expectation is over the controller's randomization. For fixed H, α, L_x, L_u, the upper and lower bounds therefore match in their $\sqrt{T}$ dependence on the horizon T. Gaps remain in their dependence on memory length H and retention coefficient α, as quantified in Section 6.4.

1.2.5 Rate adaptation from binary feedback

Chapter 7 develops and analyzes a transmission-rate controller that applies UCB1 to a finite grid of rates using binary success or failure feedback. A known lower bound $\varepsilon > 0$ on service slack determines the grid resolution. For independent i.i.d. arrival and capacity sequences and an initially empty queue, the controller achieves time-average expected queue length $O((1 + \log(1/\varepsilon))/\varepsilon^2)$ uniformly over all horizons. A matching $\Omega(\varepsilon^{-2} \log(1/\varepsilon))$ lower bound on a suitable deterministic instance and horizon shows that the logarithmic factor is necessary for this UCB1 controller.

1.3 Notation and conventions

We write $\mathbb{R}_+$ for the nonnegative reals, $[n] \coloneqq \{1, \dots, n\}$, $\mathbf{1}$ for the all-ones vector, and $[v]_+$ for the componentwise positive part. Vector inequalities are componentwise. All logarithms are natural unless a base is shown. Empty sums are zero. For an event E, its probability is $\mathbb{P}(E)$, and its indicator is $\mathbf{1}_{\{E\}}$. We use $O(\cdot)$ for asymptotic upper bounds, $\Omega(\cdot)$ for asymptotic lower bounds, and $\Theta(\cdot)$ for matching upper and lower orders. The notation $a \coloneqq b$ defines a as b.

Scalar variables and scalar coordinates are italic. In the abstract optimization and geometry of Chapters 2–4, plain italic letters also denote vectors, matrices, and linear maps. Their domains specify their type. In the resource models of Chapters 5 and 6, bold letters distinguish whole vectors and matrices from their scalar coordinates, as in $\mathbf{m}_t = (m_{t,1}, \dots, m_{t,H})^\top$. Randomness is specified by the probability model and conditioning, rather than by case or boldface alone.

Coordinate lists for column vectors are written $(v_1, \dots, v_n)^\top$. Scalar coordinates use subscripts: v_i for coordinate $i \in [n]$ and $v_{t,i}$ for that coordinate at round t. When time and coordinate indices occur together, time comes first. For vectors, $\langle x, y \rangle \coloneqq x^\top y$. An unqualified norm in a Euclidean argument is the Euclidean norm.

The unqualified symbol Π_Q denotes Euclidean projection onto Q.

We write $\mathrm{Reg}_T(\mathcal{C})$ for the controller's cumulative cost minus the infimum comparator cost over $\mathcal{C}$, evaluated in hindsight on the same realized inputs. Expected regret is $\mathbb{E}[\mathrm{Reg}_T(\mathcal{C})]$.

1.4 Organization of the thesis

Chapter 2 collects the convex-analysis, Bregman-divergence, and mirror-descent tools used in the research chapters outlined above. The final chapter brings the results together and identifies directions for further work.

Chapter 2

Mathematical Preliminaries

This chapter collects the tools used across the optimization and control analyses. It presents standard material for which we make no claim of novelty. This chapter was generated primarily using AI.

2.1 Notation and online comparisons

We follow the notation of Section 1.3. In this chapter, $Q \subseteq \mathbb{R}^n$ is a convex feasible set and T is the horizon.

In online convex optimization, $z_t \in Q$ is selected before the convex loss ℓ_t is revealed. Static regret on a fixed convex set Q [2, 3] is

$$\mathrm{Reg}_T(Q) \coloneqq \sum_{t=1}^{T} \ell_t(z_t) - \inf_{z \in Q} \sum_{t=1}^{T} \ell_t(z). \tag{2.1}$$

On a nonempty, closed, bounded convex domain with uniformly bounded subgradients, projected subgradient descent with suitable step sizes gives $O(\sqrt{T})$ regret.

In the storage-control chapter, z specifies a fixed policy. To define $\ell_t(z)$, we apply that policy from the first round using the same arrivals and cost functions as the online controller. Here, $\ell_t(z)$

is the resulting cost in round t. The online controller changes its policy over time, so its current state also reflects earlier policies. Its actual cost can therefore differ from $\ell_t(z_t)$.

For a norm $\|\cdot\|$, its dual is

$$\|g\|_* \coloneqq \sup_{\|x\|\le 1} \langle g, x\rangle . \tag{2.2}$$

In particular, ℓ_1 and ℓ_∞ are dual, and ℓ_2 is self-dual.

Lemma 2.1 (Hölder and Young inequalities; cf. [4, Example 5.11])**.** *For any dual pair of norms and any $\gamma > 0$,*

$$|\langle g, v\rangle| \le \|g\|_* \|v\|, \qquad \langle g, v\rangle - \frac{\gamma}{2}\|v\|^2 \le \frac{\|g\|_*^2}{2\gamma}. \tag{2.3}$$

For the norm induced by the inner product, the first inequality is Cauchy–Schwarz. For $a, b \ge 0$ and conjugate exponents $p, q > 1$, meaning $1/p + 1/q = 1$, Young's inequality also gives

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}. \tag{2.4}$$

Proof. For $v \ne 0$, apply (2.2) to both $v/\|v\|$ and $-v/\|v\|$. The case $v = 0$ is immediate. With $r \coloneqq \|v\|$, this gives

$$\begin{aligned}\langle g, v\rangle - \frac{\gamma}{2}\|v\|^2 &\le \|g\|_* r - \frac{\gamma}{2} r^2 \\ &= -\frac{\gamma}{2}\left(r - \frac{\|g\|_*}{\gamma}\right)^2 + \frac{\|g\|_*^2}{2\gamma} \le \frac{\|g\|_*^2}{2\gamma}.\end{aligned}$$

For (2.4), the case $b = 0$ is immediate. If $b > 0$, the derivative of $br - r^p/p$ is $b - r^{p-1}$, so its maximum over $r \ge 0$ occurs at $r = b^{1/(p-1)}$. Since $q = p/(p-1)$,

$$ab - \frac{a^p}{p} \le \max_{r\ge 0}\left\{br - \frac{r^p}{p}\right\} = b^q - \frac{b^q}{p} = \frac{b^q}{q}. \qquad \square$$

2.2 Convexity, subgradients, and affine parameterizations

A set Q is convex if it contains every segment between its points. A function $f : Q \to \mathbb{R}$ is convex if

$$f(\theta x + (1-\theta)y) \le \theta f(x) + (1-\theta)f(y), \qquad x, y \in Q, \quad 0 \le \theta \le 1. \tag{2.5}$$

A subgradient $g \in \partial f(x)$ satisfies

$$f(y) \ge f(x) + \langle g, y - x \rangle, \qquad y \in Q. \tag{2.6}$$

When f is differentiable at x, we use $\nabla f(x)$ as a subgradient.

Lemma 2.2 (Convex comparison and affine composition [5, 4])**.** *Let f be convex on Q. For $x_i \in Q$, $\theta_i \ge 0$, and $\sum_i \theta_i = 1$,*

$$f\Big(\sum_i \theta_i x_i\Big) \le \sum_i \theta_i f(x_i). \tag{2.7}$$

For affine composition, let $P \subseteq \mathbb{R}^p$ and $Q \subseteq \mathbb{R}^q$ be convex, with $A \in \mathbb{R}^{q \times p}$ and $b \in \mathbb{R}^q$. If $F(z) \coloneqq Az + b$ maps P into Q, then $\ell(z) \coloneqq f(F(z))$ is convex and

$$g \in \partial f(F(z)) \quad \Longrightarrow \quad A^\top g \in \partial \ell(z). \tag{2.8}$$

Proof. For (2.7), discard zero weights and induct on the number of remaining weights. If there is more than one, choose a weight $0 < \theta_j < 1$, normalize the others by $1 - \theta_j$, and apply the induction hypothesis to their convex combination, followed by (2.5) with weights θ_j and $1 - \theta_j$. For $z, z' \in P$ and $0 \le \lambda \le 1$,

$$\ell(\lambda z + (1-\lambda)z') = f(\lambda F(z) + (1-\lambda)F(z')) \overset{\text{(A)}}{\le} \lambda \ell(z) + (1-\lambda)\ell(z'),$$

where (A) follows from (2.5), since $F(z), F(z') \in Q$. For every $z' \in P$,

$$\ell(z') - \ell(z) = f(F(z')) - f(F(z)) \overset{\text{(A)}}{\geq} \langle g, F(z') - F(z) \rangle = \left\langle A^\top g, z' - z \right\rangle.$$

Here (A) follows from (2.6), proving (2.8). □

Lemma 2.3 (Jensen's inequality for fractional powers)**.** *For a nonnegative random variable Z with $\mathbb{E}[Z] < \infty$ and $0 < \theta < 1$,*

$$\mathbb{E}[Z^\theta] \leq (\mathbb{E}[Z])^\theta. \tag{2.9}$$

Proof. Put $\mu \coloneqq \mathbb{E}[Z]$. If $\mu = 0$, then $Z = 0$ almost surely and the conclusion is immediate. If $\mu > 0$, concavity of $z \mapsto z^\theta$ on $[0, \infty)$ gives

$$z^\theta \leq \mu^\theta + \theta \mu^{\theta - 1}(z - \mu), \qquad z \geq 0.$$

Taking expectations with $z = Z$ proves (2.9). □

Lemma 2.4 (Bounded subgradient selections and Lipschitz continuity; cf. [4, Theorem 2.24])**.** *Suppose that at every $x \in Q$ a subgradient $g_x \in \partial f(x)$ can be selected with $\|g_x\|_* \leq L$. Then*

$$|f(x) - f(y)| \leq L \|x - y\|, \qquad x, y \in Q. \tag{2.10}$$

Proof. For $x, y \in Q$,

$$f(x) - f(y) \leq \langle g_x, x - y \rangle \leq \|g_x\|_* \|x - y\| \leq L \|x - y\|.$$

Interchanging x and y proves (2.10). □

Inequality (2.10) defines L-Lipschitz continuity.

In the mirror-descent updates below, strong convexity of the regularizer bounds the distance between successive iterates. See (2.28).

Definition 2.5 (Strong convexity [4, Section 4.1.1])**.** A function $f : Q \to \mathbb{R}$ is σ-strongly convex with respect to $\|\cdot\|$ if, for $x, y \in Q$ and $0 \le \theta \le 1$,

$$f(\theta x + (1-\theta)y) \le \theta f(x) + (1-\theta) f(y) - \frac{\sigma}{2}\theta(1-\theta)\|x-y\|^2 . \tag{2.11}$$

If f is σ-strongly convex on Q, then for $x \in Q$ and $g \in \partial f(x)$,

$$f(y) \ge f(x) + \langle g, y-x\rangle + \frac{\sigma}{2}\|y-x\|^2 , \qquad y \in Q. \tag{2.12}$$

Proof. For $x_\theta \coloneqq (1-\theta)x + \theta y$ and $0 < \theta < 1$,

$$\begin{aligned} f(x) + \theta\langle g, y-x\rangle &\overset{\text{(A)}}{\le} f(x_\theta) \\ &\overset{\text{(B)}}{\le} (1-\theta)f(x) + \theta f(y) - \frac{\sigma}{2}\theta(1-\theta)\|x-y\|^2 . \end{aligned}$$

Here (A) uses (2.6) with comparison point $x_\theta \in Q$. Step (B) uses (2.11) with weights $1-\theta$ and θ. Rearranging, dividing by $\theta > 0$, and letting $\theta \downarrow 0$ proves (2.12). □

2.3 Bregman divergence and the three-point inequality

A Bregman divergence serves as a proximity term in mirror descent [6, 7, 8]. The three-point inequality below converts the minimization that defines an update into a bound relative to any feasible comparison point [9, 10, 11, 12].

Let C be a convex domain and $\Phi : C \to \mathbb{R}$ a convex function, differentiable on $\mathrm{ri}(C)$, the interior relative to its affine hull. Derivatives are taken in that affine hull. The constraint set $Q \subseteq C$ can have smaller dimension. Define $Q_{\mathrm{ref}} \coloneqq Q \cap \mathrm{ri}(C)$. For $x \in Q$, $y \in Q_{\mathrm{ref}}$, the Bregman divergence [13] is

$$D_\Phi(x, y) \coloneqq \Phi(x) - \Phi(y) - \langle \nabla\Phi(y), x - y \rangle . \tag{2.13}$$

The restriction on the second argument ensures that $\nabla\Phi(y)$ is defined. The first argument may lie on the boundary, because the formula requires only the value $\Phi(x)$ there. For entropy, this allows comparison points with zero coordinates while requiring positive coordinates at the reference point. Applying (2.6) to Φ at y gives $D_\Phi(x, y) \geq 0$. If Φ is σ-strongly convex on Q, then (2.12) with base point y, comparison point x, and subgradient $\nabla\Phi(y)$ gives

$$D_\Phi(x, y) \geq \frac{\sigma}{2} \|x - y\|^2 , \qquad x \in Q, \quad y \in Q_{\mathrm{ref}}. \tag{2.14}$$

Expanding each divergence using (2.13) gives the three-point identity

$$\langle \nabla\Phi(x) - \nabla\Phi(y), z - x \rangle = D_\Phi(z, y) - D_\Phi(z, x) - D_\Phi(x, y), \tag{2.15}$$

for $x, y \in Q_{\mathrm{ref}}$ and $z \in Q$.

Lemma 2.6 (Bregman three-point inequality; cf. [12, Lemma 2.1])**.** *Let $Q \subseteq C$ be convex, let $f : Q \to \mathbb{R}$ be convex, and fix $\alpha > 0$ and $y \in Q_{\mathrm{ref}}$. Suppose*

$$\hat{x} \in \operatorname*{arg\,min}_{x \in Q} \{ f(x) + \alpha D_\Phi(x, y) \}, \qquad \hat{x} \in Q_{\mathrm{ref}}. \tag{2.16}$$

Then, for all $z \in Q$,

$$f(\hat{x}) + \alpha D_\Phi(\hat{x}, y) \le f(z) + \alpha D_\Phi(z, y) - \alpha D_\Phi(z, \hat{x}). \tag{2.17}$$

Proof. For $x_\theta \coloneqq (1-\theta)\hat{x} + \theta z$ and $0 < \theta < 1$, convexity of Q gives $x_\theta \in Q$, so

$$\begin{aligned} f(\hat{x}) + \alpha D_\Phi(\hat{x}, y) &\overset{(A)}{\le} f(x_\theta) + \alpha D_\Phi(x_\theta, y) \\ &\overset{(B)}{\le} (1-\theta) f(\hat{x}) + \theta f(z) + \alpha D_\Phi(x_\theta, y). \end{aligned}$$

Here (A) is optimality in (2.16), and (B) is (2.5) for f with weights $1-\theta$ and θ. Rearranging and dividing by $\theta > 0$ gives

$$f(\hat{x}) - f(z) \le \alpha \frac{D_\Phi(x_\theta, y) - D_\Phi(\hat{x}, y)}{\theta}.$$

By (2.13), the difference quotient on the right tends to $\langle \nabla\Phi(\hat{x}) - \nabla\Phi(y), z - \hat{x} \rangle$ as $\theta \downarrow 0$. Thus

$$f(\hat{x}) - f(z) \le \alpha \langle \nabla\Phi(\hat{x}) - \nabla\Phi(y), z - \hat{x} \rangle = \alpha \{ D_\Phi(z, y) - D_\Phi(z, \hat{x}) - D_\Phi(\hat{x}, y) \}. \qquad \square$$

2.4 Quadratic and entropy geometries

We use quadratic regularizers for Euclidean updates, and entropy regularizers for allocation fractions on a simplex.

For the Euclidean norm, set

$$\Phi(x) \coloneqq \tfrac{1}{2} \|x\|^2, \qquad D_\Phi(x, y) = \tfrac{1}{2} \|x - y\|^2. \tag{2.18}$$

The function Φ is 1-strongly convex in the Euclidean norm, whose dual norm is also Euclidean.

For entropy, define the probability and subprobability simplices

$$\Delta_d \coloneqq \{p \geq 0 : \mathbf{1}^\top p = 1\}, \qquad S_d \coloneqq \{p \geq 0 : \mathbf{1}^\top p \leq 1\}, \tag{2.19}$$

and the unnormalized negative entropy

$$\Psi(p) \coloneqq \sum_{i=1}^{d} (p_i \log p_i - p_i), \qquad 0 \log 0 \coloneqq 0. \tag{2.20}$$

Its Bregman divergence is the generalized Kullback–Leibler (KL) divergence: for $p \geq 0$ and $q > 0$,

$$D_\Psi(p, q) = \sum_i \left(p_i \log \frac{p_i}{q_i} - p_i + q_i \right). \tag{2.21}$$

When $p, q \in \Delta_d$ with $q > 0$, the linear terms cancel, giving the ordinary KL divergence between probability vectors.

Lemma 2.7 (Bounds on entropy divergence; cf. [14, Proposition 5.1])**.** *For $p \in S_d$ and $q \in S_d \cap (0, \infty)^d$,*

$$D_\Psi(p, q) \geq \tfrac{1}{2} \|p - q\|_1^2 . \tag{2.22}$$

The uniform initial points satisfy

$$\sup_{p \in \Delta_d} D_\Psi(p, \mathbf{1}/d) \leq \log d, \qquad \sup_{p \in S_d} D_\Psi(p, \mathbf{1}/(d+1)) \leq \log(d+1). \tag{2.23}$$

Proof. For positive $p, q \in S_d$, put $v \coloneqq p - q$ and $q_s \coloneqq q + sv$ for $0 \le s \le 1$. Then $q_s > 0$, $q_s \in S_d$, and differentiating (2.20) gives $\nabla^2 \Psi(q_s) = \operatorname{diag}(1/q_{s,i})$. Hence

$$\|v\|_1^2 \overset{(A)}{\le} \left(\sum_i \frac{v_i^2}{q_{s,i}} \right) \sum_i q_{s,i} \overset{(B)}{\le} v^\top \nabla^2 \Psi(q_s) v. \tag{2.24}$$

Step (A) is Cauchy–Schwarz from Lemma 2.1, applied to the vectors with components $|v_i|/\sqrt{q_{s,i}}$ and $\sqrt{q_{s,i}}$. Step (B) uses $\sum_i q_{s,i} \le 1$ from (2.19). Taylor's formula with integral remainder yields

$$D_\Psi(p, q) = \int_0^1 (1 - s) v^\top \nabla^2 \Psi(q_s) v \, \mathrm{d}s \overset{(A)}{\ge} \|v\|_1^2 \int_0^1 (1 - s) \, \mathrm{d}s = \frac{1}{2} \|p - q\|_1^2 .$$

Here (A) integrates (2.24) against the nonnegative weight $1 - s$. The same bound holds when p has zero coordinates by continuity, using $0 \log 0 = 0$. For the initial bounds, $D_\Psi(\cdot, q)$ is convex for fixed $q > 0$. Writing e_i for the coordinate vectors, every point of Δ_d is a convex combination of the e_i, and every point of S_d is a convex combination of $0, e_1, \ldots, e_d$. Thus (2.7) reduces the bounds to the vertex values from (2.21):

$$\begin{aligned} D_\Psi(e_i, \mathbf{1}/d) &= \log d, \\ D_\Psi(0, \mathbf{1}/(d+1)) &= \frac{d}{d+1}, \\ D_\Psi(e_i, \mathbf{1}/(d+1)) &= \log(d+1) - \frac{1}{d+1}. \end{aligned}$$

Since $d/(d+1) \le \int_1^{d+1} t^{-1} \, \mathrm{d}t = \log(d+1)$, these values prove (2.23), including $d = 1$. □

For the uniform initial points in (2.23), the term involving the initial Bregman divergence in the mirror-descent bound (2.29) is at most $\log d/\eta$ on Δ_d and $\log(d+1)/\eta$ on S_d, where $\eta > 0$ is the step size. We measure subgradients in the ℓ_∞ norm, dual to ℓ_1.

Lemma 2.8 (Entropy mirror-descent updates; cf. [14, Section 5])**.** *Let q be positive in $Q = \Delta_d$ or $Q = S_d$, let $\eta > 0$, and set $a_i := q_i e^{-\eta g_i}$. The unique minimizer of $\eta \langle g, p\rangle + D_\Psi(p, q)$ over $p \in Q$ is*

$$p^+ := \begin{cases} a/(\sum_i a_i), & Q = \Delta_d, \\ a/\max\{1, \sum_i a_i\}, & Q = S_d. \end{cases} \tag{2.25}$$

Both updates preserve positivity.

Proof. Let $A := \sum_i a_i > 0$ and let p^+ be the candidate in (2.25). Write $p^+ = a/Z$, where $Z := A$ on Δ_d and $Z := \max\{1, A\}$ on S_d, and put $\lambda := \log Z$. Both candidates are positive and feasible. Since $\log a_i = \log q_i - \eta g_i$, (2.21) gives

$$\eta \langle g, p\rangle + D_\Psi(p, q) = D_\Psi(p, a) + \sum_i (q_i - a_i).$$

The last sum is independent of p. For every feasible p, expansion gives

$$D_\Psi(p, a) - D_\Psi(p^+, a) = D_\Psi(p, p^+) + \lambda(\mathbf{1}^\top p^+ - \mathbf{1}^\top p).$$

This identity also holds when p has zero coordinates, with $0 \log 0 = 0$. On Δ_d, the last term vanishes because both masses equal one. On S_d, either $A \le 1$ and $\lambda = 0$, or $A > 1$, $\lambda > 0$, and $\mathbf{1}^\top p^+ = 1 \ge \mathbf{1}^\top p$. The difference is therefore at least $D_\Psi(p, p^+) \ge \frac{1}{2} \|p - p^+\|_1^2$ by (2.22). Hence p^+ is the unique minimizer. □

2.5 Mirror descent

The movement bound below is used in the storage-control chapter to account for changes in the controller's policy.

Lemma 2.9 (Mirror descent: comparison, regret, and movement [14, 15])**.** *Suppose* Φ *is* σ*-strongly convex on a convex set* Q*, where* $\sigma > 0$*, and* $x, x^+ \in Q_{\mathrm{ref}}$ *satisfy*

$$x^+ \in \arg\min_{v \in Q} \{\eta \langle g, v\rangle + D_\Phi(v, x)\}, \qquad \eta > 0. \tag{2.26}$$

For every $u \in Q$*,*

$$\langle g, x - u\rangle \le \frac{D_\Phi(u, x) - D_\Phi(u, x^+)}{\eta} + \frac{\eta}{2\sigma} \|g\|_*^2, \tag{2.27}$$

$$\left\|x^+ - x\right\| \le \frac{\eta}{\sigma} \|g\|_*. \tag{2.28}$$

For a sequence of these updates with a common step size and $g_t \in \partial \ell_t(x_t)$*, convexity of* ℓ_t *gives*

$$\sum_{t=1}^{T} (\ell_t(x_t) - \ell_t(u)) \le \frac{D_\Phi(u, x_1)}{\eta} + \frac{\eta}{2\sigma} \sum_{t=1}^{T} \|g_t\|_*^2. \tag{2.29}$$

The same bound holds for the sum of $\langle g_t, x_t - u\rangle$*.*

Proof. Apply Lemma 2.6 with $f(v) \coloneqq \eta \langle g, v\rangle$, $\alpha = 1$, $y = x$, $\hat{x} = x^+$, and comparison point $z = u$. Both reference points are admissible by $x, x^+ \in Q_{\mathrm{ref}}$, so

$$\eta \left\langle g, x^+ - u \right\rangle \le D_\Phi(u, x) - D_\Phi(u, x^+) - D_\Phi(x^+, x). \tag{2.30}$$

Adding $\eta \langle g, x - x^+\rangle$ and applying Hölder's inequality, the curvature bound (2.14), and Young's inequality (2.3) with $\gamma = \sigma$ gives

$$\begin{aligned} \eta \langle g, x - u\rangle &\le D_\Phi(u, x) - D_\Phi(u, x^+) + \eta \|g\|_* \left\|x - x^+\right\| - \frac{\sigma}{2} \left\|x - x^+\right\|^2 \\ &\le D_\Phi(u, x) - D_\Phi(u, x^+) + \frac{\eta^2}{2\sigma} \|g\|_*^2. \end{aligned}$$

Dividing by $\eta > 0$ proves (2.27). For movement, set $u = x$ in (2.30) and apply (2.14) in both directions:

$$\sigma \left\|x^+ - x\right\|^2 \le D_\Phi(x, x^+) + D_\Phi(x^+, x) \le \eta \left\langle g, x - x^+ \right\rangle \le \eta \left\|g\right\|_* \left\|x^+ - x\right\|.$$

The last inequality is Hölder's. If $x^+ = x$, the movement bound is immediate. Otherwise, divide by $\sigma \left\|x^+ - x\right\| > 0$ to obtain (2.28).

For the sequence of updates, convexity and summation of (2.27) with a common step size give

$$\begin{aligned}\sum_{t=1}^{T} (\ell_t(x_t) - \ell_t(u)) &\le \sum_{t=1}^{T} \langle g_t, x_t - u \rangle \\ &\le \frac{D_\Phi(u, x_1) - D_\Phi(u, x_{T+1})}{\eta} + \frac{\eta}{2\sigma} \sum_{t=1}^{T} \left\|g_t\right\|_*^2 \\ &\le \frac{D_\Phi(u, x_1)}{\eta} + \frac{\eta}{2\sigma} \sum_{t=1}^{T} \left\|g_t\right\|_*^2.\end{aligned}$$

This proves (2.29) and the same bound for the sum of inner products. □

2.6 Projection and positive-part bounds

The projection condition is used to derive the domain-separation update. The positive-part bounds below control clipped multipliers and storage trajectories.

Lemma 2.10 (Euclidean projection [15])**.** *Let $Q \subseteq \mathbb{R}^n$ be nonempty, closed, and convex, and define $\Pi_Q(v) \coloneqq \arg\min_{q \in Q} \left\|q - v\right\|^2$. For $v \in \mathbb{R}^n$ and $p \in Q$,*

$$p = \Pi_Q(v) \quad \Longleftrightarrow \quad \langle p - v, z - p \rangle \ge 0 \qquad \text{for every } z \in Q. \tag{2.31}$$

Proof. If $p = \Pi_Q(v)$, the right directional derivative of $\|p + \theta(z-p) - v\|^2$ at $\theta = 0$ is nonnegative for every $z \in Q$, giving $2\langle p - v, z - p\rangle \geq 0$. Conversely, the stated inequality gives

$$\|z - v\|^2 - \|p - v\|^2 = 2\langle p - v, z - p\rangle + \|z - p\|^2 \geq \|z - p\|^2 \geq 0, \qquad z \in Q.$$

Thus p is the unique minimizer. □

Lemma 2.11 (Positive-part bounds)**.** *For $a, b \in \mathbb{R}$,*

$$|[a]_+ - [b]_+| \leq |a - b|. \tag{2.32}$$

For $v \in \mathbb{R}^n$, with the Euclidean norm and the positive part taken componentwise,

$$\|[v]_+\| \leq \|v\|. \tag{2.33}$$

Proof. By symmetry, assume $a \geq b$. Then

$$[a]_+ - [b]_+ = \begin{cases} 0, & a \leq 0, \\ a - b, & b \geq 0, \\ a, & b < 0 < a, \end{cases} \qquad 0 \leq [a]_+ - [b]_+ \leq a - b.$$

This proves (2.32). Applying it with $b = 0$ in each coordinate and summing squares gives $\|[v]_+\|^2 \leq \|v\|^2$, hence (2.33). □

Chapter 3

Projection-Free Nonsmooth Optimization over Convex Domains

Source and author contributions. This chapter is based on joint work with Michael J. Neely, reported in [16, 17]. I formulated the research question concerning projection-free optimization for nonsmooth convex objectives. Neely proposed the core algorithm. I introduced a correction to the proposed algorithm and proved its convergence guarantees. Neely reviewed the resulting work.

Material from the 2024 article by Asgari and Neely [17] (© The Author(s) 2024) is adapted under the Creative Commons Attribution 4.0 International license. Notation, presentation, and analysis have been modified.

3.1 Introduction

We study the minimization of a nonsmooth convex function over a compact convex domain:

$$\min_{x\in\mathcal{X}} f(x), \qquad f^* \coloneqq \min_{x\in\mathcal{X}} f(x). \tag{3.1}$$

On some domains, minimizing a linear function is substantially cheaper than computing a Euclidean projection [18, 19]. For smooth objectives, Frank–Wolfe methods avoid domain projection by using linear optimization and a smoothness bound on the linear-model error [20, 19].

3.1.1 Failure of direct subgradient substitution

For nonsmooth objectives, replacing the gradient in a Frank–Wolfe step with a subgradient need not give convergence. Nesterov's counterexample [21, Example 1] gives valid subgradient directions for which the objective gap remains bounded away from zero.

3.1.2 Method and scope

The method of Asgari and Neely [17] makes one ordinary linear-oracle call over $\mathcal{X}$, one subgradient query on an auxiliary set $\mathcal{Y} \supseteq \mathcal{X}$, and one projection onto $\mathcal{Y}$ per update. Here, projection-free refers to avoiding projection onto $\mathcal{X}$. The method uses a single loop, without Moreau smoothing or nested subproblems.

Under the assumptions below, let $D > 0$ bound the diameter of $\mathcal{X}$ and let $L > 0$ bound the subgradients returned throughout $\mathcal{Y}$. With parameters chosen for a query budget T, the returned point satisfies $f(\bar{x}_T) - f^* \leq 2LD/\sqrt{T}$ (Corollary 3.3). Both oracle counts are $O(\epsilon^{-2})$ for objective accuracy ϵ, with L, D fixed.

The results below use deterministic subgradients and an exact linear oracle. The published analysis also permits stochastic subgradients and inexact linear optimization [17]. Those variants use the same main proof ideas, with expectations and additional error terms. The numerical experiments are also outside this chapter's scope.

3.2 Related work

Chapter 4 reviews methods that also handle functional inequalities.

3.2.1 From smooth conditional gradients to nonsmooth optimization

To handle nonsmooth objectives, existing methods use richer local models, smoothing, or dual formulations. Other methods separate feasible points from subgradient query points.

White gives an early subgradient extension of Frank–Wolfe to concave nondifferentiable maximization [22]. Ravi et al. use approximate subdifferential sets to choose search directions [23]. Cheung and Li construct uniform affine approximations for separable objectives [24]. The latter two methods require additional structure or information beyond a single subgradient.

Argyriou et al. and Yurtsever et al. combine smoothing of a nonsmooth objective term with conditional gradient steps over the domain [25, 26]. Their methods require an efficiently computable proximity operator for that term.

3.2.2 Nonsmooth methods with linear and Fenchel oracles

Lan combines randomized smoothing with the conditional gradient method for nonsmooth Lipschitz convex objectives [27]. For fixed dimension, the method attains expected objective accuracy ϵ using $O(\epsilon^{-2})$ linear oracle calls and $O(\epsilon^{-4})$ subgradient calls. The subgradient queries may lie outside the original domain.

Hazan and Kale combine randomized smoothing with online Frank–Wolfe updates for nonsmooth Lipschitz convex losses [28, Theorem 4.3]. For losses sampled independently from a fixed distribution, their average iterate has a gap in expected loss of $O(T^{-1/3})$ with high probability, for fixed dimension and confidence level. Each round uses one linear oracle call and the gradient of the average of the smoothed losses observed so far.

The MOLES method of Thekumparampil et al. combines Moreau smoothing with approximate projections computed by Frank–Wolfe steps [29]. For nonsmooth Lipschitz convex objectives, it

attains expected objective accuracy ϵ using $O(\epsilon^{-2})$ subgradient calls and $O(\epsilon^{-2})$ linear oracle calls. The method requires subgradient access on a larger set containing the original domain.

Cox, Juditsky, and Nemirovski apply subgradient methods to a dual problem obtained from a Fenchel-type representation of the objective [30]. Each dual subgradient is computed using a linear oracle call over the original domain. Weights supplied by the dual method combine the returned points into an approximate primal solution. The approach requires the representation and efficient proximal updates on the dual domain.

Yurtsever, Tran-Dinh, and Cevher use a Fenchel oracle that minimizes the objective plus a linear term over the domain [31]. This requires more than an ordinary linear oracle. Their primal–dual framework handles additional linear inclusion constraints through proximal updates on the dual variables.

The domain-separation method of Asgari and Neely [17] uses a single loop and matches the oracle-complexity order of MOLES, with a smaller leading constant in the published worst-case linear-oracle bound under common Lipschitz and diameter bounds [29, Appendix C.2.2]. Its convergence proof is more direct, using a telescoping argument without smoothing or inner-loop error estimates. Section 3.4 gives its updates. Chapter 4 extends this method to functional inequality constraints [17].

Table 3.1: Selected approaches for general nonsmooth convex objectives. The accuracy parameter is ϵ, with dimension and other problem parameters fixed. The counts refer to subgradient and linear oracle calls.

Method	Domain access	Oracle complexity
Lan's randomized smoothing [27]	Ordinary linear oracle.	$O(\epsilon^{-2})$ linear calls and $O(\epsilon^{-4})$ subgradient calls.
MOLES [29]	Ordinary linear oracle and easy auxiliary projection.	$O(\epsilon^{-2})$ for each oracle.
Domain separation [17]	Ordinary linear oracle and easy auxiliary projection.	$O(\epsilon^{-2})$ for each oracle.

For general Lipschitz convex objectives, the classical worst-case lower bound is $\Omega(\epsilon^{-2})$ first-order oracle calls to reach objective accuracy ϵ, provided the dimension is sufficiently large [32, 15]. MOLES and the method of Asgari and Neely match this order in their subgradient counts under their respective assumptions.

Lan proves that $\Omega(\epsilon^{-2})$ linear oracle calls are necessary in the worst case for general Lipschitz convex objectives in sufficiently high dimension [27, Section 2.3]. His model requires the output to be a convex combination of the initial point and the points returned by the linear oracle. His randomized smoothing method attains the optimal order in ϵ for linear oracle calls, with an additional dimension factor. MOLES and our general convex method have the required output form and match the linear oracle lower bound without the additional dimension factor, using $O(\epsilon^{-2})$ calls to each oracle.

For the domain-separation method, the bound $f(\bar{x}_T) - f^* \leq 2LD/\sqrt{T}$ gives at most $\lceil 4L^2D^2/\epsilon^2 \rceil$ linear-oracle calls, including initialization, to attain objective accuracy ϵ, for $L, D, \epsilon > 0$. Its leading constant is smaller than that of the corresponding worst-case MOLES bound [29, Appendix C.2.2].

3.3 Problem and oracle assumptions

We follow the notation of Section 1.3. We work in $\mathbb{R}^n$ with the Euclidean inner product and norm $\|\cdot\|$. The function $f : \mathbb{R}^n \to \mathbb{R}$ is finite-valued and convex, and $\mathcal{X} \subseteq \mathcal{Y} \subseteq \mathbb{R}^n$. The vectors $x_t, y_t, q_t \in \mathbb{R}^n$ denote the domain point, auxiliary query point, and equality multiplier. Throughout this chapter, the integer $T \geq 1$ is a query budget for each oracle: the algorithm uses at most T calls to each oracle, including any linear-oracle call used for initialization.

Assumption 3.1 (Domain and ordinary oracles)**.** *The following conditions hold.*

(i) The set $\mathcal{X}$ is nonempty, compact, and convex, with Euclidean diameter

$$\operatorname{diam}(\mathcal{X}) \coloneqq \sup_{x,z\in\mathcal{X}} \|x - z\| \le D,$$

where $0 \le D < \infty$ is a known upper bound, not necessarily the exact diameter. The closed convex set $\mathcal{Y}$ contains $\mathcal{X}$ and has an available Euclidean projection $\Pi_{\mathcal{Y}}$.

(ii) A linear optimization oracle is available. For each direction $v \in \mathbb{R}^n$, it returns

$$\mathrm{LOO}_{\mathcal{X}}(v) \in \arg\min_{x\in\mathcal{X}} \langle v, x\rangle .$$

(iii) At every query $y \in \mathcal{Y}$, the subgradient oracle returns s with

$$s \in \partial f(y), \qquad \|s\| \le L,$$

where $0 \le L < \infty$ is known.

An initial point can be obtained by calling $\mathrm{LOO}_{\mathcal{X}}(0)$, or any available point in $\mathcal{X}$ can be used. The numerical values of L and D are used to choose the algorithm parameters and evaluate the convergence guarantee.

Lemma 2.4, with the subgradient norm bound in Assumption 3.1, gives

$$|f(y) - f(w)| \le L\|y - w\|, \qquad y, w \in \mathcal{Y}. \tag{3.2}$$

Let $x^* \in \mathcal{X}$ denote a minimizer of f over $\mathcal{X}$.

3.3.1 Choosing the auxiliary set

The set $\mathcal{Y}$ must contain $\mathcal{X}$, admit an available Euclidean projection, and support the required bounded subgradient queries. It need not be bounded: if the subgradient bound holds throughout

$\mathbb{R}^n$, we can take $\mathcal{Y} \coloneqq \mathbb{R}^n$ and the projection is the identity. Given $x_1 \in \mathcal{X}$ and the diameter bound D, two bounded choices are

$$\mathcal{Y} \coloneqq \{y : \|y - x_1\| \le D\} \quad \text{or} \quad \mathcal{Y} \coloneqq \{y : |y_i - x_{1,i}| \le D \text{ for all } i\}. \tag{3.3}$$

Both contain $\mathcal{X}$: for $x \in \mathcal{X}$, Assumption 3.1(i) gives $|x_i - x_{1,i}| \le \|x - x_1\| \le D$ for every coordinate i. Projection onto the closed ball rescales a displacement that exceeds its radius, while projection onto the box clips each coordinate [18, Section 3.2].

A subgradient bound on $\mathcal{X}$ alone is not enough. This bound must hold on the chosen $\mathcal{Y}$. In the guarantee $f(\bar{x}_T) - f^* \le 2LD/\sqrt{T}$ of Corollary 3.3, D bounds the diameter of $\mathcal{X}$. For fixed $\mathcal{X}$ and T, enlarging $\mathcal{Y}$ therefore weakens this guarantee only if a larger uniform bound L is needed. Section 3.5.3 gives an example and a Lipschitz extension. An extension's existence does not supply an inexpensive subgradient oracle: both the bound and oracle access on $\mathcal{Y}$ are required.

3.4 Derivation and algorithm

The failure in Section 3.1.1 motivates a different linear-oracle direction. Here the oracle uses a multiplier that accumulates disagreement between feasible points and auxiliary query points.

Introducing $y = x$ gives the equivalent problem

$$\min_{x \in \mathcal{X},\, y \in \mathcal{Y}} \{f(y) : y = x\}.$$

The Lagrangian is

$$\mathcal{L}(x, y, q) \coloneqq f(y) + \langle q, x - y \rangle .$$

For a fixed multiplier q, minimizing the Lagrangian separates into minimizing $\langle q, x\rangle$ over $\mathcal{X}$ and $f(y)-\langle q, y\rangle$ over $\mathcal{Y}$. Even at an optimal multiplier, these subproblems can have multiple minimizers, so independently selected minimizers need not satisfy $x = y$ [5, Section 5.5.5]. The multiplier update below gives (3.18). The linear oracle chooses x_{t+1} by minimizing $\langle q_t, x\rangle$ over $\mathcal{X}$. At iteration t, let s_t denote the subgradient returned by the oracle at y_t, satisfying Assumption 3.1(iii). The relevant subgradient and derivative are

$$\partial_y \mathcal{L}(x, y, q) = \partial f(y) - q, \qquad \nabla_q \mathcal{L}(x, y, q) = x - y.$$

These expressions suggest descent in y and ascent in q. For $\alpha, \eta > 0$, take a projected subgradient step in y of size $1/\alpha$, using the updated multiplier q_{t+1}, and a gradient ascent step in q of size η, using the new points x_{t+1} and y_{t+1}. This gives the coupled updates

$$y_{t+1} = \Pi_{\mathcal{Y}}\left(y_t + \frac{q_{t+1} - s_t}{\alpha}\right), \qquad q_{t+1} = q_t - \eta(y_{t+1} - x_{t+1}). \tag{3.4}$$

Eliminating q_{t+1} gives an explicit update with one projection. For a nonempty closed convex set $\mathcal{Y}$, Lemma 2.10 characterizes $p = \Pi_{\mathcal{Y}}(v)$ by $\langle p - v, w - p\rangle \geq 0$ for every $w \in \mathcal{Y}$. Taking $p = y_{t+1}$ and $v = y_t + (q_{t+1} - s_t)/\alpha$ in this condition and multiplying by $\alpha > 0$ gives

$$\langle s_t - q_{t+1} + \alpha(y_{t+1} - y_t), w - y_{t+1}\rangle \geq 0 \qquad (w \in \mathcal{Y}). \tag{3.5}$$

Substituting the multiplier update (3.4) into the first argument of the inner product in (3.5) gives

$$(\alpha + \eta)y_{t+1} - (\alpha y_t + \eta x_{t+1} + q_t - s_t).$$

Dividing the resulting condition by $\alpha + \eta > 0$ and applying the same projection characterization gives the explicit update

$$y_{t+1} \coloneqq \Pi_{\mathcal{Y}} \left(\frac{\alpha y_t + \eta x_{t+1} + q_t - s_t}{\alpha + \eta} \right). \tag{3.6}$$

By Assumption 3.1(ii), comparing the linear-oracle minimizer x_{t+1} with $x^* \in \mathcal{X}$ gives

$$\langle q_t, x_{t+1} - x^* \rangle \leq 0. \tag{3.7}$$

Algorithm 1 Domain separation for a nonsmooth convex objective

Require: Integer query budget $T \geq 1$, $\alpha, \eta > 0$, $x_1 \in \mathcal{X}$

$y_1 \leftarrow x_1$, $q_1 \leftarrow 0$
for $t = 1, \ldots, T-1$ **do**
  $x_{t+1} \leftarrow \mathrm{LOO}_{\mathcal{X}}(q_t)$
  Query s_t at y_t
  $y_{t+1} \leftarrow \Pi_{\mathcal{Y}} \left(\frac{\alpha y_t + \eta x_{t+1} + q_t - s_t}{\alpha + \eta} \right)$
  $q_{t+1} \leftarrow q_t - \eta(y_{t+1} - x_{t+1})$
end for
return $\bar{x}_T \coloneqq \frac{1}{T} \sum_{t=1}^{T} x_t$

3.5 Convergence analysis

3.5.1 Objective bound and parameter choice

Theorem 3.2 (General convex objective bound)**.** *Under Assumption 3.1, Algorithm 1 with any positive α, η and integer $T \geq 1$ satisfies*

$$0 \leq f(\bar{x}_T) - f^* \leq \frac{L^2}{2\alpha} + \frac{\alpha D^2}{2T} + \frac{(T-1)\eta D^2}{2T} + \frac{L^2}{2\eta T}. \tag{3.8}$$

Corollary 3.3 (Convex convergence rate with explicit parameter choices)**.** *Under Assumption 3.1, assume $L, D > 0$, fix an integer $T \geq 1$, and take*

$$\eta := \frac{L}{D\sqrt{T}}, \qquad \alpha := T\eta. \tag{3.9}$$

Then Algorithm 1 satisfies

$$f(\bar{x}_T) - f^* \leq \frac{2LD}{\sqrt{T}}. \tag{3.10}$$

Proof. The choices in (3.9) are positive. Substituting them into (3.8) gives

$$f(\bar{x}_T) - f^* \leq \frac{LD}{\sqrt{T}} + \left(\frac{T-1}{T} + 1\right)\frac{LD}{2\sqrt{T}} \leq \frac{2LD}{\sqrt{T}}.$$

□

The first-order and ordinary linear-oracle complexity is $O(\epsilon^{-2})$. Indeed, for $L, D, \epsilon > 0$, choose

$$T := \left\lceil \left(\frac{2LD}{\epsilon}\right)^2 \right\rceil. \tag{3.11}$$

The ceiling ensures $T \geq 1$ and $\sqrt{T} \geq 2LD/\epsilon$, so (3.10) gives $f(\bar{x}_T) - f^* \leq \epsilon$. Algorithm 1 makes $T-1$ calls to each oracle, with at most one additional linear-oracle call to initialize x_1. Thus both oracle counts respect the query budget T. Since $T \leq (2LD/\epsilon)^2 + 1$ in (3.11), these counts have the claimed order for fixed L, D as $\epsilon \downarrow 0$.

3.5.2 Proof of the objective bound

Proof of Theorem 3.2. Fix a minimizer $x^* \in \mathcal{X}$. For $t = 1, \dots, T-1$,

$$
\begin{aligned}
f(y_t) - f^* &\overset{(A)}{\le} \langle s_t, y_t - x^* \rangle \\
&= \langle s_t, y_{t+1} - x^* \rangle + \langle s_t, y_t - y_{t+1} \rangle \\
&\overset{(B)}{\le} \langle q_{t+1}, y_{t+1} - x^* \rangle + \alpha \langle y_t - y_{t+1}, y_{t+1} - x^* \rangle + \langle s_t, y_t - y_{t+1} \rangle \\
&= \langle q_{t+1}, y_{t+1} - x^* \rangle + \frac{\alpha}{2}(\|y_t - x^*\|^2 - \|y_{t+1} - x^*\|^2) \\
&\quad + \langle s_t, y_t - y_{t+1} \rangle - \frac{\alpha}{2}\|y_t - y_{t+1}\|^2 \\
&\overset{(C)}{\le} \langle q_{t+1}, y_{t+1} - x^* \rangle + \frac{\alpha}{2}(\|y_t - x^*\|^2 - \|y_{t+1} - x^*\|^2) + \frac{\|s_t\|^2}{2\alpha} \\
&\overset{(D)}{\le} \langle q_{t+1}, y_{t+1} - x^* \rangle + \frac{\alpha}{2}(\|y_t - x^*\|^2 - \|y_{t+1} - x^*\|^2) + \frac{L^2}{2\alpha}.
\end{aligned}
\tag{3.12}
$$

Here (A) uses (2.6); (B) uses (3.5) with $w = x^*$; (C) uses Young's inequality (2.3); and (D) uses Assumption 3.1(iii).

By the multiplier update (3.4),

$$
\begin{aligned}
&\langle q_{t+1}, y_{t+1} - x^* \rangle \\
&\quad = \langle q_t, x_{t+1} - x^* \rangle + \langle q_t, y_{t+1} - x_{t+1} \rangle - \eta \langle y_{t+1} - x_{t+1}, y_{t+1} - x^* \rangle \\
&\quad = \langle q_t, x_{t+1} - x^* \rangle + \frac{\|q_t\|^2 - \|q_{t+1}\|^2}{2\eta} + \frac{\eta}{2}\|y_{t+1} - x_{t+1}\|^2 \\
&\qquad - \frac{\eta}{2}(\|y_{t+1} - x_{t+1}\|^2 + \|y_{t+1} - x^*\|^2 - \|x_{t+1} - x^*\|^2) \\
&\quad = \frac{\|q_t\|^2 - \|q_{t+1}\|^2}{2\eta} + \langle q_t, x_{t+1} - x^* \rangle + \frac{\eta}{2}(\|x_{t+1} - x^*\|^2 - \|y_{t+1} - x^*\|^2) \\
&\quad \overset{(A)}{\le} \frac{\|q_t\|^2 - \|q_{t+1}\|^2}{2\eta} + \frac{\eta D^2}{2}.
\end{aligned}
\tag{3.13}
$$

Here (A) uses (3.7), the diameter bound in Assumption 3.1(i), and $\eta > 0$. Substituting (3.13) into (3.12) gives

$$\begin{aligned} &f(y_t) - f^* + \frac{\|q_{t+1}\|^2 - \|q_t\|^2}{2\eta} \\ &\quad \le \frac{\alpha}{2}(\|y_t - x^*\|^2 - \|y_{t+1} - x^*\|^2) + \frac{L^2}{2\alpha} + \frac{\eta D^2}{2}. \end{aligned} \tag{3.14}$$

Summing (3.14) and using $q_1 = 0$ gives

$$\begin{aligned} &\sum_{t=1}^{T-1}[f(y_t) - f^*] + \frac{\|q_T\|^2}{2\eta} \\ &\quad \le \frac{\alpha}{2}(\|y_1 - x^*\|^2 - \|y_T - x^*\|^2) + (T-1)\left(\frac{L^2}{2\alpha} + \frac{\eta D^2}{2}\right) \\ &\quad \overset{\text{(A)}}{\le} \frac{\alpha D^2}{2} - \frac{\alpha}{2}\|y_T - x^*\|^2 + (T-1)\left(\frac{L^2}{2\alpha} + \frac{\eta D^2}{2}\right). \end{aligned} \tag{3.15}$$

Here (A) uses $y_1 = x_1 \in \mathcal{X}$ and Assumption 3.1(i). Also,

$$f(y_T) - f^* \overset{\text{(A)}}{\le} L\|y_T - x^*\| \overset{\text{(B)}}{\le} \frac{L^2}{2\alpha} + \frac{\alpha}{2}\|y_T - x^*\|^2. \tag{3.16}$$

Here (A) uses (3.2), and (B) uses Young's inequality (2.3). Adding (3.16) to (3.15) gives

$$\begin{aligned} &\sum_{t=1}^{T}[f(y_t) - f^*] + \frac{\|q_T\|^2}{2\eta} \\ &\quad \le \frac{\alpha D^2}{2} + (T-1)\left(\frac{L^2}{2\alpha} + \frac{\eta D^2}{2}\right) + \frac{L^2}{2\alpha} \\ &\quad = \frac{TL^2}{2\alpha} + \frac{\alpha D^2}{2} + \frac{(T-1)\eta D^2}{2} =: C_T. \end{aligned} \tag{3.17}$$

For $T = 1$, the sums over $t = 1, \dots, T-1$ are empty.

Set $\bar{y}_T := T^{-1}\sum_{t=1}^{T} y_t$. By (3.4), $q_1 = 0$, and $x_1 = y_1$,

$$q_T = \sum_{t=1}^{T-1}(q_{t+1} - q_t) = \eta\sum_{t=2}^{T}(x_t - y_t) = \eta T(\bar{x}_T - \bar{y}_T). \tag{3.18}$$

Since $\bar{x}_T, \bar{y}_T \in \mathcal{Y}$,

$$\begin{aligned} T[f(\bar{x}_T) - f^*] &\overset{(A)}{\leq} T[f(\bar{y}_T) - f^*] + LT\|\bar{x}_T - \bar{y}_T\| \\ &\overset{(B)}{\leq} \sum_{t=1}^{T}[f(y_t) - f^*] + \frac{L}{\eta}\|q_T\| \\ &\overset{(C)}{\leq} C_T + \frac{L}{\eta}\|q_T\| - \frac{\|q_T\|^2}{2\eta} \\ &= C_T + \frac{L^2}{2\eta} - \frac{1}{2\eta}(\|q_T\| - L)^2 \\ &\overset{(D)}{\leq} C_T + \frac{L^2}{2\eta}. \end{aligned} \tag{3.19}$$

Here (A) uses (3.2); (B) uses Jensen's inequality (2.7) and (3.18); (C) uses (3.17); and (D) uses $\eta > 0$. Thus,

$$f(\bar{x}_T) - f^* \leq \frac{C_T}{T} + \frac{L^2}{2\eta T} = \frac{L^2}{2\alpha} + \frac{\alpha D^2}{2T} + \frac{(T-1)\eta D^2}{2T} + \frac{L^2}{2\eta T}.$$

Convexity of $\mathcal{X}$ gives $\bar{x}_T \in \mathcal{X}$, so $f(\bar{x}_T) \geq f^*$. □

3.5.3 Subgradient bounds on auxiliary sets

Consider

$$f(x) \coloneqq \max\{e^{-x}, e^x\}, \qquad \mathcal{X} \coloneqq [-1, 1].$$

This function has subgradients bounded by e on $\mathcal{X}$, but has no finite global subgradient bound. If $x_1 \coloneqq 1/2$ and $D \coloneqq 2$, the enclosing ball in (3.3) is $\mathcal{Y} = [-3/2, 5/2]$, where the required bound increases to $L = e^{5/2}$. Indeed, $f(x) = e^{|x|}$ has derivative magnitude $e^{|x|}$ for $x \neq 0$ and subdifferential $[-1, 1]$ at zero. The largest $|x|$ in this auxiliary interval is $5/2$.

For a convex L-Lipschitz function on the convex set $\mathcal{X}$, the construction

$$\widetilde{f}(y) \coloneqq \inf_{z \in \mathcal{X}}\{f(z) + L\|y - z\|\} \tag{3.20}$$

is convex and L-Lipschitz on $\mathbb{R}^n$ and agrees with f on $\mathcal{X}$ by the McShane–Whitney extension theorem [33, Theorem 2.3 and Proposition 3.2]. See also [17, Appendix D] for its use in projection-free optimization. In the preceding example, taking $L \coloneqq e$ in (3.20) gives $\widetilde{f}(x) = f(x)$ for $|x| \leq 1$ and $\widetilde{f}(x) = e|x|$ otherwise. Indeed, if $x > 1$ and $z \in [-1, 1]$, the subgradient inequality (2.6) at 1 with subgradient e gives $f(z) \geq e + e(z - 1) = ez$. Hence $f(z) + e|x - z| \geq ex$, with equality at $z = 1$. Symmetry gives the case $x < -1$. The extension preserves the values on $\mathcal{X}$, but evaluating it involves optimization over $\mathcal{X}$. Its existence does not provide an inexpensive subgradient oracle.

3.6 Discussion and conclusion

The method returns a feasible average while querying subgradients in the auxiliary set. It is useful when projection onto this set and linear optimization over $\mathcal{X}$ are inexpensive. The guarantee depends on a uniform subgradient norm bound on the auxiliary set.

Chapter 4 extends this construction to functional inequalities. It keeps the ordinary linear oracle for $\mathcal{X}$ and uses additional multipliers for those constraints.

Chapter 4

Projection-Free Nonsmooth Optimization with Functional Constraints

Source and author contributions. This chapter is based on the algorithm published jointly with Michael J. Neely [17]. Neely suggested extending the original projection-free method to functional constraints and directed me to the work of Neely and Yu [34]. Their treatment of Lagrangian methods served as a starting point and source of inspiration for my development of the multiplier update in the published paper. I proved the guarantees of the resulting method. Neely reviewed the method and analysis. The original update rules remain unchanged. Only the constraints are replaced by their positive parts. GPT-6 Astra suggested this modification and developed the corresponding proofs.

Material from the 2024 article by Asgari and Neely [17] (© The Author(s) 2024) is adapted under the Creative Commons Attribution 4.0 International license. Notation, presentation, and analysis have been modified.

4.1 Problem and method

This chapter extends the general convex method in Section 3.4 to minimization of f over $\mathcal{X}$ subject to $h_i(x) \leq 0$, $i = 1, \ldots, m$. Write $h : \mathbb{R}^n \to \mathbb{R}^m$ for the constraint map, $h(x) := (h_1(x), \ldots, h_m(x))^\top$. Here $\mathcal{X}$ is compact and convex, and the objective and constraints may be nonsmooth. Linear optimization over $\mathcal{X}$ is assumed inexpensive, whereas Euclidean projection may be costly.

The linear oracle exploits the structure of $\mathcal{X}$. Linear minimization over the intersection $\{x \in \mathcal{X} : h(x) \leq 0\}$ may be more costly, and no inexpensive oracle over this intersection is assumed. The algorithm therefore retains linear minimization over $\mathcal{X}$ and handles the functional inequalities through multiplier updates, allowing controlled violation. Its returned point belongs to $\mathcal{X}$ exactly, with bounds on its objective gap and functional-constraint violation.

4.2 Related work

4.2.1 Primal–dual and penalty methods

The constraint extrapolation method, ConEx, uses linearized constraints [35, Section 2.1]. It covers convex nonsmooth objectives and constraints and permits noisy constraint values and subgradients. Its multiplier update avoids projection onto a bounded set. Its primal step still requires a proximal operation over the original domain.

Exact penalty methods incorporate constraint violation into the objective [36]. In the convex setting considered here, if a Lagrange multiplier exists, a sufficiently large coefficient makes the penalized and constrained problems have the same minimizers. Choosing a coefficient based on this guarantee a priori requires a suitable bound on the norm of a Lagrange multiplier, and a larger coefficient can increase the Lipschitz bound used in the analysis.

4.2.2 Projection-free methods with functional constraints

Lan, Romeijn, and Zhou combine constraint extrapolation with conditional gradients in CoexCG and CoexDurCG [37]. Their nonsmooth extension constructs smooth approximations from explicit max representations of the functions, requiring more information than subgradients alone.

The Level Conditional Gradient method, LCG, solves a sequence of subproblems that seek a point satisfying a prescribed set of inequalities, using conditional gradients [38]. Its nonsmooth extension also requires max representations and smoothing.

4.2.3 Oracle requirements and iteration bounds

The algorithm of Asgari and Neely [17] treats general nonsmooth convex objectives and functional constraints. The published paper permits stochastic subgradients and inexact linear minimization, while constraint values are exact. It uses linear optimization over $\mathcal{X}$ and projection onto an auxiliary set $\mathcal{Y} \supseteq \mathcal{X}$. The exact deterministic specialization used here is compared with the published updates in Section 4.5.1.

For general convex nonsmooth problems, ConEx [35, Theorem 2.3] and the method of Asgari and Neely [17, Theorems 1–2] give $O(\epsilon^{-2})$ iteration bounds, with constraint violation bounds depending on the norm of a Lagrange multiplier. For structured nonsmooth problems, CoexCG and CoexDurCG have the same $O(\epsilon^{-2})$ iteration order, and their constraint violation bounds also depend on this norm [37, Corollaries 2.7 and 3.7]. LCG gives $O(\epsilon^{-2}\log(1/\epsilon))$ total inner iterations for structured nonsmooth problems, with a bound independent of the norm of a Lagrange multiplier [38, Theorem 2.4]. These bounds concern both objective accuracy and constraint violation, with guarantees interpreted in expectation for stochastic variants. Each iteration order above holds under the problem and oracle hypotheses of the theorem or corollary cited with that result, with

all constants other than ϵ held fixed. The cost of an iteration differs between methods, so these bounds alone do not compare running time.

4.3 Contributions and scope

This chapter proves a joint bound on the objective gap and squared constraint violation at the averaged output for the algorithm of Asgari and Neely [17], applied to clipped constraints with exact deterministic oracles. The method uses subgradients and constraint values without requiring smoothness, a special smoothing representation, or proximal operators for the objective and constraint functions. All update coefficients depend on the query budget T and the known bounds D, L, G.

The objective gap has an $O(T^{-1/2})$ upper bound. Under Assumption 4.3, the constraint violation norm also has an $O(T^{-1/2})$ upper bound, as stated in (4.24).

The method uses at most T calls to each oracle: T constraint-vector evaluations, $T-1$ joint subgradient queries, and $T-1$ linear-oracle calls. It also performs $T-1$ projections onto the auxiliary set. These counts assume that $x_1 \in \mathcal{X}$ is supplied. Obtaining x_1 with one additional linear-oracle call keeps the linear-oracle count within the budget T. The published paper's stochastic and inexact-oracle results, application examples, and numerical experiments are outside the scope of the present chapter.

4.4 Problem formulation and oracle assumptions

We follow the notation of Section 1.3. All norms are Euclidean. Vector inequalities and positive parts are componentwise. Let $f, h_1, \ldots, h_m : \mathbb{R}^n \to \mathbb{R}$ be finite-valued and convex. We consider

$$f^* \coloneqq \min_{x\in\mathcal{X}}\{f(x) : h(x) \le 0\}, \qquad h(x) \coloneqq (h_1(x), \dots, h_m(x))^\top. \tag{4.1}$$

Assumption 4.1 (Domain and exact deterministic oracles)**.** *Assume $\mathcal{X}$ is nonempty, compact, and convex; $\mathcal{Y}$ is closed and convex; and a feasible minimizer x^* exists. Fix known positive upper bounds D, L, G such that*

$$\begin{gathered} \mathcal{X} \subseteq \mathcal{Y}, \qquad \sup_{x,x'\in\mathcal{X}} \|x - x'\| \le D, \\ x^* \in \mathcal{X}, \qquad h(x^*) \le 0, \qquad f(x^*) = f^*. \end{gathered} \tag{4.2}$$

For every $v \in \mathbb{R}^n$, an exact linear oracle returns a minimizer of $\langle v, x\rangle$ over $x \in \mathcal{X}$, and the Euclidean projection $\Pi_{\mathcal{Y}}(v)$ is available. At every $y \in \mathcal{Y}$, the exact oracle supplies the constraint values $h(y)$ and subgradient selections satisfying

$$\begin{aligned} &s(y) \in \partial f(y), && \|s(y)\| \le L, \\ &g_i(y) \in \partial h_i(y), && \|g_i(y)\| \le G_i, \qquad \sum_{i=1}^m G_i^2 \le G^2. \end{aligned} \tag{4.3}$$

The constants $G_i \ge 0$ need not be individually known.

For computational efficiency, choose $\mathcal{Y}$ with an inexpensive projection, such as an enclosing ball or box. The bounds L and G must hold throughout that set. Section 3.3.1 explains how enlarging the query set can increase these bounds.

Define the clipped constraints and their oracle selections by

$$c_i(y) \coloneqq [h_i(y)]_+ = \max\{h_i(y), 0\}, \qquad c(y) \coloneqq (c_1(y), \dots, c_m(y))^\top, \qquad y \in \mathbb{R}^n. \tag{4.4}$$

$$k_i(y) \coloneqq \begin{cases} g_i(y), & h_i(y) > 0, \\ 0, & h_i(y) \le 0, \end{cases} \qquad y \in \mathcal{Y}. \tag{4.5}$$

Lemma 4.2 (Clipped constraints and Lipschitz bounds)**.** *Under Assumption 4.1, each c_i is convex and*

$$\{x \in \mathcal{X} : h(x) \le 0\} = \{x \in \mathcal{X} : c(x) = 0\} = \{x \in \mathcal{X} : c(x) \le 0\}. \tag{4.6}$$

At every $y \in \mathcal{Y}$,

$$k_i(y) \in \partial c_i(y), \qquad \|k_i(y)\| \le G_i, \qquad \sum_{i=1}^{m} \|k_i(y)\|^2 \le G^2. \tag{4.7}$$

On $\mathcal{Y}$, f is L-Lipschitz, and h, c, and $y \mapsto \|c(y)\|$ are G-Lipschitz. The function $y \mapsto \|c(y)\|^2$ is convex on $\mathbb{R}^n$.

Proof. Each $c_i = \max\{h_i, 0\}$ is convex as the maximum of two convex functions. Since $c_i \ge 0$ and $c_i(x) = 0$ exactly when $h_i(x) \le 0$, the feasible sets in (4.6) coincide.

Fix $y \in \mathcal{Y}$ and $x \in \mathbb{R}^n$. If $h_i(y) > 0$, then

$$\begin{aligned} c_i(x) &\ge h_i(x) \ge h_i(y) + \langle g_i(y), x - y \rangle \\ &= c_i(y) + \langle k_i(y), x - y \rangle. \end{aligned} \tag{4.8}$$

If $h_i(y) \le 0$, including $h_i(y) = 0$, then $c_i(y) = 0$ is a global minimum of c_i, so the choice $k_i(y) = 0$ is a subgradient. Thus $k_i(y) \in \partial c_i(y)$ in both cases. Its norm bounds in (4.7) follow from $k_i(y) \in \{g_i(y), 0\}$ and (4.3).

For $x, y \in \mathcal{Y}$,

$$\begin{aligned} |f(x) - f(y)| &\le L\|x - y\|, \\ |h_i(x) - h_i(y)| &\le G_i\|x - y\|, \\ |c_i(x) - c_i(y)| &\le G_i\|x - y\|. \end{aligned} \tag{4.9}$$

These inequalities use Lemma 2.4 with (4.3) and (4.7). Squaring and summing the coordinate bounds, using $\sum_i G_i^2 \le G^2$, and taking square roots gives

$$\begin{aligned} \|h(x) - h(y)\| &\le G\|x - y\|, \\ \|c(x) - c(y)\| &\le G\|x - y\|. \end{aligned} \tag{4.10}$$

The reverse triangle inequality then gives

$$\big|\|c(x)\| - \|c(y)\|\big| \le \|c(x) - c(y)\| \le G\|x - y\|. \tag{4.11}$$

Finally, for $x, y \in \mathbb{R}^n$ and $0 \le \lambda \le 1$,

$$\begin{aligned} \|c(\lambda x + (1-\lambda)y)\|^2 &\overset{\text{(A)}}{\le} \|\lambda c(x) + (1-\lambda)c(y)\|^2 \\ &\overset{\text{(B)}}{\le} \lambda\|c(x)\|^2 + (1-\lambda)\|c(y)\|^2. \end{aligned} \tag{4.12}$$

Here (A) uses convexity and nonnegativity of each c_i and monotonicity of the squared norm on the nonnegative orthant. Step (B) uses convexity of the squared norm. □

4.5 Projection-free algorithm

The method maintains a domain point $x_t \in \mathcal{X}$ and a query point $y_t \in \mathcal{Y}$. Linear optimization produces the x_t, while subgradients and constraint values are evaluated at the y_t. The equality

multiplier $q_t \in \mathbb{R}^n$ couples the two sequences. The inequality multiplier $w_t \in \mathbb{R}^m_+$ is updated using the linearized clipped constraints.

Let $T \geq 1$ be an integer query budget for each oracle. We count a full constraint-vector evaluation as one query and a joint evaluation of the objective and constraint subgradients as one subgradient query. For problem (4.1), use the update coefficients

$$a := \frac{2L\sqrt{T}}{D}, \qquad \beta := \frac{a}{2G^2}, \qquad \eta := \frac{a}{T}. \tag{4.13}$$

Initialize

$$y_1 := x_1 \in \mathcal{X}, \qquad q_1 := 0, \qquad w_1 := 0. \tag{4.14}$$

Evaluate $h(y_1)$ and form $c(y_1)$. For $t = 1, \ldots, T-1$, use

$$s_t := s(y_t), \qquad K_t := \begin{bmatrix} k_1(y_t)^\top \\ \vdots \\ k_m(y_t)^\top \end{bmatrix}. \tag{4.15}$$

Define the affine model of the clipped constraints by

$$\widehat{c}_t(y) := c(y_t) + K_t(y - y_t). \tag{4.16}$$

Then update

$$x_{t+1} \in \arg\min_{x \in \mathcal{X}} \langle q_t, x \rangle, \tag{4.17}$$

$$y_{t+1} := \Pi_{\mathcal{Y}} \left(\frac{a y_t + \eta x_{t+1} + q_t - s_t - \beta K_t^\top (w_t + c(y_t))}{a + \eta} \right), \tag{4.18}$$

$$
\begin{aligned}
q_{t+1} &\coloneqq q_t - \eta(y_{t+1} - x_{t+1}), \\
w_{t+1} &\coloneqq [w_t + \widehat{c}_t(y_{t+1})]_+.
\end{aligned}
\tag{4.19}
$$

Evaluate $h(y_{t+1})$ and form $c(y_{t+1})$ after the update. Equations (4.4), (4.14), and (4.19) give

$$
w_t \geq 0, \qquad c(y_t) \geq 0, \qquad w_t + c(y_t) \geq 0. \tag{4.20}
$$

The method returns the average $\bar{x}_T$, with objective gap Δ_T and constraint violation V_T:

$$
\begin{aligned}
\bar{x}_T &\coloneqq \frac{1}{T}\sum_{t=1}^{T} x_t, \\
\Delta_T \coloneqq f(\bar{x}_T) - f^*, &\qquad V_T \coloneqq \|c(\bar{x}_T)\| = \|[h(\bar{x}_T)]_+\|.
\end{aligned}
\tag{4.21}
$$

Convexity of $\mathcal{X}$ ensures that $\bar{x}_T \in \mathcal{X}$. The functional constraints need not hold exactly at this average. Their violation is measured by V_T. The objective gap Δ_T can therefore be negative when the output is infeasible.

Algorithm 2 collects (4.13)–(4.21).

Algorithm 2 Projection-free method with functional constraints

Require: Integer query budget $T \geq 1$, bounds $D, L, G > 0$, and $x_1 \in \mathcal{X}$

Set a, β, η by (4.13)
$y_1 \leftarrow x_1$, $q_1 \leftarrow 0$, $w_1 \leftarrow 0$
Evaluate $h(y_1)$ and form $c(y_1)$
for $t = 1, \ldots, T-1$ **do**
    Query $(s_t, g_1(y_t), \ldots, g_m(y_t))$ jointly
    Form K_t using (4.5) and (4.15)
    $x_{t+1} \leftarrow \mathrm{LOO}_{\mathcal{X}}(q_t)$
    $y_{t+1} \leftarrow \Pi_{\mathcal{Y}}\left(\dfrac{a y_t + \eta x_{t+1} + q_t - s_t - \beta K_t^\top (w_t + c(y_t))}{a + \eta}\right)$
    $q_{t+1} \leftarrow q_t - \eta(y_{t+1} - x_{t+1})$
    $w_{t+1} \leftarrow [w_t + \widehat{c}_t(y_{t+1})]_+$ using (4.16)
    Evaluate $h(y_{t+1})$ and form $c(y_{t+1})$
end for
return $\bar{x}_T = T^{-1}\sum_{t=1}^{T} x_t$

4.5.1 Relation to the published updates

This chapter applies the published algorithm [17] to $c_i = [h_i]_+$ with exact deterministic oracles. Relative to the published notation, $q_t = -\eta Q_t$, $w_t = W_t$, and $a = \alpha + 2G^2\beta$. Since $c_i \geq 0$, the published initialization becomes $w_1 = [-c(y_1)]_+ = 0$, and the lower bound $[-c(y_{t+1})]_+$ in its multiplier update is also zero. This gives (4.18)–(4.19).

The tuning in (4.13) sets $a = 2G^2\beta > 0$, corresponding to $\alpha = 0$ in the published notation. Although the published analysis requires a separate positive α, the proof in Section 4.6.2 bounds the term involving the objective subgradient by $L\|y_{t+1} - y_t\| - (a/4)\|y_{t+1} - y_t\|^2 \leq L^2/a$. See (4.31) and (4.34). Thus the proof covers this parameter choice.

4.6 Convergence analysis

4.6.1 Main convergence guarantee

Assumption 4.3 (Existence of a Lagrange multiplier)**.** *There is a vector μ satisfying*

$$\begin{aligned} &\mu \in \mathbb{R}^m_+, \\ &f(x) + \mu^\top h(x) \geq f^* \qquad (x \in \mathcal{X}). \end{aligned} \tag{4.22}$$

The multiplier μ is not an algorithm input.

Assumption 4.3 supplies the multiplier used in the constraint violation bound and the objective lower bound in (4.24). A sufficient condition is the existence of $x^\circ \in \operatorname{relint} \mathcal{X}$ with $h_i(x^\circ) < 0$ for every $i = 1, \ldots, m$ [5].

Theorem 4.4 (Objective gap and constraint violation bounds)**.** *Under Assumption 4.1, Algorithm 2 with (4.13) satisfies*

$$\frac{\Delta_T}{LD} + \frac{\sqrt{T}}{4}\left(\frac{V_T}{GD}\right)^2 \le \frac{3}{\sqrt{T}}, \qquad \Delta_T \le \frac{3LD}{\sqrt{T}}. \tag{4.23}$$

If Assumption 4.3 also holds, then

$$V_T \le \frac{4GD}{\sqrt{T}}\left(1 + \frac{G\|\mu\|}{L}\right), \qquad -\|\mu\|V_T \le \Delta_T \le \frac{3LD}{\sqrt{T}}. \tag{4.24}$$

4.6.2 Proof and oracle complexity

Proof of Theorem 4.4. For $t = 1, \dots, T-1$ and $v \in \mathbb{R}^n$, (4.15) gives

$$\begin{aligned} \|K_t v\|^2 &= \sum_i \langle k_i(y_t), v\rangle^2 \\ &\overset{\text{(A)}}{\le} \sum_i \|k_i(y_t)\|^2\|v\|^2 \overset{\text{(B)}}{\le} G^2\|v\|^2, \\ \widehat{c}_t(x^*) &= c(y_t) + K_t(x^* - y_t) \overset{\text{(C)}}{\le} c(x^*) = 0. \end{aligned} \tag{4.25}$$

Here (A) is Cauchy–Schwarz, (B) uses (4.7), and (C) applies the subgradient inequality to each c_i, using $k_i(y_t) \in \partial c_i(y_t)$ from (4.7).

By (4.16) and (4.19),

$$
\begin{aligned}
&\beta(w_t + c(y_t))^\top \widehat{c}_t(y_{t+1}) \\
&\quad = \beta w_t^\top \widehat{c}_t(y_{t+1}) + \frac{\beta}{2}(\|\widehat{c}_t(y_{t+1})\|^2 + \|c(y_t)\|^2 - \|K_t(y_{t+1} - y_t)\|^2) \\
&\quad = \frac{\beta}{2}(\|w_t + \widehat{c}_t(y_{t+1})\|^2 - \|w_t\|^2 + \|c(y_t)\|^2 - \|K_t(y_{t+1} - y_t)\|^2) \\
&\quad \overset{\text{(A)}}{\geq} \frac{\beta}{2}(\|[w_t + \widehat{c}_t(y_{t+1})]_+\|^2 - \|w_t\|^2 + \|c(y_t)\|^2 - \|K_t(y_{t+1} - y_t)\|^2) \\
&\quad = \frac{\beta}{2}(\|w_{t+1}\|^2 - \|w_t\|^2 + \|c(y_t)\|^2 - \|K_t(y_{t+1} - y_t)\|^2) \\
&\quad \overset{\text{(B)}}{\geq} \frac{\beta}{2}(\|w_{t+1}\|^2 - \|w_t\|^2 + \|c(y_t)\|^2) - \frac{\beta G^2}{2}\|y_{t+1} - y_t\|^2.
\end{aligned}
\tag{4.26}
$$

Here (A) uses (2.33). Step (B) uses (4.25).

The projection condition (2.31) gives

$$
\begin{aligned}
&\langle s_t + \beta K_t^\top (w_t + c(y_t)) - q_t, y_{t+1} - x^* \rangle \\
&\quad + \frac{\eta}{2}\|y_{t+1} - x_{t+1}\|^2 + \frac{a}{2}\|y_{t+1} - y_t\|^2 \\
&\quad \overset{\text{(A)}}{\leq} -a\langle y_{t+1} - y_t, y_{t+1} - x^* \rangle - \eta \langle y_{t+1} - x_{t+1}, y_{t+1} - x^* \rangle \\
&\qquad + \frac{\eta}{2}\|y_{t+1} - x_{t+1}\|^2 + \frac{a}{2}\|y_{t+1} - y_t\|^2 \\
&\quad = \frac{\eta}{2}(\|x_{t+1} - x^*\|^2 - \|y_{t+1} - x^*\|^2) + \frac{a}{2}(\|y_t - x^*\|^2 - \|y_{t+1} - x^*\|^2) \\
&\quad \overset{\text{(B)}}{\leq} \frac{\eta D^2}{2} + \frac{a}{2}(\|y_t - x^*\|^2 - \|y_{t+1} - x^*\|^2).
\end{aligned}
\tag{4.27}
$$

Here (A) applies (2.31) to (4.18) with comparison point x^*. Step (B) uses (4.2).

For the q_t terms,

$$
\begin{aligned}
&- \langle q_t, y_{t+1} - x^* \rangle + \frac{\eta}{2}\|y_{t+1} - x_{t+1}\|^2 \\
&= -\langle q_t, y_{t+1} - x_{t+1} \rangle + \frac{\eta}{2}\|y_{t+1} - x_{t+1}\|^2 - \langle q_t, x_{t+1} - x^* \rangle \\
&= \frac{\|q_t - \eta(y_{t+1} - x_{t+1})\|^2 - \|q_t\|^2}{2\eta} - \langle q_t, x_{t+1} - x^* \rangle \\
&= \frac{\|q_{t+1}\|^2 - \|q_t\|^2}{2\eta} - \langle q_t, x_{t+1} - x^* \rangle \\
&\overset{\text{(A)}}{\geq} \frac{\|q_{t+1}\|^2 - \|q_t\|^2}{2\eta}.
\end{aligned}
\tag{4.28}
$$

Step (A) uses (4.17).

Also,

$$
\begin{aligned}
&\langle s_t, y_{t+1} - x^* \rangle \\
&= \langle s_t, y_t - x^* \rangle + \langle s_t, y_{t+1} - y_t \rangle \\
&\overset{\text{(A)}}{\geq} f(y_t) - f^* + \langle s_t, y_{t+1} - y_t \rangle \\
&\overset{\text{(B)}}{\geq} f(y_t) - f^* - L\|y_{t+1} - y_t\|,
\end{aligned}
\tag{4.29}
$$

$$
\begin{aligned}
&(w_t + c(y_t))^\top K_t (y_{t+1} - x^*) \\
&= (w_t + c(y_t))^\top (\widehat{c}_t(y_{t+1}) - \widehat{c}_t(x^*)) \\
&\overset{\text{(C)}}{\geq} (w_t + c(y_t))^\top \widehat{c}_t(y_{t+1}).
\end{aligned}
$$

Here (A) is the subgradient inequality with $f(x^*) = f^*$ from (4.2). Step (B) is Cauchy–Schwarz with (4.3). Step (C) uses (4.20) and (4.25).

Substituting the multiplier identity (4.28) and the subgradient bounds (4.29) into (4.27) gives

$$\begin{aligned} & f(y_t) - f^* + \beta(w_t + c(y_t))^\top \widehat{c}_t(y_{t+1}) \\ & \quad + \frac{a}{2}(\|y_{t+1} - x^*\|^2 - \|y_t - x^*\|^2) \\ & \quad + \frac{\|q_{t+1}\|^2 - \|q_t\|^2}{2\eta} + \frac{a}{2}\|y_{t+1} - y_t\|^2 \\ & \quad \le \frac{\eta D^2}{2} + L\|y_{t+1} - y_t\|. \end{aligned} \tag{4.30}$$

For $r \ge 0$,

$$Lr - \frac{a}{4}r^2 = \frac{L^2}{a} - \frac{a}{4}\left(r - \frac{2L}{a}\right)^2 \le \frac{L^2}{a}. \tag{4.31}$$

By (4.13),

$$a = 2\beta G^2 = \eta T = \frac{2L\sqrt{T}}{D}. \tag{4.32}$$

Define

$$\begin{aligned} F(y) &\coloneqq f(y) - f^* + \frac{\beta}{2}\|c(y)\|^2, \\ E_t &\coloneqq \frac{a}{2}\|y_t - x^*\|^2 + \frac{\|q_t\|^2}{2\eta} + \frac{\beta}{2}\|w_t\|^2. \end{aligned} \tag{4.33}$$

Using (4.33),

$$\begin{aligned} F(y_t) + E_{t+1} - E_t &\overset{\text{(A)}}{\le} \frac{\eta D^2}{2} + L\|y_{t+1} - y_t\| - \frac{a - \beta G^2}{2}\|y_{t+1} - y_t\|^2 \\ &= \frac{\eta D^2}{2} + L\|y_{t+1} - y_t\| - \frac{a}{4}\|y_{t+1} - y_t\|^2 \\ &\overset{\text{(B)}}{\le} \frac{\eta D^2}{2} + \frac{L^2}{a}. \end{aligned} \tag{4.34}$$

Here (A) substitutes (4.26) into (4.30), using (4.33). Step (B) is (4.31) at $r = \|y_{t+1} - y_t\|$.

For $T = 1$, the sums over $t = 1, \dots, T-1$ are empty. By (4.33) and (4.14),

$$\begin{aligned}\sum_{t=1}^{T-1} F(y_t) + E_T &= \sum_{t=1}^{T-1} (F(y_t) + E_{t+1} - E_t) + E_1 \\ &\overset{\text{(A)}}{\le} (T-1)\left(\frac{\eta D^2}{2} + \frac{L^2}{a}\right) + \frac{a}{2}\|y_1 - x^*\|^2 \\ &\overset{\text{(B)}}{\le} \frac{aD^2}{2} + (T-1)\left(\frac{\eta D^2}{2} + \frac{L^2}{a}\right).\end{aligned} \tag{4.35}$$

Here (A) sums (4.34). Step (B) uses $y_1 = x_1 \in \mathcal{X}$ from (4.14) and the diameter bound in (4.2).

By (4.2) and (4.6), $f(x^*) = f^*$ and $c(x^*) = 0$. Thus

$$\begin{aligned}F(y_T) &= f(y_T) - f(x^*) + \frac{\beta}{2}\|c(y_T) - c(x^*)\|^2 \\ &\overset{\text{(A)}}{\le} L\|y_T - x^*\| + \frac{\beta G^2}{2}\|y_T - x^*\|^2 \\ &= \frac{a}{2}\|y_T - x^*\|^2 + L\|y_T - x^*\| - \frac{a}{4}\|y_T - x^*\|^2 \\ &\overset{\text{(B)}}{\le} \frac{a}{2}\|y_T - x^*\|^2 + \frac{L^2}{a}.\end{aligned} \tag{4.36}$$

Here (A) uses the Lipschitz bounds (4.9)–(4.10). Step (B) is (4.31) at $r = \|y_T - x^*\|$.

Consequently,

$$\begin{aligned}&\sum_{t=1}^{T} F(y_t) + \frac{\|q_T\|^2}{2\eta} \\ &\quad= \sum_{t=1}^{T-1} F(y_t) + E_T + F(y_T) - \frac{a}{2}\|y_T - x^*\|^2 - \frac{\beta}{2}\|w_T\|^2 \\ &\quad\overset{\text{(A)}}{\le} \sum_{t=1}^{T-1} F(y_t) + E_T + \frac{L^2}{a} \\ &\quad\overset{\text{(B)}}{\le} \frac{aD^2}{2} + \frac{(T-1)\eta D^2}{2} + \frac{TL^2}{a} \\ &\quad= LD\sqrt{T} + \frac{(T-1)LD}{\sqrt{T}} + \frac{LD\sqrt{T}}{2} \le \frac{5}{2}LD\sqrt{T}.\end{aligned} \tag{4.37}$$

Here (A) uses (4.36), and (B) uses (4.35).

By (4.14) and (4.19),

$$\begin{aligned} \bar{y}_T &\coloneqq \frac{1}{T}\sum_{t=1}^{T} y_t, \qquad r_T \coloneqq \|\bar{x}_T - \bar{y}_T\|, \\ q_T &= \eta \sum_{t=1}^{T-1} (x_{t+1} - y_{t+1}) \\ &= \eta \sum_{t=1}^{T} (x_t - y_t) \\ &= \eta T(\bar{x}_T - \bar{y}_T), \\ \frac{\|q_T\|^2}{2\eta T} &= \frac{\eta T}{2} r_T^2 = \frac{a}{2} r_T^2. \end{aligned} \tag{4.38}$$

The function F is convex by (4.12) and convexity of f. Convexity gives $\bar{x}_T \in \mathcal{X}$ and $\bar{y}_T \in \mathcal{Y}$. Hence

$$\begin{aligned} F(\bar{y}_T) + \frac{a}{2} r_T^2 = F(\bar{y}_T) + \frac{\|q_T\|^2}{2\eta T} \\ \overset{\text{(A)}}{\leq} \frac{1}{T}\left(\sum_{t=1}^{T} F(y_t) + \frac{\|q_T\|^2}{2\eta} \right) \overset{\text{(B)}}{\leq} \frac{5LD}{2\sqrt{T}}. \end{aligned} \tag{4.39}$$

Step (A) applies Jensen's inequality to the convex function F defined in (4.33). Step (B) divides (4.37) by T.

By (4.21) and (4.38),

$$\begin{aligned} \Delta_T = f(\bar{x}_T) - f^* &\overset{\text{(A)}}{\leq} f(\bar{y}_T) - f^* + L r_T, \\ V_T = \|c(\bar{x}_T)\| &\overset{\text{(B)}}{\leq} \|c(\bar{y}_T)\| + G r_T. \end{aligned} \tag{4.40}$$

Here (A) and (B) use (4.9) and (4.11), respectively.

Thus

$$\begin{aligned}
\Delta_T + \frac{\beta}{4} V_T^2 &\overset{(A)}{\le} f(\bar{y}_T) - f^* + L r_T + \frac{\beta}{4} \left(\|c(\bar{y}_T)\| + G r_T \right)^2 \\
&\overset{(B)}{\le} f(\bar{y}_T) - f^* + L r_T + \frac{\beta}{2} \|c(\bar{y}_T)\|^2 + \frac{\beta G^2}{2} r_T^2 \\
&= F(\bar{y}_T) + L r_T + \frac{a}{4} r_T^2 \\
&= F(\bar{y}_T) + \frac{a}{2} r_T^2 + \left(L r_T - \frac{a}{4} r_T^2 \right) \\
&\overset{(C)}{\le} F(\bar{y}_T) + \frac{a}{2} r_T^2 + \frac{L^2}{a} \\
&\overset{(D)}{\le} \frac{5LD}{2\sqrt{T}} + \frac{L^2}{a} \\
&= \frac{5LD}{2\sqrt{T}} + \frac{LD}{2\sqrt{T}} = \frac{3LD}{\sqrt{T}}.
\end{aligned} \tag{4.41}$$

Here (A) uses (4.40), (B) bounds the square of a sum by twice the sum of the squares, (C) is (4.31) at $r = r_T$, and (D) is (4.39).

By (4.41),

$$\frac{\Delta_T}{LD} + \frac{\sqrt{T}}{4} \left(\frac{V_T}{GD} \right)^2 = \frac{\Delta_T + \beta V_T^2/4}{LD} \le \frac{3}{\sqrt{T}}.$$

This proves (4.23).

Under Assumption 4.3,

$$\Delta_T \overset{(A)}{\ge} -\mu^\top h(\bar{x}_T) \overset{(B)}{\ge} -\mu^\top c(\bar{x}_T) \overset{(C)}{\ge} -\|\mu\| V_T. \tag{4.42}$$

Here (A) is (4.22) at $\bar{x}_T \in \mathcal{X}$. Step (B) uses $\mu \ge 0$ and (4.4). Step (C) is Cauchy–Schwarz with (4.21).

By (4.13),

$$\begin{aligned}\left(\frac{\sqrt{T}\,V_T}{2GD}-\frac{G\|\mu\|}{L}\right)^2 &= \left(\frac{G\|\mu\|}{L}\right)^2+\frac{\sqrt{T}}{LD}\left(\frac{\beta}{4}V_T^2-\|\mu\|V_T\right)\\ &\overset{(A)}{\le} \left(\frac{G\|\mu\|}{L}\right)^2+\frac{\sqrt{T}}{LD}\left(\frac{\beta}{4}V_T^2+\Delta_T\right)\\ &\overset{(B)}{\le} \left(\frac{G\|\mu\|}{L}\right)^2+3.\end{aligned} \tag{4.43}$$

Step (A) uses (4.42), and (B) uses (4.41). Therefore,

$$\begin{aligned}V_T &\overset{(A)}{\le} \frac{2GD}{\sqrt{T}}\left[\frac{G\|\mu\|}{L}+\sqrt{\left(\frac{G\|\mu\|}{L}\right)^2+3}\right]\\ &\overset{(B)}{\le} \frac{4GD}{\sqrt{T}}\left(1+\frac{G\|\mu\|}{L}\right).\end{aligned}$$

Here (A) follows from (4.43), and (B) uses

$$\left(\frac{G\|\mu\|}{L}+2\right)^2-\left[\left(\frac{G\|\mu\|}{L}\right)^2+3\right]=\frac{4G\|\mu\|}{L}+1\ge 0.$$

Together with (4.42) and the objective upper bound in (4.23), this proves (4.24). □

Corollary 4.5 (Oracle complexity)**.** *Under Assumptions 4.1 and 4.3, Algorithm 2 with* (4.13) *satisfies* $\Delta_T \le \epsilon$ *and* $V_T \le \epsilon$ *for any* $\epsilon > 0$ *whenever its integer query budget obeys*

$$T \ge \max\left\{1, \left\lceil\frac{9L^2D^2}{\epsilon^2}\right\rceil, \left\lceil\frac{16G^2D^2}{\epsilon^2}\left(1+\frac{G\|\mu\|}{L}\right)^2\right\rceil\right\}. \tag{4.44}$$

Proof. The budget in (4.44) gives $\sqrt{T} \ge 3LD/\epsilon$ and $\sqrt{T} \ge (4GD/\epsilon)(1 + G\|\mu\|/L)$. Substituting these inequalities into Theorem 4.4 proves both claims. □

4.7 Conclusion

Using linear optimization over $\mathcal{X}$ and projection onto the auxiliary set $\mathcal{Y}$, the method returns a point with objective gap at most $3LD/\sqrt{T}$ by (4.23). This bound is computable from the known constants.

Under Assumption 4.3, the constraint violation norm also has an $O(T^{-1/2})$ upper bound. Computing the bound in (4.24) requires a known constant $M \geq \|\mu\|$ for a multiplier μ satisfying (4.22). The returned point's violation can also be checked by evaluating $h(\bar{x}_T)$. Unless its value is already available, this check requires one additional constraint-vector evaluation beyond the stated optimization budget.

Chapter 5

Online Convex Optimization with Stochastic Energy Harvesting

Source and author contributions. This chapter is based on research published jointly with Michael J. Neely in [39] and presented at *ACM SIGMETRICS 2021*. Both authors contributed equally to the original research. Neely developed the main research idea and proposed most of the original algorithm. I contributed algorithmic refinements, including separating the allocation into total energy usage and allocation proportions. I developed most of the proofs and wrote the initial manuscript. Neely reviewed the proofs and manuscript and provided suggestions.

To simplify the presentation, this chapter specializes the general energy bounds used in the published work [39] to nonnegative arrivals and total per-slot energy usage, both bounded above by one. The core algorithm, which separately updates total energy usage and allocation proportions, is retained from the published paper. The published paper's simulation study is not reproduced here.

Most changes to the published work in this chapter were produced with AI assistance at my request, including clarifying assumptions and refining the analysis. The Preface gives details.

5.1 Introduction

Consider a system that allocates harvested energy among n subsystems over T time slots, storing unused energy in a finite battery. On slot t, the controller chooses a nonnegative vector

$$\mathbf{X}_t \coloneqq (X_{t,1}, \ldots, X_{t,n})^\top ,$$

where $X_{t,i}$ is the energy assigned to subsystem i. This decision has a cost $c_t(\mathbf{X}_t)$. The function c_t is convex, but it is unknown when $\mathbf{X}_t$ is chosen and is revealed only after the decision.

The costs need not follow a probabilistic model. Energy arrivals are random, and their distribution is unknown. Writing B_t for the energy stored at the start of slot t and A_t for the new arrival, the allocation must satisfy

$$\sum_{i=1}^{n} X_{t,i} \le B_t + A_t. \tag{5.1}$$

Unused energy is stored up to the capacity $B_{\max}$, so

$$B_{t+1} = \min \left\{ B_t + A_t - \sum_{i=1}^{n} X_{t,i}, B_{\max} \right\}, \qquad t \in [T]. \tag{5.2}$$

Here B_1 is the initial energy. Constraint (5.1) ensures that the battery update remains nonnegative.

Choosing $\mathbf{X}_t$ before observing c_t is the standard setting of online convex optimization (OCO) [40, 2, 3, 41, 42]. The battery recursion couples allocations across slots. Spending energy now leaves less for later slots, when it might be more useful. Saving too much can leave energy unused or fill the battery, causing new arrivals to be lost.

We measure performance using cumulative regret: the controller's total cost minus the cost of the best nonnegative allocation $\mathbf{X}^*$ held fixed across slots and chosen with knowledge of the full

cost sequence. In addition to the per-slot total-usage limit in Section 5.2, this comparator must satisfy

$$\sum_{i=1}^{n} X_i^* \le \overline{A}, \qquad \overline{A} \coloneqq \mathbb{E}\left[A_t\right].$$

This comparator need not satisfy the battery constraint on every slot.

The method in [1] achieves an $O(\sqrt{T})$ regret bound with an $O(\sqrt{T})$ battery capacity. Its regret bound contains a factor of $O(\sqrt{n})$. This chapter reduces that factor to $O(\sqrt{\log n})$ for $n \ge 2$. For this comparison, we keep the bound on each component of the cost subgradients fixed as n varies.

The available energy changes with arrivals and past spending, so a simplex update alone cannot enforce the total energy budget. We separate total usage from allocation proportions: the usage update enforces the battery constraint, and entropic mirror descent updates the proportions on a fixed probability simplex. The latter gives the dimension dependence stated above.

5.1.1 Wireless transmission example

For n parallel wireless channels, let $S_{t,i} \ge 0$ be the power gain, $N_{t,i} > 0$ the noise power, and $W_i > 0$ the effective bandwidth. With unit-duration slots, additive white Gaussian noise, and channel conditions fixed within each slot, define the cost as the negative of the throughput given by Shannon's formula [43, Section 5.2.1, Eq. (5.12)]:

$$c_t(\mathbf{X}_t) \coloneqq -\sum_{i=1}^{n} W_i \log_2\left(1 + \frac{X_{t,i} S_{t,i}}{N_{t,i}}\right).$$

This function is convex in the nonnegative allocation. The channel parameters are unknown when the allocation is chosen and revealed afterward. They need not be independent across channels or slots. The energy arrivals must satisfy the assumptions in Section 5.2.1, including (5.7).

For solar harvesting, the i.i.d. arrival model is a simplification: the solar trace studied in [39, Section 6.2] exhibits approximately periodic behavior and is treated there as non-i.i.d.

For the uniform subgradient bound in (5.8),

$$\left|\frac{\partial c_t}{\partial X_i}(\mathbf{X})\right| = \frac{W_i S_{t,i}}{\ln 2\,(N_{t,i} + X_i S_{t,i})} \overset{(A)}{\leq} \frac{W_i S_{t,i}}{N_{t,i} \ln 2},$$

where (A) uses $X_i, S_{t,i} \geq 0$ and $N_{t,i} > 0$. A uniform bound on $W_i S_{t,i}/N_{t,i}$ therefore suffices.

5.1.2 Related work

The OCO benchmark over a fixed feasible set and the convex regret bound are reviewed in Section 2.1. The common mirror descent analysis is in Section 2.5.

Constrained online convex optimization considers fixed constraint functions [44, 45, 46, 47, 48] or time-varying constraints, including stochastic and adversarial models [49, 50, 51], rather than the slotwise battery constraint and state recursion studied here. Bregman methods also appear in constrained OCO [12], although that model does not include the physical storage dynamics studied here.

In more general online learning models with arbitrarily varying rewards and sample-path average constraints, the best feasible performance in hindsight need not be attainable [52]. Here, the conditional-mean condition in (5.7) provides a predictable expected energy supply given the past, while the costs need not follow a probabilistic model.

The closest work, [1], combines online learning with finite energy storage. Its regret and capacity bounds are compared above. Packet-scheduling models minimize transmission completion time or maximize finite-horizon throughput [53, 54]. Other work learns policies with unknown Markov models for energy and data arrivals [55] or studies transmission under time-correlated energy

supply [56]. With i.i.d. arrivals, online power-control policies target long-term throughput or general concave utility [57, 58]. Online throughput maximization with battery overflow is studied in [59]. Network models couple energy management with traffic utility [60, 61].

Finite-capacity inventory models also couple decisions through stored resources [62, 63, 64, 65]. For example, [62] treats a finite-capacity inventory with a linear objective that is known at the decision time. Here the objective can be nonlinear and is revealed only after the decision.

5.1.3 Contributions and scope

We give explicit usage and proportion updates and a sufficient capacity under which the constraint imposed by the energy currently available does not change the usage decisions. The tuned controller uses a planned horizon and the known subgradient bound, but does not require the mean energy arrival.

5.2 Problem formulation

We follow the conventions in Section 1.3. The system has $n \geq 2$ subsystems, and regret is evaluated over slots $t \in [T]$ for an integer $T \geq 1$. The recursions, one-step bounds, and feasibility bounds also apply beyond this horizon when the parameters and bounds are unchanged.

The fixed allocation set is

$$\mathcal{X} := \left\{ \mathbf{x} \in \mathbb{R}_+^n : \sum_{i=1}^{n} x_i \leq 1 \right\}. \tag{5.3}$$

Let A_t denote the energy arriving on slot t. The battery has capacity $B_{\max}$ and starts empty, $B_1 = 0$. At the beginning of slot t, the controller observes the stored energy B_t and the new arrival A_t, then selects an allocation $\mathbf{X}_t$ satisfying

$$\sum_{i=1}^{n} X_{t,i} \leq B_t + A_t, \tag{5.4}$$

$$\mathbf{X}_t \in \mathcal{X}. \tag{5.5}$$

After the allocation is chosen, the cost function $c_t : \mathcal{X} \to \mathbb{R}$ is revealed. This function is continuous and convex. The controller incurs cost $c_t(\mathbf{X}_t)$. The battery is then updated according to (5.2). The algorithm uses a subgradient of c_t at $\mathbf{X}_t$.

5.2.1 Assumptions

The arrivals are independent and follow the same unknown distribution. Their mean is $\overline{A} \coloneqq \mathbb{E}[A_t]$. Each arrival satisfies

$$0 \leq A_t \leq 1 \qquad \text{with probability one.} \tag{5.6}$$

In particular, $0 \leq \overline{A} \leq 1$.

The costs may vary arbitrarily over time. The entire cost sequence may be fixed in advance, or costs may be chosen during operation using observations from earlier slots and random choices. For the expected regret guarantee, we require that the history through slot t does not change the expected arrival on slot $t + 1$.

Let $\mathcal{F}_t$ contain the history through the end of slot t. This includes observed arrivals, allocations, costs, battery states, and any random choices already used to generate them. The requirement is

$$\mathbb{E}[A_{t+1} \mid \mathcal{F}_t] = \overline{A}. \tag{5.7}$$

For example, this condition holds when the arrivals are independent with the same distribution, are independent of the random choices used to select costs, and costs are chosen without access to future arrivals.

The condition concerns the whole history because earlier costs could reveal information about future arrivals. Requiring only A_t and c_t to be independent on the same slot would not rule this out.

At every slot t and every point $\mathbf{x} \in \mathcal{X}$, assume that the controller can obtain a subgradient $\mathbf{g}_t(\mathbf{x}) \in \partial c_t(\mathbf{x})$ satisfying

$$\|\mathbf{g}_t(\mathbf{x})\|_\infty = \max_{1 \le i \le n} |g_{t,i}(\mathbf{x})| \le G. \tag{5.8}$$

Here, $G > 0$ is a known constant. In particular, the subgradient inequality gives

$$c_t(\mathbf{x}) - c_t(\mathbf{y}) \le \mathbf{g}_t(\mathbf{x})^\top (\mathbf{x} - \mathbf{y}) \qquad (\mathbf{x}, \mathbf{y} \in \mathcal{X}). \tag{5.9}$$

Lemma 2.4, with $Q = \mathcal{X}$, the ℓ_1 norm, its dual ℓ_∞ norm, and $L = G$ from (5.8), also gives

$$|c_t(\mathbf{x}) - c_t(\mathbf{y})| \le G\|\mathbf{x} - \mathbf{y}\|_1. \tag{5.10}$$

The bound in (5.8) applies to the subgradient the controller selects at each point. It need not hold for every possible subgradient.

5.2.2 Comparison allocation and regret

We compare the algorithm with an allocation that stays the same on every slot. Its total usage cannot exceed the mean arrival $\overline{A} = \mathbb{E}[A_t]$. This fixed allocation need not satisfy the battery constraint on every slot. It must satisfy

$$U^* := \sum_{i=1}^{n} X_i^* \le \overline{A}, \tag{5.11}$$

$$\mathbf{X}^* \in \mathcal{X}. \tag{5.12}$$

Since $\overline{A} \le 1$, the allocations satisfying (5.11)–(5.12) form the set

$$\mathcal{A} := \left\{ \mathbf{x} \in \mathbb{R}_+^n : \sum_{i=1}^{n} x_i \le \overline{A} \right\}. \tag{5.13}$$

We define regret by

$$\mathrm{Reg}_T(\mathcal{A}) := \sum_{t=1}^{T} c_t(\mathbf{X}_t) - \min_{\mathbf{X} \in \mathcal{A}} \sum_{t=1}^{T} c_t(\mathbf{X}). \tag{5.14}$$

The minimizing vector may depend on the full cost sequence, including when costs depend on past observations.

5.2.3 Main bounds

Under the assumptions of Section 5.2.1, fix a planned horizon $T \ge 1$. Algorithm 3, initialized as specified below and using (5.57) and (5.58), satisfies

$$B_{\max} = 3\sqrt{T}, \qquad \mathbb{E}[\mathrm{Reg}_T(\mathcal{A})] \le G\sqrt{T}\left[6 + \sqrt{2\log n}\right].$$

The capacity choice makes the relaxed usage update feasible on every slot by Theorem 5.1. The regret bound follows from Theorem 5.4 and (5.61). For fixed G, it is $O(\sqrt{T(1+\log n)})$, and the sufficient capacity is independent of n.

5.3 Algorithm

5.3.1 Energy usage and allocation proportions

Write an allocation as

$$\mathbf{X}_t = U_t \mathbf{P}_t, \qquad U_t \coloneqq \sum_{i=1}^{n} X_{t,i}, \qquad \mathbf{P}_t \in \Delta_n, \tag{5.15}$$

where Δ_n is the probability simplex in (2.19). The scalar U_t is total usage, and $\mathbf{P}_t$ gives the allocation proportions. We continue updating the proportions even on slots when no energy is used. The constraints become

$$\mathbf{P}_t \in \Delta_n, \qquad U_t \in [0, \min\{1, B_t + A_t\}].$$

Write the observed cost subgradient and the corresponding usage subgradient as

$$\mathbf{g}_t \coloneqq \mathbf{g}_t(\mathbf{X}_t), \qquad h_t \coloneqq \mathbf{g}_t^\top \mathbf{P}_t. \tag{5.16}$$

By Lemma 2.2, h_t is a subgradient with respect to total usage when the proportions are fixed. Since $\mathbf{P}_t$ is a probability vector,

$$|h_t| \le \sum_{i=1}^{n} P_{t,i} |g_{t,i}| \overset{(\mathrm{A})}{\le} G. \tag{5.17}$$

Here (A) uses (5.8) and $\sum_i P_{t,i} = 1$. The first inequality is the triangle inequality applied to (5.16).

Although c_t is convex in $\mathbf{X}$, the function $c_t(u\mathbf{p})$ need not be jointly convex in $(u, \mathbf{p})$. It is convex in either variable with the other fixed, by affine composition [5, Section 3.2.2]. For a fixed comparator $\mathbf{X}^* = U^*\mathbf{P}^* \in \mathcal{A}$, take $U^* = \sum_i X_i^*$ and $\mathbf{P}^* \in \Delta_n$. When $U^* = 0$, any $\mathbf{P}^* \in \Delta_n$ gives this factorization. Then

$$\mathbf{X}_t - \mathbf{X}^* = (U_t - U^*)\mathbf{P}_t + U^*(\mathbf{P}_t - \mathbf{P}^*). \tag{5.18}$$

Applying (5.9) and substituting (5.18) gives

$$c_t(\mathbf{X}_t) - c_t(\mathbf{X}^*) \overset{(A)}{\leq} \mathbf{g}_t^\top(\mathbf{X}_t - \mathbf{X}^*) = h_t(U_t - U^*) + U^*\mathbf{g}_t^\top(\mathbf{P}_t - \mathbf{P}^*). \tag{5.19}$$

Here (A) is the subgradient inequality at $\mathbf{X}_t$. The equality uses (5.16). The usage and proportion updates use h_t and $\mathbf{g}_t$, respectively. The analysis multiplies the proportion bound by U^*, which is fixed across slots. Neither update uses U^*. Direct linearization in $\mathbf{p}$ at fixed U_t would instead give $U_t\mathbf{g}_t$.

5.3.2 Two Bregman divergences

We use separate Bregman proximity terms for changes in total usage and allocation proportions. For usage, use the quadratic term $(u-v)^2/2$ from the scalar case of (2.18). For proportions, use the Kullback–Leibler (KL) divergence, obtained by restricting the generalized KL divergence in (2.21) to the probability simplex:

$$D_\Psi(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^{n} p_i \log \frac{p_i}{q_i}, \qquad \mathbf{p}, \mathbf{q} \in \Delta_n, \quad q_i > 0. \tag{5.20}$$

We use the convention $0 \log 0 = 0$. The initial proportions are uniform. Lemma 2.7 therefore gives

$$D_\Psi(\mathbf{p}, \mathbf{P}_1) \le \log n \qquad \text{for every } \mathbf{p} \in \Delta_n.$$

These Bregman proximity terms give the mirror-descent updates [32, 14, 8, 66].

5.3.3 Battery feedback and update rules

Choose positive step sizes η and λ for usage and proportions, and a price scale $\gamma > 0$. At the end of slot t, the controller knows the battery level B_{t+1} and the cost subgradient $\mathbf{g}_t$. It uses $\mathbf{g}_t$ to compute $\mathbf{P}_{t+1}$ and B_{t+1} to compute an energy price π_t. At the start of the next slot, it observes A_{t+1} and then chooses U_{t+1} using this price and the available energy.

Define the battery price by

$$\pi_t \coloneqq \gamma(B_{\max} - B_{t+1}) \ge 0. \tag{5.21}$$

The price measures the deficit from full capacity: a lower battery level gives a higher price and discourages usage, while a full battery has zero price. With the usage step size fixed, γ controls how strongly this deficit affects the usage decision. The battery recursion gives

$$\pi_{t+1} = [\pi_t + \gamma(U_{t+1} - A_{t+1})]_+, \qquad \pi_0 = \gamma B_{\max}. \tag{5.22}$$

Indeed, (5.2) and $\sum_i X_{t+1,i} = U_{t+1}$ from (5.15) give $B_{t+2} = \min\{B_{t+1} + A_{t+1} - U_{t+1}, B_{\max}\}$. Substitute this expression into (5.21), and use $b - \min\{z, b\} = \max\{b - z, 0\}$ and $\gamma > 0$.

This battery-dependent penalty is related to drift-plus-penalty energy-harvesting control in [1, Section III-B and Lemma 3]. For the battery price π_t, define

$$V_t \coloneqq \frac{\pi_t^2}{2\gamma} = \frac{\gamma}{2}(B_{\max} - B_{t+1})^2. \tag{5.23}$$

Its one-slot change, or drift, satisfies

$$V_{t+1} - V_t \le \pi_t(U_{t+1} - A_{t+1}) + \frac{\gamma}{2}(U_{t+1} - A_{t+1})^2.$$

Section 5.5.2 derives this inequality and bounds the squared term by one. With π_t fixed and A_{t+1} observed, the remaining usage-dependent term is $\pi_t U_{t+1}$.

The usage update is

$$U_{t+1} \coloneqq \operatorname*{arg\,min}_{u \in I_{t+1}} \left\{ \eta(h_t + \pi_t)u + \frac{1}{2}(u - U_t)^2 \right\}, \tag{5.24}$$

where

$$I_{t+1} \coloneqq [0, \min\{1, B_{t+1} + A_{t+1}\}].$$

Completing the square in (5.24) gives $\frac{1}{2}[u - (U_t - \eta(h_t + \pi_t))]^2$ plus a term independent of u. Thus the usage update is the Euclidean projection onto I_{t+1}:

$$U_{t+1} = [U_t - \eta(h_t + \pi_t)]_0^{\min\{1, B_{t+1} + A_{t+1}\}}, \tag{5.25}$$

where $[z]_\ell^r \coloneqq \min\{r, \max\{\ell, z\}\}$ for $\ell \le r$. Before projection, usage increases when the linearized marginal benefit $-h_t$ exceeds the battery price π_t, and decreases when it falls below the price. The

projection enforces the usage limits and available energy. The cap uses $B_{t+1} + A_{t+1}$, while π_t uses B_{t+1}.

The proportion update is

$$\mathbf{P}_{t+1} \coloneqq \arg\min_{\mathbf{p} \in \Delta_n} \{\lambda \mathbf{g}_t^\top \mathbf{p} + D_\Psi(\mathbf{p}, \mathbf{P}_t)\}. \tag{5.26}$$

Lemma 2.8, with $Q = \Delta_n$, $q = \mathbf{P}_t$, $g = \mathbf{g}_t$, and its step size set to $\lambda > 0$, yields

$$P_{t+1,i} = \frac{P_{t,i} \exp(-\lambda g_{t,i})}{\sum_j P_{t,j} \exp(-\lambda g_{t,j})}. \tag{5.27}$$

The initial proportions are positive. The update formula therefore keeps $P_{t,i} > 0$ for every slot and subsystem.

5.3.4 Algorithm specification

Algorithm 3 Energy usage, allocation proportions, and battery price

Require: Arrivals satisfying (5.6)
Require: Battery capacity $B_{\max} > 0$
Require: Step sizes $\eta, \lambda > 0$ and price scale $\gamma > 0$
$B_1 \leftarrow 0$, $\pi_0 \leftarrow \gamma B_{\max}$, $U_1 \leftarrow 0$
$\mathbf{P}_1 \leftarrow (1/n, \ldots, 1/n)^\top$
for $t = 1, \ldots, T$ **do**
    Observe A_t and the stored energy B_t
    if $t \geq 2$ **then**
        $U_t \leftarrow [U_{t-1} - \eta(h_{t-1} + \pi_{t-1})]_0^{\min\{1, B_t + A_t\}}$
    end if
    Allocate $\mathbf{X}_t \leftarrow U_t \mathbf{P}_t$
    Observe c_t, incur $c_t(\mathbf{X}_t)$, and obtain $\mathbf{g}_t \leftarrow \mathbf{g}_t(\mathbf{X}_t)$
    $h_t \leftarrow \mathbf{g}_t^\top \mathbf{P}_t$
    $B_{t+1} \leftarrow \min\{B_t + A_t - U_t, B_{\max}\}$
    $\pi_t \leftarrow \gamma(B_{\max} - B_{t+1})$
    Update $\mathbf{P}_{t+1}$ using (5.27)
end for

Each slot uses $O(n)$ arithmetic operations, apart from obtaining the subgradient. The algorithm needs $O(n)$ memory.

5.4 Feasibility of the relaxed usage update

Using the fixed interval $[0, 1]$ allows comparison with a fixed usage that may exceed the available energy on individual slots. We show that sufficient battery capacity makes the relaxed algorithm feasible, so its regret bound also applies to Algorithm 3.

Fix $\eta, \gamma, \lambda > 0$ and any sequence satisfying (5.6) and (5.8). The relaxed algorithm replaces (5.24) by

$$U_{t+1} \coloneqq \operatorname*{arg\,min}_{u \in [0,1]} \left\{ \eta(h_t + \pi_t)u + \frac{1}{2}(u - U_t)^2 \right\}, \tag{5.28}$$

and retains the other updates and initializations of Algorithm 3. Completing the square in (5.28) gives its unique minimizer:

$$U_{t+1} = [U_t - \eta(h_t + \pi_t)]_0^1 . \tag{5.29}$$

5.4.1 A lower bound on the battery reserve

We prove the invariant $B_{t+1} \geq rU_t$. The analysis parameter $r \geq 0$ will be chosen to minimize the sufficient capacity bound

$$B_{\mathrm{req}}(r) \coloneqq \frac{G}{\gamma} + 1 + r + \frac{1}{\eta\gamma(1+r)}. \tag{5.30}$$

Theorem 5.1 (Sufficient capacity and feasibility)**.** *If* $B_{\max} \geq B_{\mathrm{req}}(r)$ *in* (5.30)*,* $B_1 = 0$*, and* $U_1 = 0$*, then the relaxed algorithm satisfies*

$$0 \leq rU_t \leq B_{t+1} \leq B_{\max} \qquad \textit{for every } t \geq 1. \tag{5.31}$$

Consequently,

$$U_t \le B_t + A_t, \qquad t \ge 1, \tag{5.32}$$

and the relaxed algorithm coincides with Algorithm 3.

Proof. By $U_1 = 0$ and (5.29), $0 \le U_t \le 1$. Thus $rU_t \ge 0$, and (5.2) gives $B_{t+1} \le B_{\max}$. For $t = 1$,

$$B_2 = \min\{A_1, B_{\max}\} \ge 0 = rU_1. \tag{5.33}$$

This uses $B_1 = U_1 = 0$, (5.2), and $A_1, B_{\max} \ge 0$. Assume (5.31) at slot t.

Case 1: $B_{t+1} \ge 1 + r$. Then

$$(1+r)U_{t+1} \overset{\text{(A)}}{\le} 1 + r \overset{\text{(B)}}{\le} B_{t+1} \overset{\text{(C)}}{\le} B_{t+1} + A_{t+1}. \tag{5.34}$$

Steps (A)–(C) use $U_{t+1} \le 1$, the case assumption, and $A_{t+1} \ge 0$.

Case 2: $B_{t+1} < 1 + r$. By (5.21),

$$\begin{aligned} h_t + \pi_t &\overset{\text{(A)}}{\ge} -G + \pi_t = -G + \gamma(B_{\max} - B_{t+1}) \\ &\overset{\text{(B)}}{\ge} -G + \gamma(B_{\text{req}}(r) - B_{t+1}) \\ &= -G + \gamma\left(\frac{G}{\gamma} + 1 + r + \frac{1}{\eta\gamma(1+r)} - B_{t+1}\right) \\ &= \gamma(1 + r - B_{t+1}) + \frac{1}{\eta(1+r)} \\ &\overset{\text{(C)}}{\ge} \frac{1}{\eta(1+r)}. \end{aligned} \tag{5.35}$$

Steps (A)–(C) use (5.17), $B_{\max} \geq B_{\mathrm{req}}(r)$ with $\gamma > 0$, and $B_{t+1} < 1 + r$, respectively. Thus

$$\begin{aligned} U_{t+1} &= [U_t - \eta(h_t + \pi_t)]_0^1 \\ &\overset{(A)}{\leq} \left[U_t - \frac{\eta}{\eta(1+r)}\right]_0^1 = \left[U_t - \frac{1}{1+r}\right]_0^1 \\ &= \max\left\{U_t - \frac{1}{1+r}, 0\right\}. \end{aligned} \tag{5.36}$$

Here (A) uses the preceding bound and monotonicity of clipping with $\eta > 0$. The equalities use (5.29), $U_t \leq 1$, and $1/(1+r) > 0$. Thus

$$\begin{aligned} (1+r)U_{t+1} &\overset{(A)}{\leq} (1+r)\max\left\{U_t - \frac{1}{1+r}, 0\right\} \\ &= \max\left\{(1+r)\left(U_t - \frac{1}{1+r}\right), 0\right\} = \max\{(1+r)U_t - 1, 0\} \\ &= \max\{rU_t + U_t - 1, 0\} \overset{(B)}{\leq} \max\{rU_t, 0\} = rU_t \\ &\overset{(C)}{\leq} B_{t+1} \overset{(D)}{\leq} B_{t+1} + A_{t+1}. \end{aligned} \tag{5.37}$$

Step (A) uses (5.36) and $1 + r > 0$. Step (B) uses $U_t \leq 1$. Step (C) is the induction hypothesis. Step (D) uses $A_{t+1} \geq 0$. The maximum is monotone, and $rU_t \geq 0$.

In both cases, (5.2) gives

$$\begin{aligned} B_{t+2} - rU_{t+1} &= \min\{B_{t+1} + A_{t+1} - U_{t+1}, B_{\max}\} - rU_{t+1} \\ &= \min\{B_{t+1} + A_{t+1} - U_{t+1} - rU_{t+1}, B_{\max} - rU_{t+1}\} \\ &= \min\{B_{t+1} + A_{t+1} - (1+r)U_{t+1}, B_{\max} - rU_{t+1}\} \\ &\overset{(A)}{\geq} \min\{0, B_{\max} - rU_{t+1}\} \overset{(B)}{\geq} \min\{0, B_{\max} - r\} \\ &\overset{(C)}{\geq} \min\{0, 0\} = 0. \end{aligned} \tag{5.38}$$

By monotonicity of the minimum, (A) follows from (5.34) or (5.37). Step (B) uses $r \geq 0$ and $U_{t+1} \leq 1$. Step (C) uses $B_{\max} \geq B_{\mathrm{req}}(r) > r$ from (5.30). This completes the induction. Finally,

$$U_t \overset{\text{(A)}}{\leq} U_t + B_{t+1} \overset{\text{(B)}}{\leq} U_t + (B_t + A_t - U_t) = B_t + A_t. \tag{5.39}$$

Here (A) uses (5.31). Step (B) uses (5.2). Thus (5.32) holds, and the minimizer in (5.28) belongs to the interval in (5.24). The unique minimizers therefore coincide, and identical initialization gives identical iterates. □

5.4.2 Minimizing the sufficient battery-capacity bound

The following calculation corrects the capacity tuning in [39, Section 4.2], under the unit bounds used here.

For $0 < \eta\gamma \leq 1$, (5.30) gives

$$\begin{aligned} B_{\mathrm{req}}(r) - \frac{G}{\gamma} - \frac{2}{\sqrt{\eta\gamma}} &= 1 + r + \frac{1}{\eta\gamma(1+r)} - \frac{2}{\sqrt{\eta\gamma}} \\ &= \left(\sqrt{1+r} - \frac{1}{\sqrt{\eta\gamma(1+r)}}\right)^2 \\ &\geq 0. \end{aligned} \tag{5.40}$$

Equality in (5.40) holds at $r = 1/\sqrt{\eta\gamma} - 1 \geq 0$, where nonnegativity follows from $\eta\gamma \leq 1$. For $\eta\gamma > 1$,

$$\begin{aligned} B_{\mathrm{req}}(r) - B_{\mathrm{req}}(0) &= r + \frac{1}{\eta\gamma(1+r)} - \frac{1}{\eta\gamma} \\ &= r - \frac{r}{\eta\gamma(1+r)} \\ &= r\left(1 - \frac{1}{\eta\gamma(1+r)}\right) \\ &\overset{\text{(A)}}{\geq} r\left(1 - \frac{1}{\eta\gamma}\right) \\ &\overset{\text{(B)}}{\geq} 0. \end{aligned} \tag{5.41}$$

In (5.41), (A) uses $1 + r \geq 1$ and $r \geq 0$. Step (B) uses $\eta\gamma > 1$ and $r \geq 0$. Equality holds at $r = 0$. Hence (5.40)–(5.41) give

$$r^* := \left[\frac{1}{\sqrt{\eta\gamma}} - 1\right]_+. \tag{5.42}$$

Substituting (5.42) into (5.30) yields

$$\begin{aligned} B^\star_{\mathrm{req}} &:= \min_{r \geq 0} B_{\mathrm{req}}(r) \\ &= \begin{cases} \dfrac{G}{\gamma} + \dfrac{2}{\sqrt{\eta\gamma}}, & 0 < \eta\gamma \leq 1, \\ \dfrac{G}{\gamma} + 1 + \dfrac{1}{\eta\gamma}, & \eta\gamma > 1. \end{cases} \end{aligned} \tag{5.43}$$

Thus $B_{\max} \geq B^\star_{\mathrm{req}}$ in (5.43) suffices for Theorem 5.1 with $r = r^*$.

5.5 Regret analysis of the relaxed algorithm

We analyze the regret in (5.14) against the comparison set $\mathcal{A}$ in (5.13).

Throughout this section, assume that $B_{\max} \geq B_{\mathrm{req}}(r)$ in (5.30) for some $r \geq 0$. Under this condition, Theorem 5.1 ensures that Algorithm 3 uses the relaxed update (5.28).

The price and subgradients in (5.21) and (5.16) satisfy

$$0 \le \pi_t \le \gamma B_{\max}, \qquad \|\mathbf{g}_t\|_\infty \le G, \qquad |h_t| \le G. \tag{5.44}$$

The price bound follows from (5.21) and (5.31). The other two bounds are (5.8) and (5.17). For the proof only, extend the usage and battery recursions to slot $T+1$ using the next arrival A_{T+1} under the same arrival model. The resulting values U_{T+1} and B_{T+2} let us sum the one-step bounds through slot T. No decision through slot T uses this future arrival, and the regret in (5.14) includes costs only through slot T.

5.5.1 Bounds for the two mirror-descent updates

Lemma 5.2 (Updating allocation proportions)**.** *For every* $\mathbf{P}^* \in \Delta_n$ *and every* $t \ge 1$,

$$\mathbf{g}_t^\top(\mathbf{P}_t - \mathbf{P}^*) \le \frac{\lambda G^2}{2} + \frac{D_\Psi(\mathbf{P}^*, \mathbf{P}_t) - D_\Psi(\mathbf{P}^*, \mathbf{P}_{t+1})}{\lambda}. \tag{5.45}$$

Proof. Apply Lemma 2.9 to (5.26) with $Q = \Delta_n$, $\Phi = \Psi$, $\sigma = 1$, step size $\lambda > 0$, and $(x, x^+, u) = (\mathbf{P}_t, \mathbf{P}_{t+1}, \mathbf{P}^*)$ in the ℓ_1/ℓ_∞ norm pair. The curvature follows from (2.22). The reference points are positive by (5.27), while the comparator may lie on the boundary:

$$\begin{aligned}\mathbf{g}_t^\top(\mathbf{P}_t - \mathbf{P}^*) &\overset{\text{(A)}}{\le} \frac{D_\Psi(\mathbf{P}^*, \mathbf{P}_t) - D_\Psi(\mathbf{P}^*, \mathbf{P}_{t+1})}{\lambda} + \frac{\lambda}{2}\|\mathbf{g}_t\|_\infty^2 \\ &\overset{\text{(B)}}{\le} \frac{D_\Psi(\mathbf{P}^*, \mathbf{P}_t) - D_\Psi(\mathbf{P}^*, \mathbf{P}_{t+1})}{\lambda} + \frac{\lambda G^2}{2}.\end{aligned}$$

Step (A) is (2.27). Step (B) uses (5.8). □

Lemma 5.3 (Usage update)**.** *For every $U^* \in [0,1]$ and every $t \geq 1$,*

$$h_t(U_t - U^*) + \pi_t(U_{t+1} - U^*) \leq \frac{\eta G^2}{2} + \frac{(U^* - U_t)^2 - (U^* - U_{t+1})^2}{2\eta}. \tag{5.46}$$

Proof. Projection optimality in (5.29) gives

$$\begin{aligned} \eta(h_t + \pi_t)(U_{t+1} - U^*) &\leq (U_t - U_{t+1})(U_{t+1} - U^*) \\ &= \frac{(U^* - U_t)^2 - (U^* - U_{t+1})^2 - (U_t - U_{t+1})^2}{2}. \end{aligned} \tag{5.47}$$

Hence

$$\begin{aligned} &\eta h_t(U_t - U^*) + \eta\pi_t(U_{t+1} - U^*) \\ &\quad = \eta(h_t + \pi_t)(U_{t+1} - U^*) + \eta h_t(U_t - U_{t+1}) \\ &\quad \overset{\text{(A)}}{\leq} \frac{(U^* - U_t)^2 - (U^* - U_{t+1})^2}{2} + \eta h_t(U_t - U_{t+1}) - \frac{1}{2}(U_t - U_{t+1})^2 \\ &\quad = \frac{(U^* - U_t)^2 - (U^* - U_{t+1})^2}{2} + \frac{\eta^2 h_t^2}{2} - \frac{1}{2}(U_t - U_{t+1} - \eta h_t)^2 \\ &\quad \overset{\text{(B)}}{\leq} \frac{(U^* - U_t)^2 - (U^* - U_{t+1})^2}{2} + \frac{\eta^2 G^2}{2}. \end{aligned} \tag{5.48}$$

Step (A) uses (5.47). Step (B) uses $|h_t| \leq G$ from (5.17). Divide by $\eta > 0$ to obtain (5.46). □

5.5.2 Regret bound

For $V_t = \pi_t^2/(2\gamma)$ in (5.23), the price recursion (5.22) and $[z]_+^2 \leq z^2$ give

$$\begin{aligned} V_{t+1} - V_t &\overset{\text{(A)}}{\leq} \pi_t(U_{t+1} - A_{t+1}) + \frac{\gamma}{2}(U_{t+1} - A_{t+1})^2 \\ &\overset{\text{(B)}}{\leq} \pi_t(U_{t+1} - A_{t+1}) + \frac{\gamma}{2}. \end{aligned} \tag{5.49}$$

Step (A) expands $[\pi_t + \gamma(U_{t+1} - A_{t+1})]^2 - \pi_t^2$ and divides by $2\gamma > 0$. For (B), both A_{t+1} and U_{t+1} belong to $[0,1]$, so $(A_{t+1} - U_{t+1})^2 \leq 1$.

Theorem 5.4 (Expected regret)**.** *Fix positive parameters η, λ, and γ. Suppose the subgradient and energy bounds in Section 5.2.1 and the conditional-mean assumption (5.7) hold. Choose $B_{\max} \geq B_{\mathrm{req}}(r)$ from (5.30) for some $r \geq 0$. Initialize Algorithm 3 with $B_1 = 0$, $U_1 = 0$, and $\mathbf{P}_1 = (1/n, \dots, 1/n)^\top$. Then, for every $T \geq 1$,*

$$\mathbb{E}[\mathrm{Reg}_T(\mathcal{A})] \leq \frac{(\eta + \lambda \overline{A})G^2 + \gamma}{2} T + \frac{\overline{A}^2}{2\eta} + \frac{\overline{A} \log n}{\lambda} + \frac{\gamma B_{\max}^2}{2}. \tag{5.50}$$

Proof. Fix a sample path and any comparison allocation $\mathbf{X}^* = U^* \mathbf{P}^* \in \mathcal{A}$. Define

$$\mathcal{H}_t(\mathbf{X}^*) := \frac{(U^* - U_t)^2}{2\eta} + \frac{U^*}{\lambda} D_\Psi(\mathbf{P}^*, \mathbf{P}_t) + \frac{\pi_t^2}{2\gamma}. \tag{5.51}$$

Then

$$\begin{aligned} & c_t(\mathbf{X}_t) - c_t(\mathbf{X}^*) + \mathcal{H}_{t+1}(\mathbf{X}^*) - \mathcal{H}_t(\mathbf{X}^*) \\ & \overset{\text{(A)}}{\leq} \frac{(\eta + \lambda U^*)G^2 + \gamma}{2} + \pi_t(U^* - U_{t+1}) + \pi_t(U_{t+1} - A_{t+1}) \\ & = \frac{(\eta + \lambda U^*)G^2 + \gamma}{2} + \pi_t(U^* - A_{t+1}) \\ & \overset{\text{(B)}}{\leq} \frac{(\eta + \lambda U^*)G^2 + \gamma}{2} + \pi_t(\overline{A} - A_{t+1}). \end{aligned} \tag{5.52}$$

For (A), substitute (5.45), multiplied by $U^* \geq 0$, and (5.46) into (5.19), and add (5.49). Step (B) uses $\pi_t \geq 0$ from (5.44) and $U^* \leq \overline{A}$ from (5.11). Summing (5.52) gives

$$\begin{aligned} \sum_{t=1}^{T} [c_t(\mathbf{X}_t) - c_t(\mathbf{X}^*)] \leq \frac{(\eta + \lambda U^*)G^2 + \gamma}{2} T + \mathcal{H}_1(\mathbf{X}^*) - \mathcal{H}_{T+1}(\mathbf{X}^*) \\ + \sum_{t=1}^{T} \pi_t(\overline{A} - A_{t+1}). \end{aligned} \tag{5.53}$$

Here $\mathcal{H}_{T+1}(\mathbf{X}^*) \geq 0$ since $U^* \geq 0$ and KL divergence is nonnegative (Lemma 2.7). Uniform initialization and (5.20) give

$$D_\Psi(\mathbf{P}^*, \mathbf{P}_1) = \log n + \sum_{i=1}^{n} P_i^* \log P_i^* \leq \log n,$$

because $0 \leq P_i^* \leq 1$, with $0 \log 0 = 0$. Consequently,

$$\begin{aligned} \mathcal{H}_1(\mathbf{X}^*) &\overset{(A)}{\leq} \frac{\overline{A}^2}{2\eta} + \frac{\overline{A} \log n}{\lambda} + \frac{\pi_1^2}{2\gamma} \\ &\overset{(B)}{\leq} \frac{\overline{A}^2}{2\eta} + \frac{\overline{A} \log n}{\lambda} + \frac{\gamma B_{\max}^2}{2}. \end{aligned} \tag{5.54}$$

Step (A) uses $U_1 = 0$, $0 \leq U^* \leq \overline{A}$ from (5.13), and the entropy bound above in (5.51), with $\eta, \lambda > 0$. Step (B) uses $0 \leq \pi_1 \leq \gamma B_{\max}$ from (5.44) and $\gamma > 0$. Therefore,

$$\begin{aligned} \sum_{t=1}^{T} [c_t(\mathbf{X}_t) - c_t(\mathbf{X}^*)] &\overset{(A)}{\leq} \frac{(\eta + \lambda \overline{A})G^2 + \gamma}{2} T + \frac{\overline{A}^2}{2\eta} \\ &\quad + \frac{\overline{A} \log n}{\lambda} + \frac{\gamma B_{\max}^2}{2} + \sum_{t=1}^{T} \pi_t(\overline{A} - A_{t+1}). \end{aligned} \tag{5.55}$$

Step (A) uses $U^* \leq \overline{A}$, $\mathcal{H}_{T+1}(\mathbf{X}^*) \geq 0$, and (5.54) in (5.53). The bound holds simultaneously for every $\mathbf{X}^* \in \mathcal{A}$, and its right side is independent of $\mathbf{X}^*$. Taking the supremum over $\mathbf{X}^* \in \mathcal{A}$ gives the same bound for $\mathrm{Reg}_T(\mathcal{A})$, including a comparator chosen in hindsight.

By (5.10), (5.3), (5.6), and (5.44),

$$|\mathrm{Reg}_T(\mathcal{A})| \leq 2GT, \qquad |\pi_t(\overline{A} - A_{t+1})| \leq \gamma B_{\max}.$$

Since π_t is $\mathcal{F}_t$-measurable, (5.7) gives

$$\mathbb{E}\left[\pi_t(\overline{A} - A_{t+1}) \mid \mathcal{F}_t\right] = \pi_t(\overline{A} - \mathbb{E}\left[A_{t+1} \mid \mathcal{F}_t\right]) = 0. \tag{5.56}$$

Taking expectations in the regret bound and applying the tower property gives (5.50). □

5.5.3 Choosing the parameters

Fix a horizon $T \geq 1$ before running the controller. The following parameter choice gives the stated regret rate without requiring the mean arrival $\overline{A}$. Set the usage step size and price scale to

$$\eta \coloneqq \frac{1}{G\sqrt{T}}, \qquad \gamma \coloneqq \frac{G}{\sqrt{T}}, \qquad \eta\gamma = \frac{1}{T}. \tag{5.57}$$

For $n \geq 2$, set the proportion step size to

$$\lambda^* \coloneqq \frac{\sqrt{2\log n}}{G\sqrt{T}}. \tag{5.58}$$

Choose the sufficient battery capacity

$$B_{\max} \coloneqq 3\sqrt{T}. \tag{5.59}$$

Since $T \geq 1$, these choices satisfy $0 < \eta\gamma \leq 1$. Equation (5.42) gives $r^* = \sqrt{T} - 1 \geq 0$, and the first branch of (5.43) gives the capacity above: $G/\gamma + 2/\sqrt{\eta\gamma} = \sqrt{T} + 2\sqrt{T} = 3\sqrt{T}$.

The choice $\lambda = \lambda^*$ balances the two proportion terms in (5.50): $\lambda^* G^2 T/2 = \log n/\lambda^*$. Their sum is therefore

$$\overline{A}\left(\frac{\lambda^* G^2 T}{2} + \frac{\log n}{\lambda^*}\right) = \overline{A}G\sqrt{2T\log n}. \tag{5.60}$$

Substituting these choices into (5.50) gives

$$\begin{aligned}\mathbb{E}[\mathrm{Reg}_T(\mathcal{A})] &\overset{(A)}{\leq} G\sqrt{T}\left[\frac{11+\overline{A}^2}{2}+\overline{A}\sqrt{2\log n}\right] \\ &\overset{(B)}{\leq} G\sqrt{T}\left[6+\sqrt{2\log n}\right].\end{aligned} \tag{5.61}$$

Step (A) uses (5.57) in the usage and price terms, (5.60) for the proportion terms, and $B_{\max}^2 = 9T$ from (5.59). Step (B) uses $0 \leq \overline{A} \leq 1$. With G fixed,

$$\mathbb{E}[\mathrm{Reg}_T(\mathcal{A})] = O\left(\sqrt{T(1+\log n)}\right), \qquad B_{\max} = O(\sqrt{T}).$$

5.6 Discussion and conclusion

The regret guarantee compares the controller with the best fixed nonnegative allocation that uses no more than the mean energy arrival per slot. This allocation may require more energy than is available on some slots. Chapter 6 studies policies that assign fractions of each arrival for use in later slots. Each comparison policy respects its own energy availability. Its cost is evaluated using the battery levels and allocations produced by its decisions.

Chapter 6

Online Storage Control via Simplex Disturbance-Action Policies

Source and author contributions. This chapter is based on and extends the preprint "Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies" [67], coauthored with Michael J. Neely. I formulated the storage-control problem, developed the SDAC policy class and the original algorithm, and wrote most of the original proofs of the scalar upper bounds. Neely reviewed the original manuscript and proofs.

The lower-bound construction and proofs in Section 6.4 were developed primarily using ChatGPT, as disclosed in the source manuscript.

6.1 Introduction

Chapter 5 studies online allocation with stochastic energy arrivals in $[0, 1]$ and a fixed-action benchmark. Here, arrivals may follow any sequence in $[0, 1]$, each action is selected before the current arrival is observed, and convex costs may depend jointly on storage and allocation. Each fixed comparator policy is evaluated on its own storage trajectory. Its actions may vary with the arrival history.

The scalar model has storage decay and enough capacity to avoid saturation. A *Simplex Disturbance-Action Control* (SDAC) policy assigns nonnegative fractions of each arrival to withdrawal delays, with total fraction at most one. Lemmas 6.2–6.4 establish affine fixed-policy trajectories, feasibility, and convexity of the resulting costs in the policy parameters.

The *entropic SDAC controller* uses entropic mirror descent and clips requested allocations to the resource currently available. Its regret bound combines surrogate regret with a tracking bound for the executed trajectory. After each action, the controller uses the revealed full cost function to obtain a subgradient at the current fixed-policy state and action.

6.1.1 Contributions and scope

The entropic SDAC controller is feasible on every round. For fixed retention coefficient and cost bounds, parameters chosen for a horizon of T rounds give $O(\sqrt{T\log(H+1)})$ regret against policies with memory H. Truncating infinite-memory policies gives $O(\sqrt{T\log\log T})$ regret against the best fixed infinite-memory SDAC policy. This class includes every feasible scalar fixed-fraction policy under the stated dynamics.

For $G_\alpha := \alpha(L_x/(1-\alpha)+L_u)$ and horizons satisfying (6.50), Theorem 6.10 gives a worst-case expected regret lower bound of $G_\alpha\sqrt{T}/8$ for every causal feasible controller, including randomized controllers. For fixed H, α, L_x, L_u, this matches the upper bound's $\sqrt{T}$ dependence on the horizon T. For fixed $L_x, L_u > 0$, as $\alpha \uparrow 1$ and with the same horizon condition, the lower bound is $\Omega((1-\alpha)^{-1}\sqrt{T})$. The upper and lower bounds still differ by factors of $\sqrt{\log(H+1)}$ in memory dependence and $(1-\alpha)^{-1/2}$ in retention dependence. Closing these gaps remains open.

For energy-harvesting storage, the state, arrivals, and actions represent stored, harvested, and allocated energy, respectively. The model also applies to other resources with proportional storage loss.

The joint manuscript [67] introduces the scalar SDAC policy class and entropic controller. It also gives the equivalence between clipping and projection, the finite- and infinite-memory upper bounds, the fixed-fraction inclusion, and the regret lower bound.

Sections 6.2 and 6.3 give the controller and its finite- and infinite-memory guarantees. Section 6.4 gives the lower bound and compares its parameter dependence with the upper bound. Section 6.6 proves the projection interpretation of action clipping.

6.1.2 Related work

Online control Goulart, Kerrigan, and Maciejowski [68] express control actions as affine functions of past disturbances. For convex costs, this makes finite-horizon policy optimization convex while enforcing constraints involving both the state and the action for every allowed bounded disturbance sequence. Agarwal et al. combine a stabilizing linear controller with learned responses to past disturbances and obtain $\widetilde{O}(\sqrt{T})$ regret against strongly stable linear controllers [69]. Golowich et al. [70] adapt policies based on past disturbances to population dynamics, using nonnegative policy parameters subject to normalization constraints. They use entropy-based policy updates and bound the cost difference caused by changing the policy over time. Their state records population proportions, and the available actions do not depend on the current state. Here, the simplex weights distribute arrivals across withdrawal times, and feasibility requires $0 \le u_t \le \alpha x_t$.

Li et al. consider separate affine constraints on the state and action and assume a linear policy that always maintains a fixed positive margin from every constraint boundary [71]. They project onto a restricted set of policy parameters that preserves feasibility. Such a margin need not exist here: when storage is empty, the only feasible allocation is zero.

Liu et al. [72] allow convex constraints involving both the state and the action. They bound accumulated violation for changing constraints and violation at each round for fixed constraints.

Projection can satisfy fixed affine constraints exactly [72, Table 1]. Our constraint is fixed and affine. Fixed SDAC policies and the clipping rule (6.25) satisfy it exactly.

When costs depend on past decisions, the analysis can first consider the cost of repeatedly using the current decision, then bound the additional cost caused by changing decisions over time [73, 74]. Our tracking estimate bounds this additional cost using nonnegative arrivals and decay in the storage dynamics, while accounting for allocations clipped to the available resource. Kumar, Dean, and Kleinberg [74] also give a lower bound for costs whose dependence on past decisions decreases geometrically, using a Euclidean norm on decision histories. For sufficiently long horizons and discount factors approaching one, their bound scales as $\sqrt{T}$ divided by one minus the discount factor. This parallels the retention dependence of Theorem 6.10 under (6.50). The storage result applies to causal feasible controllers under the state-dependent action constraint and compares with fixed SDAC policies on their own trajectories.

Energy harvesting Under i.i.d. arrivals, Shaviv and Özgür [57] allocate a fixed fraction of stored energy each round and keep the long-run expected transmission rate within a constant of the optimum. Arafa et al. [75] extend such policies to increasing, concave utility functions under i.i.d. arrivals. For Bernoulli arrivals that either fully recharge the battery or add no energy, they characterize allocations that decrease or stay constant between recharges. Zibaeenejad and Chen [76] derive an optimal policy under Bernoulli arrivals with advance knowledge of a fixed number of future arrivals. The policy uses a decreasing allocation sequence when none of those arrivals recharges the battery. These policies and SDAC both distribute arrivals across future allocations. The exact inclusion in Corollary 6.8 concerns feasible fixed-fraction policies under the chapter's own dynamics.

Yu and Neely [1] study utility maximization when current operating conditions are revealed only after the allocation decision. Their journal version allows conditions other than energy arrivals to be correlated over time and to have changing distributions, while energy arrivals are independent and have a common mean [1, Sec. V, Thm. 4]. Asgari and Neely [39] report expected regret $O(\sqrt{T})$ with battery capacity $O(\sqrt{T})$ for fixed dimension, with i.i.d. energy arrivals and time-varying convex losses under their stated independence assumptions. Chapter 5 makes the conditional-mean condition (5.7) explicit for history-dependent costs. Both papers compare expected performance against constant allocations whose total energy use per round does not exceed the mean energy arrival. These comparison allocations need not satisfy the battery constraint on every realized arrival sequence. Neither objective depends directly on the battery state. Lin et al. [77] allow arbitrary energy arrivals and changing objectives that measure the freshness of received information. They observe the current arrival and reward function before allocation. They bound the worst-case ratio between the optimal reward with the entire input sequence known in advance and their controller's reward, under the same battery-capacity and allocation-rate constraints. Our setting combines arbitrary bounded arrivals, decisions made before the current arrival and cost are revealed, and convex costs depending jointly on storage and allocation. We measure additive regret against fixed SDAC policies evaluated on their own trajectories.

Storage management Qin et al. [78] study storage control under uncertain prices and net demand. Their controller balances current operating cost against a term depending on stored energy. Under their assumptions on the input process, it respects the storage constraints and bounds how far its long-run expected average cost exceeds the optimal value. Kim et al. [79] study energy routing in networks with storage that loses energy over time. They give competitive guarantees for the value of meeting demand minus production cost. Their guarantee including

production costs assumes shared storage and advance knowledge of demands. These papers bound long-run average cost or ratios of total performance. This chapter bounds the cumulative cost difference relative to the best fixed policy chosen in hindsight.

Queueing and network optimization Classical queueing control studies throughput and queue stability through dynamic service and scheduling decisions [80, 81, 82], while stochastic network optimization studies tradeoffs between time-average performance and average queue backlog [83].

Khojastepour and Sabharwal [84] study schedulers that split each arrival among service times within a maximum delay. For their class with fixed allocation fractions, the service rate is a fixed weighted sum of current and past arrivals. The fractions are nonnegative and sum to one. SDAC permits their sum to be at most one. Under i.i.d. arrivals, spreading each arrival equally across the allowed service slots minimizes long-run average transmit power over a Gaussian channel within this class of policies with fixed allocation fractions. Their objective uses a fixed cost function.

Inventory and reservoir management Yang et al. [62] study purchasing to meet demand under inventory constraints, observing the current demand and price before acting. They obtain the best possible worst-case ratios between their cost and the minimum cost with all future demands and prices known. Hihat et al. [85] study inventory systems where unsold stock carries over between periods and may expire. They establish feasibility and $O(\sqrt{T})$ expected regret relative to the best fixed replenishment target. Their guarantee requires a uniformly positive conditional probability, given the past, that demand for every product is at least a fixed positive amount. Hihat and Fermanian [86] subsequently develop a method for learning replenishment targets for more general inventory dynamics, without a regret guarantee for that method. Classical reservoir models study

how random inflows and prescribed release rules determine the long-run distribution of stored water [87].

6.2 Model, policies, and algorithm

We follow the shared conventions in Section 1.3.

6.2.1 Model and regret benchmark

Fix a positive integer T and a known retention coefficient $\alpha \in (0, 1)$. The state $x_t \in \mathbb{R}_+$ is the resource level at the beginning of time t. The system starts empty:

$$x_1 = 0. \tag{6.1}$$

At each time $t \in [T]$, the controller chooses an action $u_t \in \mathbb{R}_+$ satisfying

$$0 \le u_t \le \alpha x_t. \tag{6.2}$$

The system then receives an arrival $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{6.3}$$

Here, $1 - \alpha$ is the fraction lost per period, and (6.2) limits allocation to the resource remaining after decay.

At time t, the controller chooses u_t before observing w_t or c_t. After the action is selected, the arrival and the entire cost function are revealed, the controller incurs $c_t(x_t, u_t)$, and the state is updated according to (6.3).

For every feasible trajectory, (6.3), (6.2), and $w_t \in [0,1]$ give $0 \le x_{t+1} \le \alpha x_t + 1$. Iterating from (6.1) gives

$$0 \le x_t \le \sum_{j=0}^{t-2} \alpha^j = \frac{1-\alpha^{t-1}}{1-\alpha} \le \frac{1}{1-\alpha}, \qquad t \in [T+1]. \tag{6.4}$$

The sum is empty at $t = 1$. Together with (6.2), this bound places every feasible state–action pair in the compact set

$$\mathcal{D}_\alpha := \left\{ (x,u) : 0 \le x \le \frac{1}{1-\alpha}, 0 \le u \le \alpha x \right\}. \tag{6.5}$$

Remark 6.1 (Storage capacity)**.** *A physical storage capacity $B \ge 1/(1-\alpha)$ suffices to avoid overflow on every feasible trajectory starting at zero. Thus the same results hold for $x_{t+1} = \min\{B, \alpha x_t - u_t + w_t\}$ at these capacities. For smaller capacities, overflow introduces a minimum operation into the storage dynamics. Fixed-policy trajectories may then cease to be affine in the policy parameters, so the analysis below does not directly apply.*

Each revealed cost function $c_t : \mathbb{R}_+^2 \to \mathbb{R}$ is convex in (x,u) and satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there are constants $L_x, L_u > 0$ such that

$$|c_t(x,u) - c_t(x',u')| \le L_x|x-x'| + L_u|u-u'| \tag{6.6}$$

for all $(x,u),(x',u') \in \mathcal{D}_\alpha$. At every $(x,u) \in \mathcal{D}_\alpha$, we also assume access to a cost subgradient

$$\boldsymbol{\zeta}_t(x,u) := \left(\zeta_t^x(x,u), \zeta_t^u(x,u)\right)^\top \in \partial c_t(x,u) \tag{6.7}$$

satisfying

$$|\zeta_t^x(x,u)| \le L_x, \qquad |\zeta_t^u(x,u)| \le L_u. \tag{6.8}$$

These subgradient bounds hold, for example, when c_t has a convex extension to an open neighborhood of $\mathcal{D}_\alpha$ satisfying (6.6). See [88, Thm. 9.13].

We use x_t and u_t without parameter arguments for the trajectory generated by the online controller. Its cumulative cost is

$$J_T^{\mathrm{on}} \coloneqq \sum_{t=1}^{T} c_t(x_t, u_t). \tag{6.9}$$

Unless stated otherwise, SDAC policies have finite memory, with parameter domain $\mathcal{M}_H$ defined in Subsection 6.2.2. The *fixed-policy trajectory* $\{(x_t(\mathbf{m}), u_t(\mathbf{m}))\}_{t=1}^{T}$ is the unique trajectory obtained by using $\mathbf{m}$ at every time. Its cumulative cost is

$$J_T(\mathbf{m}) \coloneqq \sum_{t=1}^{T} c_t(x_t(\mathbf{m}), u_t(\mathbf{m})). \tag{6.10}$$

For a fixed positive integer H, the regret against the best fixed SDAC policy is

$$\mathrm{Reg}_T(\mathcal{M}_H) \coloneqq J_T^{\mathrm{on}} - \min_{\mathbf{m}\in\mathcal{M}_H} J_T(\mathbf{m}). \tag{6.11}$$

The comparison uses the same realized arrival and cost sequence for the online controller and all fixed policies.

Our objective is to satisfy (6.2) at every time and achieve sublinear regret.

6.2.2 Finite-memory SDAC policies

DAC uses a finite linear filter of past disturbances [69], building on affine disturbance-feedback parameterizations [68]. In our scalar storage model, this filter has the following form:

$$u_t \coloneqq \sum_{i=1}^{H} \theta_i w_{t-i}, \tag{6.12}$$

where H is a positive integer (the memory length) and $\boldsymbol{\theta} \coloneqq (\theta_1, \ldots, \theta_H)^\top \in \mathbb{R}^H$ is the vector of filter coefficients.

An unrestricted filter can request a negative allocation or more than αx_t. SDAC restricts the coefficients so that each fixed policy is feasible and its surrogate costs are convex.

Define the subprobability simplex

$$\mathcal{M}_H \coloneqq \left\{ \mathbf{m} \in \mathbb{R}_+^H : \|\mathbf{m}\|_1 \leq 1 \right\}. \tag{6.13}$$

With the convention $w_\tau \coloneqq 0$ for $\tau \leq 0$, each $\mathbf{m} \in \mathcal{M}_H$ defines an SDAC policy through

$$u_t(\mathbf{m}) \;\coloneqq\; \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle, \tag{6.14}$$

where

$$\boldsymbol{\phi}_t^u \coloneqq \left(\alpha w_{t-1}, \alpha^2 w_{t-2}, \ldots, \alpha^H w_{t-H}\right)^\top \in \mathbb{R}^H. \tag{6.15}$$

Here, $m_i \alpha^i w_s$ is the amount from arrival w_s used i periods later.

Lemma 6.2 (State decomposition under SDAC)**.** *Fix a positive integer H. For $t \in [T]$ and $i \in [H]$, define*

$$r_t(i) \coloneqq \sum_{j=1}^{i} \alpha^{j-1} w_{t-j}, \qquad r_t \coloneqq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j}, \tag{6.16}$$

where the empty sum defining r_1 is zero. Then, for every $\mathbf{m} \in \mathcal{M}_H$ and $t \in [T]$, the fixed-policy state associated with $\mathbf{m}$ satisfies

$$x_t(\mathbf{m}) = \sum_{i=1}^{H} m_i r_t(i) + \left(1 - \sum_{i=1}^{H} m_i\right) r_t. \tag{6.17}$$

Equivalently, with

$$\boldsymbol{\phi}_t^x \coloneqq \big(r_t(1) - r_t, \dots, r_t(H) - r_t\big)^\top, \tag{6.18}$$

the state and action satisfy

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle\,, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle\,. \tag{6.19}$$

Thus, both are affine in $\mathbf{m}$.

Proof. Shifting the index in (6.16) gives, for $1 \le t < T$ and $i \in [H]$,

$$\begin{aligned} r_{t+1} &= w_t + \alpha r_t, \\ r_{t+1}(i) &= w_t + \sum_{j=1}^{i-1} \alpha^j w_{t-j} = w_t + \alpha r_t(i) - \alpha^i w_{t-i}. \end{aligned}$$

With empty sums equal to zero and $w_\tau = 0$ for $\tau \le 0$, the recurrences also hold at $t = 1$ and $i = 1$. Since the system starts empty, $x_1(\mathbf{m}) = r_1(i) = r_1 = 0$. If (6.17) holds at t, then (6.3) and (6.14) give

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \alpha x_t(\mathbf{m}) - u_t(\mathbf{m}) + w_t \\ &= \sum_{i=1}^{H} m_i(\alpha r_t(i) - \alpha^i w_{t-i}) + \left(1 - \sum_{i=1}^{H} m_i\right) \alpha r_t + w_t \\ &= \sum_{i=1}^{H} m_i(r_{t+1}(i) - w_t) + \left(1 - \sum_{i=1}^{H} m_i\right) (r_{t+1} - w_t) + w_t \\ &= \sum_{i=1}^{H} m_i r_{t+1}(i) + \left(1 - \sum_{i=1}^{H} m_i\right) r_{t+1}. \end{aligned}$$

This completes the induction. □

Lemma 6.3 (Feasibility of SDAC policies)**.** *For every* $\mathbf{m} \in \mathcal{M}_H$*, the fixed SDAC policy indexed by* $\mathbf{m}$ *is feasible:*

$$0 \le u_t(\mathbf{m}) \le \alpha x_t(\mathbf{m}) \qquad \text{for every } t \in [T].$$

Proof. By (6.16), for $t \in [T]$ and $i \in [H]$,

$$\alpha r_t(i) - \alpha^i w_{t-i} = \sum_{j=1}^{i-1} \alpha^j w_{t-j} \ge 0, \qquad r_t = \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j} \ge 0.$$

Substituting (6.17) and (6.14),

$$\begin{aligned} u_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \ge 0, \\ \alpha x_t(\mathbf{m}) - u_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i (\alpha r_t(i) - \alpha^i w_{t-i}) + \alpha \left(1 - \sum_{i=1}^{H} m_i\right) r_t \overset{\text{(A)}}{\ge} 0. \end{aligned}$$

Here (A) uses the nonnegative residuals above and the nonnegative weights m_i and $1 - \sum_i m_i$ from (6.13). □

Lemma 6.4 (Convexity and Lipschitz continuity of the surrogate cost)**.** *For* $t \in [T]$*, define the fixed-policy surrogate cost*

$$\ell_t(\mathbf{m}) \coloneqq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H, \tag{6.20}$$

and

$$G_\alpha \coloneqq \alpha \left(\frac{L_x}{1-\alpha} + L_u\right). \tag{6.21}$$

Then ℓ_t is convex on $\mathcal{M}_H$ and, for all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \le G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.$$

For $\mathbf{m} \in \mathcal{M}_H$, use (6.7) to define

$$\begin{aligned} (\zeta_t^x, \zeta_t^u)^\top &:= \boldsymbol{\zeta}_t(x_t(\mathbf{m}), u_t(\mathbf{m})), \\ \mathbf{g}_t &:= \zeta_t^x \boldsymbol{\phi}_t^x + \zeta_t^u \boldsymbol{\phi}_t^u. \end{aligned} \tag{6.22}$$

Then

$$\mathbf{g}_t \in \partial \ell_t(\mathbf{m}), \qquad \|\mathbf{g}_t\|_\infty \le G_\alpha.$$

Proof. By (6.16) and $w_\tau = 0$ for $\tau \le 0$, we have $0 \le r_t(i) \le r_t$, with $r_t(i) = r_t$ when $i \ge t-1$. Thus (6.18) gives, for $i \in [H]$,

$$\begin{aligned} |\phi_{t,i}^x| = |r_t(i) - r_t| = r_t - r_t(i) &= \sum_{j=i+1}^{t-1} \alpha^{j-1} w_{t-j} \\ &\le \sum_{j=i+1}^{\infty} \alpha^{j-1} = \frac{\alpha^i}{1-\alpha} \le \frac{\alpha}{1-\alpha}, \\ 0 \le \phi_{t,i}^u = \alpha^i w_{t-i} &\le \alpha^i \le \alpha. \end{aligned}$$

Here $w_s \in [0,1]$, $0 < \alpha < 1$, and the finite sum is empty when $i \ge t-1$. Hence

$$\|\boldsymbol{\phi}_t^x\|_\infty \le \frac{\alpha}{1-\alpha}, \qquad \|\boldsymbol{\phi}_t^u\|_\infty \le \alpha. \tag{6.23}$$

Lemma 6.3 and (6.4) place the fixed-policy pairs in $\mathcal{D}_\alpha$, where the cost and subgradient bounds apply. The affine formula (6.19) and Lemma 2.2 give convexity. For $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$\begin{aligned}
\ell_t(\mathbf{m}') - \ell_t(\mathbf{m}) &\overset{(A)}{\geq} \zeta_t^x[x_t(\mathbf{m}') - x_t(\mathbf{m})] + \zeta_t^u[u_t(\mathbf{m}') - u_t(\mathbf{m})] \\
&\overset{(B)}{=} \zeta_t^x \left\langle \boldsymbol{\phi}_t^x, \mathbf{m}' - \mathbf{m} \right\rangle + \zeta_t^u \left\langle \boldsymbol{\phi}_t^u, \mathbf{m}' - \mathbf{m} \right\rangle \\
&= \left\langle \zeta_t^x \boldsymbol{\phi}_t^x + \zeta_t^u \boldsymbol{\phi}_t^u, \mathbf{m}' - \mathbf{m} \right\rangle \\
&\overset{(C)}{=} \left\langle \mathbf{g}_t, \mathbf{m}' - \mathbf{m} \right\rangle.
\end{aligned}$$

Here (A) uses the subgradient in (6.7); (B) substitutes (6.19); (C) is (6.22). Thus $\mathbf{g}_t \in \partial \ell_t(\mathbf{m})$, as in (2.8), and

$$\begin{aligned}
\|\mathbf{g}_t\|_\infty &= \|\zeta_t^x \boldsymbol{\phi}_t^x + \zeta_t^u \boldsymbol{\phi}_t^u\|_\infty \\
&\leq |\zeta_t^x| \, \|\boldsymbol{\phi}_t^x\|_\infty + |\zeta_t^u| \, \|\boldsymbol{\phi}_t^u\|_\infty \\
&\overset{(A)}{\leq} L_x \|\boldsymbol{\phi}_t^x\|_\infty + L_u \|\boldsymbol{\phi}_t^u\|_\infty \\
&\overset{(B)}{\leq} L_x \frac{\alpha}{1-\alpha} + L_u \alpha = G_\alpha.
\end{aligned}$$

Here (A) uses (6.8), (B) uses (6.23), and the equality is (6.21). Therefore

$$\begin{aligned}
\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}') &\leq \left\langle \mathbf{g}_t, \mathbf{m} - \mathbf{m}' \right\rangle \\
&\leq \|\mathbf{g}_t\|_\infty \|\mathbf{m} - \mathbf{m}'\|_1 \\
&\leq G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.
\end{aligned}$$

Interchanging $\mathbf{m}$ and $\mathbf{m}'$ gives the Lipschitz bound, as in Lemma 2.4. □

6.2.3 Entropic SDAC controller

The controller updates the allocation fractions $\mathbf{m}_t \in \mathcal{M}_H = S_H$ using entropic online mirror descent (OMD) [14] with step size $\eta > 0$, the unnormalized negentropy Ψ, and generalized KL

divergence D_Ψ from (2.20)–(2.21). The memory H and step size η are supplied parameters. The divergence is defined for $\mathbf{m} \in \mathcal{M}_H$ and $\mathbf{q} > \mathbf{0}$. Initialize

$$\mathbf{m}_1 \coloneqq \frac{1}{H+1}\mathbf{1}. \tag{6.24}$$

Action selection Fixed-policy feasibility need not hold when $\mathbf{m}_t$ varies: the requested action $\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle$ can exceed αx_t. For $H = 2$, $w_1 = 1$, and $w_2 = 0$, take $\mathbf{m}_2 \coloneqq (2/3, 1/3)^\top$ and $\mathbf{m}_3 \coloneqq (1/3, 2/3)^\top$ in $\mathcal{M}_2$. Then

$$x_2 = 1, \qquad u_2 = \frac{2\alpha}{3}, \qquad x_3 = \frac{\alpha}{3}, \qquad \langle \boldsymbol{\phi}_3^u, \mathbf{m}_3 \rangle = \frac{2\alpha^2}{3} > \alpha x_3 = \frac{\alpha^2}{3}.$$

The controller therefore clips the requested action:

$$u_t \coloneqq \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle, \alpha x_t\}. \tag{6.25}$$

By (6.15), $\phi_{t,i}^u = \alpha^i w_{t-i} \geq 0$ for every $i \in [H]$, because $\alpha > 0$ and the arrivals are nonnegative. Thus (6.25) and $\mathbf{m}_t \in \mathcal{M}_H$ give, whenever $x_t \geq 0$,

$$0 \leq u_t \leq \alpha x_t, \qquad x_{t+1} = \alpha x_t - u_t + w_t \geq w_t \geq 0.$$

Since $x_1 = 0$, the online trajectory is feasible. The learning update uses the surrogate subgradient at $\mathbf{m}_t$. Clipping only changes the executed action. Lemma 6.11 shows that the same action results from projecting the policy parameters onto the set compatible with the current storage level.

Surrogate cost and OMD update After w_t and c_t are revealed, the controller evaluates the surrogate subgradient $\mathbf{g}_t = \zeta_t^x \boldsymbol{\phi}_t^x + \zeta_t^u \boldsymbol{\phi}_t^u$ from (6.22), with the cost subgradient evaluated at $(x_t(\mathbf{m}_t), u_t(\mathbf{m}_t))$, and sets

$$\mathbf{m}_{t+1} := \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} \{\eta \langle \mathbf{g}_t, \mathbf{m} \rangle + D_\Psi(\mathbf{m}, \mathbf{m}_t)\}. \tag{6.26}$$

Apply Lemma 2.8 with $d = H$, $Q = S_H = \mathcal{M}_H$, $q = \mathbf{m}_t$, and $g = \mathbf{g}_t$. Its conditions are $\eta > 0$ and $\mathbf{m}_t > \mathbf{0}$. The latter holds initially by (6.24) and is preserved by the following two-step update:

$$\widetilde{m}_{t+1,i} := m_{t,i} \exp\bigl(-\eta g_{t,i}\bigr), \qquad i \in [H]. \tag{6.27}$$

Then project onto $\mathcal{M}_H$:

$$\mathbf{m}_{t+1} = \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} D_\Psi(\mathbf{m}, \widetilde{\mathbf{m}}_{t+1}) = \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}. \tag{6.28}$$

6.3 Performance guarantees

6.3.1 Finite-memory regret guarantee

For every $\mathbf{m} \in \mathcal{M}_H$, the fixed-policy cost (6.10) and surrogate (6.20) give the decomposition

$$\begin{aligned} J_T^{\mathrm{on}} - J_T(\mathbf{m}) = &\sum_{t=1}^{T} \bigl[c_t(x_t, u_t) - \ell_t(\mathbf{m}_t)\bigr] \\ &+ \sum_{t=1}^{T} \bigl[\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\bigr]. \end{aligned} \tag{6.29}$$

The first sum is the tracking difference. The second is surrogate regret. Bounding these terms gives the following guarantee.

Theorem 6.5 (Regret against fixed SDAC policies)**.** *Fix $\alpha \in (0,1)$ and positive integers H, T. Assume the model and feedback of Subsection 6.2.1, with $x_1 = 0$, $w_t \in [0,1]$, and convex costs satisfying the Lipschitz and accessible subgradient bounds with constants $L_x, L_u > 0$. For any step size $\eta > 0$, the controller in Subsection 6.2.3, initialized by (6.24), satisfies*

$$\mathrm{Reg}_T(\mathcal{M}_H) \le \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha, \tag{6.30}$$

where

$$\Gamma_\alpha := \frac{G_\alpha^2}{2} + \frac{\alpha G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}, \tag{6.31}$$

and $G_\alpha = \alpha(\frac{L_x}{1-\alpha} + L_u)$ as in (6.21).

If the horizon T and cost bounds L_x, L_u are supplied, compute G_α and Γ_α from the expressions above and choose $\eta^* := \sqrt{\log(H+1)/(T\Gamma_\alpha)}$. Then

$$\mathrm{Reg}_T(\mathcal{M}_H) \le 2\sqrt{\Gamma_\alpha T \log(H+1)}. \tag{6.32}$$

The proof combines surrogate regret with the loss caused by policy changes, as in the analysis of online learning with memory in [74, App. C, proof of Thm. 3.1].

Proof of Theorem 6.5. Fix $\mathbf{m} \in \mathcal{M}_H = S_H$ and use the ℓ_1/ℓ_∞ dual pair. Lemma 6.4 gives convex ℓ_t and $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$ with $\|\mathbf{g}_t\|_\infty \le G_\alpha$. By Lemma 2.7, Ψ is 1-strongly convex and $D_\Psi(\mathbf{m}, \mathbf{m}_1) \le$

$\log(H+1)$ at (6.24). The updates (6.27)–(6.28) give $\mathbf{m}_t > \mathbf{0}$ for $t \in [T+1]$. Lemma 2.9 and convexity therefore give

$$\begin{aligned}
\sum_{t=1}^{T}[\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})] &\overset{\text{(A)}}{\le} \sum_{t=1}^{T} \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m} \rangle \\
&\overset{\text{(B)}}{\le} \frac{1}{\eta}\sum_{t=1}^{T}[D_\Psi(\mathbf{m}, \mathbf{m}_t) - D_\Psi(\mathbf{m}, \mathbf{m}_{t+1})] + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\
&= \frac{D_\Psi(\mathbf{m}, \mathbf{m}_1) - D_\Psi(\mathbf{m}, \mathbf{m}_{T+1})}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\
&\overset{\text{(C)}}{\le} \frac{D_\Psi(\mathbf{m}, \mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\
&\overset{\text{(D)}}{\le} \frac{\log(H+1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T} G_\alpha^2 \\
&= \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2}.
\end{aligned} \tag{6.33}$$

Here (A) is the subgradient inequality, (B) sums (2.27), (C) uses $D_\Psi \ge 0$, and (D) uses the bounds above. Also, (2.28) gives

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \le \eta\|\mathbf{g}_t\|_\infty \le \eta G_\alpha, \qquad t \in [T]. \tag{6.34}$$

For any $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$, the affine state formula (6.19) and Hölder's inequality give

$$|x_t(\mathbf{m}) - x_t(\mathbf{m}')| = |\langle \boldsymbol{\phi}_t^x, \mathbf{m} - \mathbf{m}' \rangle| \le \|\boldsymbol{\phi}_t^x\|_\infty \|\mathbf{m} - \mathbf{m}'\|_1 \le \frac{\alpha}{1-\alpha}\|\mathbf{m} - \mathbf{m}'\|_1.$$

The final inequality is (6.23). Define $v_t \coloneqq u_t(\mathbf{m}_t)$ and $e_t \coloneqq |x_t - x_t(\mathbf{m}_t)|$ for $t \in [T]$. Then $e_1 = 0$. Fixed-policy feasibility and (6.25) give $0 \le v_t \le \alpha x_t(\mathbf{m}_t)$ and, for $1 \le t < T$,

$$x_{t+1} = [\alpha x_t - v_t]_+ + w_t, \qquad x_{t+1}(\mathbf{m}_t) = [\alpha x_t(\mathbf{m}_t) - v_t]_+ + w_t.$$

By (2.32),

$$\begin{aligned}|x_{t+1} - x_{t+1}(\mathbf{m}_t)| &= |[\alpha x_t - v_t]_+ - [\alpha x_t(\mathbf{m}_t) - v_t]_+| \\ &\le |(\alpha x_t - v_t) - (\alpha x_t(\mathbf{m}_t) - v_t)| \\ &= \alpha |x_t - x_t(\mathbf{m}_t)| = \alpha e_t.\end{aligned}$$

Hence

$$\begin{aligned}e_{t+1} &= |x_{t+1} - x_{t+1}(\mathbf{m}_{t+1})| \\ &\le |x_{t+1} - x_{t+1}(\mathbf{m}_t)| + |x_{t+1}(\mathbf{m}_t) - x_{t+1}(\mathbf{m}_{t+1})| \\ &\le \alpha e_t + |x_{t+1}(\mathbf{m}_t) - x_{t+1}(\mathbf{m}_{t+1})| \\ &\overset{(A)}{\le} \alpha e_t + \frac{\alpha}{1-\alpha}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \\ &\overset{(B)}{\le} \alpha e_t + \frac{\alpha\eta G_\alpha}{1-\alpha}.\end{aligned} \tag{6.35}$$

Here (A) applies the inequality for $|x_t(\mathbf{m}) - x_t(\mathbf{m}')|$ above at time $t+1$; (B) uses (6.34). Since $e_1 = 0$,

$$\begin{aligned}(1-\alpha)\sum_{t=1}^{T} e_t + \alpha e_T &= \left(\sum_{t=1}^{T} e_t - e_1\right) - \alpha\left(\sum_{t=1}^{T} e_t - e_T\right) \\ &= \sum_{t=1}^{T-1} e_{t+1} - \alpha\sum_{t=1}^{T-1} e_t \\ &= \sum_{t=1}^{T-1}(e_{t+1} - \alpha e_t) \\ &\overset{(C)}{\le} \sum_{t=1}^{T-1}\frac{\alpha\eta G_\alpha}{1-\alpha} = \frac{\alpha\eta G_\alpha(T-1)}{1-\alpha}.\end{aligned}$$

Here (C) uses (6.35). Subtracting αe_T and dividing by $1-\alpha > 0$ gives

$$\sum_{t=1}^{T} e_t \le \frac{\alpha\eta G_\alpha(T-1)}{(1-\alpha)^2} - \frac{\alpha e_T}{1-\alpha} \le \frac{\alpha\eta G_\alpha(T-1)}{(1-\alpha)^2}. \tag{6.36}$$

The last inequality uses $e_T \ge 0$. Clipping and fixed-policy feasibility also give

$$0 \le v_t - u_t = [v_t - \alpha x_t]_+ \le [\alpha x_t(\mathbf{m}_t) - \alpha x_t]_+ \le \alpha e_t. \tag{6.37}$$

The first inequality between positive parts uses monotonicity and $v_t \le \alpha x_t(\mathbf{m}_t)$. The last uses $[a]_+ \le |a|$. Both state–action pairs lie in $\mathcal{D}_\alpha$ by feasibility and (6.4). Apply the cost Lipschitz bound (6.6) at (x_t, u_t) and $(x_t(\mathbf{m}_t), v_t)$, and sum:

$$\begin{aligned}\sum_{t=1}^{T} |c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)| &\le \sum_{t=1}^{T} \big(L_x e_t + L_u |u_t - v_t|\big) \\ &\le \sum_{t=1}^{T} \big(L_x e_t + \alpha L_u e_t\big) \\ &= (L_x + \alpha L_u) \sum_{t=1}^{T} e_t \\ &\le \frac{\alpha\eta G_\alpha (L_x + \alpha L_u)(T-1)}{(1-\alpha)^2}.\end{aligned} \tag{6.38}$$

The last two inequalities use (6.37) and (6.36), respectively.

Insert (6.33) and (6.38) into the decomposition (6.29). Since $T - 1 \le T$ and $\sum_{t=1}^{T} \ell_t(\mathbf{m}) = J_T(\mathbf{m})$, this gives

$$\begin{aligned}J_T^{\mathrm{on}} - J_T(\mathbf{m}) &\le \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2} + \frac{\alpha\eta G_\alpha (L_x + \alpha L_u)(T-1)}{(1-\alpha)^2} \\ &\le \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2} + \frac{\alpha\eta G_\alpha (L_x + \alpha L_u) T}{(1-\alpha)^2} \\ &= \frac{\log(H+1)}{\eta} + \eta T \left[\frac{G_\alpha^2}{2} + \frac{\alpha G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}\right] \\ &= \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha.\end{aligned}$$

Taking the supremum over $\mathbf{m} \in \mathcal{M}_H$ gives (6.30) by (6.11). Since $\log(H+1) > 0$ and $\Gamma_\alpha > 0$, $\eta^* = \sqrt{\log(H+1)/(T\Gamma_\alpha)} > 0$, and

$$\frac{\log(H+1)}{\eta^*} = \eta^* T \Gamma_\alpha = \sqrt{\Gamma_\alpha T \log(H+1)},$$

which gives (6.32). □

Remark 6.6 (Dependence on the retention coefficient)**.** *Fix $L_x, L_u > 0$ and let $\alpha \uparrow 1$ through $(0, 1)$. Equation (6.21) gives*

$$(1-\alpha)G_\alpha = \alpha\big(L_x + (1-\alpha)L_u\big) \longrightarrow L_x.$$

Multiplying (6.31) by $(1-\alpha)^3$ yields

$$\begin{aligned}(1-\alpha)^3\Gamma_\alpha &= \frac{1-\alpha}{2}\big((1-\alpha)G_\alpha\big)^2 + \alpha\big((1-\alpha)G_\alpha\big)(L_x + \alpha L_u)\\ &\longrightarrow L_x(L_x + L_u) > 0.\end{aligned}$$

Consequently, the coefficient $2\sqrt{\Gamma_\alpha}$ in the upper bound (6.32) is $\Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$.

6.3.2 Infinite-memory policies and fixed-fraction comparisons

The infinite-memory policy class is

$$\mathcal{M}_\infty \coloneqq \left\{\mathbf{m} = (m_i)_{i\geq 1} : m_i \geq 0, \quad \sum_{i=1}^{\infty} m_i \leq 1\right\}, \tag{6.39}$$

with the feasible action

$$u_t(\mathbf{m}) \coloneqq \sum_{i=1}^{t-1} m_i \alpha^i w_{t-i}. \tag{6.40}$$

The state follows (6.3) from $x_1(\mathbf{m}) = 0$, and (6.10) defines $J_T(\mathbf{m})$ for $\mathbf{m} \in \mathcal{M}_\infty$ as well.

Lemma 6.7 (Approximation of infinite-memory SDAC policies)**.** *Fix positive integers H, T under the model and cost assumptions of Subsection 6.2.1. For every $\mathbf{m} \in \mathcal{M}_\infty$, the infinite-memory policy is feasible. Its truncation $\mathbf{m}^{[H]} \coloneqq (m_1, \ldots, m_H)^\top \in \mathcal{M}_H$ satisfies*

$$\left|J_T(\mathbf{m}^{[H]}) - J_T(\mathbf{m})\right| \leq T G_\alpha \alpha^H. \tag{6.41}$$

If $H \geq T-1$, the two trajectories agree through round T, so their cost difference is zero.

Proof. For $t \geq 2$, the policy agrees through time t with its truncation at $t-1$, since $w_{s-i} = 0$ for $i \geq s$. The truncated vector belongs to $\mathcal{M}_{t-1}$ by (6.39) and (6.13), so Lemma 6.3 applies with memory $t-1$. At $t = 1$, both state and action are zero by (6.1) and (6.40). This proves feasibility of every $\mathbf{m} \in \mathcal{M}_\infty$.

For $t \in [T]$, define

$$\begin{aligned} e_t^u &\coloneqq u_t(\mathbf{m}) - u_t(\mathbf{m}^{[H]}) = \sum_{i=H+1}^{t-1} \alpha^i m_i w_{t-i}, \\ e_t^x &\coloneqq x_t(\mathbf{m}^{[H]}) - x_t(\mathbf{m}). \end{aligned} \tag{6.42}$$

Since $w_s \in [0,1]$, $m_i \geq 0$, $\sum_{i \geq 1} m_i \leq 1$, and $0 < \alpha < 1$, we have

$$0 \leq e_t^u \leq \alpha^{H+1} \sum_{i=H+1}^{t-1} m_i \leq \alpha^{H+1}.$$

Subtracting the two state recursions (6.3) gives

$$e_{t+1}^x = \alpha e_t^x + e_t^u, \qquad 1 \leq t < T, \qquad e_1^x = 0. \tag{6.43}$$

Iterating and using the action bound yields

$$0 \leq e_t^x = \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s^u \leq \alpha^{H+1} \sum_{j=0}^{t-2} \alpha^j \leq \frac{\alpha^{H+1}}{1-\alpha}, \qquad t \in [T]. \tag{6.44}$$

Both trajectories lie in $\mathcal{D}_\alpha$ by feasibility and (6.4). The triangle inequality and (6.6) therefore give

$$\begin{aligned} |J_T(\mathbf{m}^{[H]}) - J_T(\mathbf{m})| &\leq \sum_{t=1}^{T} (L_x e_t^x + L_u e_t^u) \\ &\leq T \left(\frac{L_x}{1-\alpha} + L_u \right) \alpha^{H+1} = T G_\alpha \alpha^H. \end{aligned}$$

The equality is (6.21). If $H \geq T-1$, the sum defining e_t^u is empty, and the recursion for e_t^x gives $e_t^u = e_t^x = 0$ for all $t \in [T]$. □

For $k \in [0, \alpha]$, the fixed-fraction policy chooses $u_t(k) \coloneqq k x_t(k)$ and starts from $x_1(k) = 0$. Substituting this action into (6.3) gives

$$x_{t+1}(k) = (\alpha - k) x_t(k) + w_t, \qquad t \in [T]. \tag{6.45}$$

Corollary 6.8 (Fixed-fraction policies as infinite-memory SDAC policies)**.** *For $k \in [0, \alpha]$, let*

$$m_1(k) \coloneqq \frac{k}{\alpha}, \qquad m_{i+1}(k) \coloneqq \left(1 - \frac{k}{\alpha}\right) m_i(k), \qquad i \geq 1. \tag{6.46}$$

Then $\mathbf{m}(k) \in \mathcal{M}_\infty$, and its trajectory equals that of the feasible fixed-fraction policy $u_t(k) = k x_t(k)$ for every arrival sequence with entries in $[0, 1]$:

$$(x_t(\mathbf{m}(k)), u_t(\mathbf{m}(k))) = (x_t(k), u_t(k)), \qquad t \in [T].$$

Proof. Put $q \coloneqq 1 - k/\alpha \in [0, 1]$. The recurrence (6.46) gives $m_i(k) = (1-q) q^{i-1} \geq 0$, with $q^0 = 1$ also when $q = 0$. For every $N \geq 1$,

$$\sum_{i=1}^{N} m_i(k) = \sum_{i=1}^{N} (q^{i-1} - q^i) = 1 - q^N \leq 1.$$

Taking $N \to \infty$ shows $\sum_{i \geq 1} m_i(k) \leq 1$. Thus $\mathbf{m}(k) \in \mathcal{M}_\infty$ by (6.39). Since $\alpha - k \geq 0$ and $w_t \geq 0$, the recursion (6.45) from $x_1(k) = 0$ gives nonnegative states. The condition $0 \leq k \leq \alpha$ then gives

feasibility in (6.2). Iterating the state recursion and using $\alpha^i m_i(k) = k(\alpha - k)^{i-1}$ from (6.46) in (6.40) yields

$$u_t(\mathbf{m}(k)) = k \sum_{i=1}^{t-1} (\alpha - k)^{i-1} w_{t-i} = k x_t(k) = u_t(k).$$

The two state trajectories start at zero and have identical actions and arrivals in (6.3), so they also coincide. □

Corollary 6.9 (No-regret against infinite-memory SDAC policies)**.** *Under the assumptions of Theorem 6.5, run the entropic SDAC controller with step size* $\eta^* = \sqrt{\log(H+1)/(T\Gamma_\alpha)}$ *and memory*

$$H = H_T \coloneqq \max\left\{1, \left\lceil \frac{\log T}{\log(1/\alpha)} \right\rceil \right\}. \tag{6.47}$$

This choice gives $T\alpha^{H_T} \le 1$*. For*

$$\mathrm{Reg}_T(\mathcal{M}_\infty) \coloneqq J_T^{\mathrm{on}} - \inf_{\mathbf{m} \in \mathcal{M}_\infty} J_T(\mathbf{m}), \tag{6.48}$$

we have

$$\mathrm{Reg}_T(\mathcal{M}_\infty) \le 2\sqrt{\Gamma_\alpha T \log(H_T + 1)} + T G_\alpha \alpha^{H_T}. \tag{6.49}$$

For fixed α, L_x, L_u*, the regret is* $O(\sqrt{T \log\log T}) = o(T)$*. The same bound holds against the best fixed-fraction policy:*

$$J_T^{\mathrm{on}} - \min_{k \in [0,\alpha]} \sum_{t=1}^{T} c_t(x_t(k), u_t(k)) \le \mathrm{Reg}_T(\mathcal{M}_\infty).$$

Proof. For every $\mathbf{m} \in \mathcal{M}_\infty$, $\mathbf{m}^{[H_T]} \in \mathcal{M}_{H_T}$ and

$$\begin{aligned} J_T^{\mathrm{on}} - J_T(\mathbf{m}) &= J_T^{\mathrm{on}} - J_T(\mathbf{m}^{[H_T]}) + J_T(\mathbf{m}^{[H_T]}) - J_T(\mathbf{m}) \\ &\overset{\text{(A)}}{\le} J_T^{\mathrm{on}} - \min_{\mathbf{m}' \in \mathcal{M}_{H_T}} J_T(\mathbf{m}') + J_T(\mathbf{m}^{[H_T]}) - J_T(\mathbf{m}) \\ &= \mathrm{Reg}_T(\mathcal{M}_{H_T}) + J_T(\mathbf{m}^{[H_T]}) - J_T(\mathbf{m}) \\ &\overset{\text{(B)}}{\le} 2\sqrt{\Gamma_\alpha T \log(H_T+1)} + J_T(\mathbf{m}^{[H_T]}) - J_T(\mathbf{m}) \\ &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} + |J_T(\mathbf{m}^{[H_T]}) - J_T(\mathbf{m})| \\ &\overset{\text{(C)}}{\le} 2\sqrt{\Gamma_\alpha T \log(H_T+1)} + T G_\alpha \alpha^{H_T}. \end{aligned}$$

Here (A) uses $\mathbf{m}^{[H_T]} \in \mathcal{M}_{H_T}$, and the next equality is (6.11). Step (B) applies (6.32) with $H = H_T$ and step size η^*; (C) is (6.41). By (6.48),

$$\begin{aligned} \mathrm{Reg}_T(\mathcal{M}_\infty) &= \sup_{\mathbf{m} \in \mathcal{M}_\infty} [J_T^{\mathrm{on}} - J_T(\mathbf{m})] \\ &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} + T G_\alpha \alpha^{H_T}, \end{aligned}$$

which is (6.49). The choice (6.47) gives

$$\frac{\log T}{\log(1/\alpha)} \le H_T \le 1 + \frac{\log T}{\log(1/\alpha)}.$$

Since $0 < \alpha < 1$,

$$\begin{aligned} T G_\alpha \alpha^{H_T} &\le T G_\alpha \alpha^{\log T / \log(1/\alpha)} \\ &= T G_\alpha \exp\!\left(\frac{\log T}{\log(1/\alpha)} \log \alpha \right) \\ &= T G_\alpha e^{-\log T} = G_\alpha. \end{aligned}$$

For fixed α, L_x, L_u, $H_T = O(\log T)$ and $\log(H_T + 1) = O(\log \log T)$, so the regret bound is $O(\sqrt{T \log \log T}) = o(T)$. Also,

$$\begin{aligned}\inf_{\mathbf{m} \in \mathcal{M}_\infty} J_T(\mathbf{m}) &\overset{(A)}{\leq} \min_{k \in [0,\alpha]} J_T(\mathbf{m}(k)) \\ &= \min_{k \in [0,\alpha]} \sum_{t=1}^{T} c_t(x_t(k), u_t(k)).\end{aligned}$$

Here (A) uses $\{\mathbf{m}(k) : k \in [0, \alpha]\} \subseteq \mathcal{M}_\infty$, and the equality follows from Corollary 6.8. Consequently,

$$\begin{aligned}J_T^{\text{on}} - \min_{k \in [0,\alpha]} \sum_{t=1}^{T} c_t(x_t(k), u_t(k)) &\leq J_T^{\text{on}} - \inf_{\mathbf{m} \in \mathcal{M}_\infty} J_T(\mathbf{m}) \\ &= \text{Reg}_T(\mathcal{M}_\infty).\end{aligned}$$

□

6.4 Regret lower bound

We now show that the entropic SDAC controller's $\sqrt{T}$ regret dependence is optimal for fixed H, α, L_x, L_u.

Theorem 6.10 (SDAC regret lower bound)**.** *Fix $\alpha \in (0, 1)$, $L_x, L_u > 0$, and an integer $H \geq 1$. Assume the dynamics, initial state, and feedback of Subsection 6.2.1, and let the integer horizon satisfy*

$$T \geq 2 \left\lceil \frac{\log 2}{\log(1/\alpha)} \right\rceil + 2. \tag{6.50}$$

For every causal feasible online controller $\mathcal{A}$, possibly randomized, there is a deterministic sequence of arrivals and costs, fixed before the interaction, that satisfies the model and cost assumptions with constants L_x, L_u and gives

$$\mathbb{E}_{\mathcal{A}}[\text{Reg}_T(\mathcal{M}_H)] \geq \frac{G_\alpha}{8} \sqrt{T}. \tag{6.51}$$

Here $G_\alpha = \alpha(L_x/(1-\alpha) + L_u)$. The expectation is over the controller's internal randomization. The sequence may depend on $\mathcal{A}$, but not on its realized random choices.

The upper bound (6.32) is $2\sqrt{\Gamma_\alpha T \log(H+1)}$. For fixed H, α, L_x, L_u, both bounds therefore scale as $\sqrt{T}$. Their parameter dependences differ: the upper bound contains $\sqrt{\log(H+1)}$, whereas the lower bound is independent of H. For fixed $L_x, L_u > 0$, as $\alpha \uparrow 1$ and with T satisfying (6.50), the lower bound is $\Omega((1-\alpha)^{-1}\sqrt{T})$, while the upper bound is $O((1-\alpha)^{-3/2}\sqrt{T\log(H+1)})$ by Remark 6.6. Thus optimality in H and α remains unresolved.

Proof. Fix a causal feasible controller $\mathcal{A}$. We use the random-sign argument of Hazan, Agarwal, and Kale [89, Sec. 2.3, footnote 3]. See also [3, Sec. 3.2, Thm. 3.2].

Let $\xi_1, \dots, \xi_T$ be independent random signs, each equal to 1 or -1 with probability $1/2$, and independent of the controller's randomization. Write $\boldsymbol{\xi} := (\xi_1, \dots, \xi_T)$ and set

$$w_t := 1, \qquad c_t(x,u) := \xi_t(L_x x - L_u u), \qquad t \in [T].$$

Each cost is affine with gradient $\xi_t(L_x, -L_u)$, so it is convex and satisfies both the Lipschitz and subgradient bounds (6.6)–(6.8).

Since $H \geq 1$, the two parameters

$$\mathbf{m}^{(0)} := \mathbf{0}, \qquad \mathbf{m}^{(1)} := \mathbf{e}_1 = (1, 0, \dots, 0)^\top$$

belong to $\mathcal{M}_H$ and are feasible by Lemma 6.3. Both trajectories start with state and action zero. For $t = 2, \dots, T$, the policy formula (6.14) and dynamics (6.3) give

$$u_t(\mathbf{m}^{(0)}) = 0, \quad x_t(\mathbf{m}^{(0)}) = \sum_{s=0}^{t-2} \alpha^s = \frac{1-\alpha^{t-1}}{1-\alpha},$$
$$u_t(\mathbf{m}^{(1)}) = \alpha, \quad x_t(\mathbf{m}^{(1)}) = 1.$$

Both policies incur zero cost at $t = 1$. Define

$$\beta_t \coloneqq \alpha \left(L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha} \right), \qquad t = 2, \dots, T. \tag{6.52}$$

Subtracting their costs gives

$$\begin{aligned} J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)}) &= \sum_{t=2}^{T} \xi_t \left[L_x \left(\frac{1-\alpha^{t-1}}{1-\alpha} - 1 \right) + \alpha L_u \right] \\ &= \sum_{t=2}^{T} \xi_t \left[L_x \frac{\alpha - \alpha^{t-1}}{1-\alpha} + \alpha L_u \right] \\ &= \sum_{t=2}^{T} \xi_t \alpha \left[L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha} \right] \\ &= \sum_{t=2}^{T} \xi_t \beta_t. \end{aligned}$$

By causality, (x_t, u_t) depends only on past signs and the controller's randomization, and is independent of ξ_t. The two fixed-policy trajectories depend only on the arrivals. Since $\mathbb{E}[\xi_t] = 0$,

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[J_T^{\mathrm{on}}] &= \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[\xi_t (L_x x_t - L_u u_t)] \\ &= \sum_{t=1}^{T} \mathbb{E}[\xi_t]\, \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[L_x x_t - L_u u_t] = 0, \\ \mathbb{E}_{\boldsymbol{\xi}}[J_T(\mathbf{m}^{(j)})] &= \sum_{t=1}^{T} \mathbb{E}[\xi_t]\, [L_x x_t(\mathbf{m}^{(j)}) - L_u u_t(\mathbf{m}^{(j)})] = 0, \quad j \in \{0, 1\}. \end{aligned}$$

Using $\min\{a,b\} = (a+b-|a-b|)/2$,

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[\mathrm{Reg}_T(\mathcal{M}_H)] &\overset{(A)}{\geq} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[J_T^{\mathrm{on}} - \min\{J_T(\mathbf{m}^{(0)}), J_T(\mathbf{m}^{(1)})\}] \\
&= \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[J_T^{\mathrm{on}}] - \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}[J_T(\mathbf{m}^{(0)}) + J_T(\mathbf{m}^{(1)})] \\
&\quad + \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left|J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)})\right| \\
&\overset{(B)}{=} \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left|J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)})\right| \\
&= \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left|\sum_{t=2}^{T}\xi_t\beta_t\right| \\
&\overset{(C)}{\geq} \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2}.
\end{aligned}$$

Here (A) uses (6.11) and $\{\mathbf{m}^{(0)}, \mathbf{m}^{(1)}\} \subseteq \mathcal{M}_H$; (B) uses the zero expected costs, and the next equality substitutes their difference above; (C) is the $p = 1$ Khintchine inequality [90] for the deterministic coefficients $\beta_2, \ldots, \beta_T$ and independent signs.

Let $q_\alpha := \lceil \log 2 / \log(1/\alpha) \rceil$. For $t \geq q_\alpha + 2$, $0 < \alpha < 1$ gives

$$\alpha^{t-2} \leq \alpha^{q_\alpha} \leq \alpha^{\log 2/\log(1/\alpha)} = \frac{1}{2}.$$

Thus (6.52) gives

$$\begin{aligned}
\beta_t &= \alpha\left(L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha}\right) \\
&\geq \alpha L_u + \frac{\alpha L_x}{2(1-\alpha)} \\
&\geq \frac{\alpha L_u}{2} + \frac{\alpha L_x}{2(1-\alpha)} = \frac{G_\alpha}{2}.
\end{aligned}$$

By (6.50), $q_\alpha + 1 \le T/2$, so

$$\begin{aligned}\sum_{t=2}^{T} \beta_t^2 &\ge \sum_{t=q_\alpha+2}^{T} \beta_t^2 \\ &\ge (T - q_\alpha - 1)\left(\frac{G_\alpha}{2}\right)^2 \\ &\ge \frac{T}{2}\left(\frac{G_\alpha}{2}\right)^2.\end{aligned}$$

Consequently,

$$\frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T} \beta_t^2} \ge \frac{1}{2\sqrt{2}}\sqrt{\frac{T}{2}\left(\frac{G_\alpha}{2}\right)^2} = \frac{G_\alpha}{8}\sqrt{T}.$$

The expectation over $\boldsymbol{\xi}$ averages over 2^T equally likely sign sequences. At least one sequence attains or exceeds this average. Fixing it before the interaction gives

$$\mathbb{E}_{\mathcal{A}}[\mathrm{Reg}_T(\mathcal{M}_H)] \ge \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T} \beta_t^2} \ge \frac{G_\alpha}{8}\sqrt{T}, \tag{6.53}$$

Equation (6.53) proves (6.51). □

6.5 Conclusion

Theorem 6.5 bounds the regret of the feasible entropic SDAC controller by $2\sqrt{\Gamma_\alpha T \log(H+1)}$ with the optimized step size. With $H = H_T$ from (6.47), Corollary 6.9 gives $O(\sqrt{T \log\log T})$ regret against infinite-memory SDAC policies for fixed α, L_x, L_u, including every feasible scalar fixed-fraction policy.

Theorem 6.10 shows that the $\sqrt{T}$ horizon dependence is optimal for fixed H, α and $L_x, L_u > 0$. The factors of $\sqrt{\log(H+1)}$ in memory dependence and $(1-\alpha)^{-1/2}$ in retention dependence as $\alpha \uparrow 1$ remain unresolved, with the latter comparison subject to (6.50).

The upper bounds require revelation of the full cost function and enough capacity to avoid saturation, as specified in Subsection 6.2.1. Feedback only at the executed state–action pair need

not determine the surrogate subgradient, and active saturation can destroy the affine fixed-policy trajectories used to establish convexity in Lemma 6.4.

6.6 Action clipping as a projection

Lemma 6.11 (Action clipping as a Bregman projection)**.** *For $t \in [T]$, define the set of parameters compatible with the current state by*

$$\mathcal{B}_t \coloneqq \{\mathbf{m} \in \mathcal{M}_H : \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle \leq \alpha x_t\}. \tag{6.54}$$

Let $\mathbf{m}_t^{\text{exec}}$ be the Bregman projection of $\mathbf{m}_t$ onto $\mathcal{B}_t$:

$$\mathbf{m}_t^{\text{exec}} \in \arg\min_{\mathbf{m} \in \mathcal{B}_t} D_\Psi(\mathbf{m}, \mathbf{m}_t). \tag{6.55}$$

Then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}} \rangle = \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle, \alpha x_t\} = u_t.$$

Proof. Write $a_t \coloneqq \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle$ and $b_t \coloneqq \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}} \rangle$. If $a_t \leq \alpha x_t$, then $\mathbf{m}_t \in \mathcal{B}_t$. In this case, (2.22) and $D_\Psi(\mathbf{m}_t, \mathbf{m}_t) = 0$ give $\mathbf{m}_t^{\text{exec}} = \mathbf{m}_t$. Otherwise $a_t > \alpha x_t \geq b_t$. Suppose $b_t < \alpha x_t$ and choose $0 < \theta < (\alpha x_t - b_t)/(a_t - b_t) < 1$. The point $\mathbf{m}_\theta \coloneqq (1-\theta)\mathbf{m}_t^{\text{exec}} + \theta \mathbf{m}_t$ belongs to $\mathcal{M}_H$ by convexity, and $\langle \boldsymbol{\phi}_t^u, \mathbf{m}_\theta \rangle = b_t + \theta(a_t - b_t) < \alpha x_t$, so $\mathbf{m}_\theta \in \mathcal{B}_t$. Convexity of $D_\Psi(\cdot, \mathbf{m}_t)$ gives

$$\begin{aligned} D_\Psi(\mathbf{m}_\theta, \mathbf{m}_t) &\leq (1-\theta) D_\Psi(\mathbf{m}_t^{\text{exec}}, \mathbf{m}_t) + \theta D_\Psi(\mathbf{m}_t, \mathbf{m}_t) \\ &= (1-\theta) D_\Psi(\mathbf{m}_t^{\text{exec}}, \mathbf{m}_t) < D_\Psi(\mathbf{m}_t^{\text{exec}}, \mathbf{m}_t). \end{aligned}$$

The strict inequality uses $\theta > 0$ and (2.22), since $a_t \neq b_t$. This contradicts optimality, so $b_t = \alpha x_t$, as required. $\square$

Chapter 7

Bandit Rate Adaptation with a Known Lower Bound on Service Slack

Source and author contributions. This chapter is based on the results for a known lower bound on service slack in "Bandit-Based Rate Adaptation for a Single-Server Queue," joint work with Mevan Wijewardena and Michael J. Neely [91]. I developed the rate-adaptation method using this slack lower bound and was primarily responsible for its proofs and presentation. Wijewardena developed the original paper's lower-bound construction and proof. That result is restated here as Theorem 7.8 without proof. Neely contributed primarily through advising and reviewing our work on the original paper. The paper's adaptive algorithm for unknown slack is not included here. The cited preprint is distributed under the Creative Commons Attribution 4.0 International license. This chapter adapts its presentation and analysis.

The statement and proof of Theorem 7.9 were generated by AI for this dissertation and are not part of the original joint paper.

7.1 Introduction

Consider a transmitter that serves a queue over a channel whose capacity is sampled independently each slot from an unknown, fixed distribution. In each slot, it chooses a transmission rate without observing the current channel capacity, then receives a binary acknowledgment indicating whether the transmission succeeded. A successful transmission supplies service to the queue. A failed transmission supplies none. We seek a bound on time-average expected queue length that holds uniformly over all horizons, including the learning period.

Chapter 6 studies feasible storage policies under full cost feedback. Here, feedback is limited to binary acknowledgments, so service characteristics must be learned as the queue evolves.

The action space is the interval $[0, 1]$. For a fixed rate, mean *offered service* is the rate multiplied by its probability of success. It measures available service even when the queue contains insufficient work to use it. Both the arrival distribution and the channel distribution are unknown. We assume that the transmitter is given a positive lower bound ε on service slack, the difference between the best achievable mean offered service and the mean arrival rate. This known slack lower bound determines the rate grid.

7.1.1 Related work

Scheduling and resource allocation in networks have been studied with objectives such as power efficiency, utility, and fairness [92, 93, 94]. Lyapunov drift provides a standard way to relate the margin by which mean service exceeds mean arrivals to queue bounds [83].

This chapter applies the classical UCB1 policy [95] to a finite grid of transmission rates. The policy selects rates using their estimated mean service together with an allowance for uncertainty. Background on bandit learning is given in [96, 66, 97, 98].

In queueing systems, the controller must learn service performance while jobs continue to arrive. Krishnasamy et al. [99, 100] study how learning unknown service probabilities increases the expected queue length at each time, compared with a controller that knows those probabilities. Fu and Modiano [101] study routing jobs among servers to maximize the reward earned when jobs finish. Yang et al. [102] study scheduling when service rates must be learned, including settings where those rates change over time. Other work also considers changing arrival and service conditions [103, 104].

We adapt the upper-bound analysis of Freund, Lykouris, and Weng [105] to workloads in $[0, 1]$ and service amounts determined by the selected transmission rate. Their analysis bounds the increase in time-average expected queue length caused by learning, relative to a controller that knows the service characteristics, uniformly over all horizons. Our bound controls the total time-average expected queue length.

Combes et al. [106] learn which transmission mode and rate provide the highest throughput. Tang et al. [107] combine rate selection with detection of changes in channel conditions. Other studies develop and evaluate bandit methods for wireless rate control [108, 109, 110]. Li et al. [111] survey related methods for allocating frequency bands. Our model permits rates throughout a continuous interval, and our performance criterion is time-average expected queue length.

Bandit learning with actions chosen from a continuous interval has been studied under continuity assumptions on mean rewards [112]. Other work constructs and refines finite sets of actions using bounds on how much rewards can differ between nearby actions [113, 114]. In our model, lowering a transmission rate cannot reduce its probability of success. This structure and the known slack lower bound let us choose a fixed grid containing a rate whose mean service exceeds the mean arrival rate, even when mean service has jumps.

Learning to set prices provides a useful interpretation [115]. A purchase occurs when the buyer's value is at least the offered price, just as a transmission succeeds when channel capacity is at least the selected rate. Revenue corresponds to offered service. Rounding a price or rate down to the nearest grid point preserves success and limits the loss from using a finite grid. The pricing construction in [115] chooses the grid using the horizon. Here, the given lower bound on service slack determines a grid that remains fixed for all horizons.

Audibert, Munos, and Szepesvári [116, Appendix A.2, Proposition 1 and Lemma 4] show that UCB1 continues selecting a suboptimal arm even when both arms have deterministic rewards. Their proof compares the two UCB indices. Here, an analogous comparison bounds how often UCB1 selects rates that supply service. The resulting service deficit gives a queue-length lower bound with constant arrivals and channel capacity.

7.1.2 Contributions and scope

We assume that arrivals are i.i.d., channel capacities are i.i.d., and the two sequences are independent. Starting from an empty queue, UCB1 on a suitably chosen finite rate grid achieves a time-average expected queue bound of order

$$O\Big(\frac{1+\log(1/\varepsilon)}{\varepsilon^2}\Big).$$

This bound holds uniformly over all horizons. Theorem 7.8, restated without proof from [91], shows that every causal rate-selection policy faces some admissible channel and horizon for which the time-average expected queue length is $\Omega(\varepsilon^{-2})$. For the specific UCB1 policy studied here, Theorem 7.9 proves the stronger lower bound $\Omega(\varepsilon^{-2}\log(1/\varepsilon))$, even with deterministic arrivals and

capacities. The logarithmic factor is therefore necessary for this algorithm in the worst case. These bounds do not determine whether another causal policy can remove this factor.

7.2 Queueing model and performance criterion

We use the conventions in Section 1.3, with time written as a parenthesized argument.

7.2.1 Arrivals, service, and observations

In a wireless link, modulation and coding determine how much data the transmitter attempts to send in each slot. We model this choice using a continuous range of rates. Workload is normalized so that the maximum service per slot is 1. Arrivals are also bounded by 1 in the same units.

Time is divided into slots $t = 1, 2, \ldots$. Let $A(t) \in [0, 1]$ denote the arriving workload and $C(t) \in [0, 1]$ the channel capacity in slot t. The transmitter chooses an attempted rate $r(t) \in [0, 1]$ without observing $C(t)$. The resulting binary acknowledgment and offered service are

$$B(t) \coloneqq \mathbf{1}_{\{r(t) \le C(t)\}}, \qquad S(t) \coloneqq r(t)B(t). \tag{7.1}$$

The transmitter observes $B(t)$ and computes $S(t)$ from its chosen rate. Let $Q(t)$ be the queue length at the beginning of slot t, before that slot's arrival and service. It satisfies

$$Q(1) = 0, \qquad Q(t+1) = [Q(t) + A(t) - S(t)]_+, \quad \forall t \ge 1. \tag{7.2}$$

Arrivals can be served in the same slot. Actual departures equal $\min\{Q(t) + A(t), S(t)\}$, whereas (7.1) records offered service. The acknowledgment is observed in every slot, even when the queue is empty.

A causal policy can use the queue, its past attempted rates and acknowledgments, and the arrivals observed so far. We allow the current arrival $A(t)$ to be observed before the decision. Randomized decisions use internal randomness independent of the arrivals and capacities. The UCB policy developed below requires only observations through slot $t-1$. It does not use $A(t)$ when choosing $r(t)$.

Assumption 7.1 (Stochastic environment)**.** *The arrival sequence is i.i.d., the capacity sequence is i.i.d., and the two sequences are independent. Their distributions are supported on* $[0,1]$ *and are unknown to the transmitter. Write* $\lambda \coloneqq \mathbb{E}[A(1)]$.

For a fixed transmission rate r, define its mean offered service by

$$\mu(r) \coloneqq r\,\mathbb{P}(C(1) \geq r), \qquad r \in [0,1]. \tag{7.3}$$

For any causal selection of $r(t)$, independence of the current capacity from the information used to select the rate gives $\mathbb{E}[S(t) \mid r(t)] = \mu(r(t))$. Equality $C(t) = r(t)$ counts as a successful transmission. We use "mean service" as shorthand for mean offered service below.

Assumption 7.2 (Known lower bound on service slack)**.** *There is a rate* $r^* \in \arg\max_{r\in[0,1]} \mu(r)$ *and a given number* $\varepsilon > 0$ *such that*

$$\mu(r^*) \geq \lambda + \varepsilon. \tag{7.4}$$

The transmitter knows ε*, but need not know* r^**,* $\mu(r^*)$*, or* λ.

Necessarily $0 < \varepsilon \leq 1$: (7.4) gives $\varepsilon \leq \mu(r^*) - \lambda$, while (7.3) gives $\mu(r^*) \leq 1$ and Assumption 7.1 gives $\lambda \geq 0$. The supplied ε may be smaller than the actual slack $\mu(r^*) - \lambda$. A smaller supplied bound gives a finer grid and a weaker worst-case guarantee.

7.2.2 Performance criterion and fixed-rate benchmark

For an integer horizon $T \geq 1$, let

$$\overline{Q}_T := \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[Q(t)]. \tag{7.5}$$

Thus $\overline{Q}_T$ already includes expectation over the environment and any controller randomization. Our objective is to bound $\sup_{T \geq 1} \overline{Q}_T$ using only ε and numerical constants, uniformly over the environments satisfying Assumptions 7.1 and 7.2.

The policy that always selects r^* provides a fixed-rate benchmark when the service characteristics are known.

Lemma 7.3 (Queue bound for the benchmark policy)**.** *Under Assumptions 7.1 and 7.2, the policy $r(t) = r^*$ for all t, started from $Q(1) = 0$, satisfies*

$$\overline{Q}_T \leq \frac{1}{2\varepsilon} \qquad \text{for every integer } T \geq 1. \tag{7.6}$$

Proof. For $q \geq 0$ and $d \in \mathbb{R}$,

$$[q + d]_+^2 - q^2 \leq (q + d)^2 - q^2 = 2qd + d^2. \tag{7.7}$$

Applying (7.7) to the queue update (7.2) gives

$$Q(t+1)^2 - Q(t)^2 \leq 2Q(t)(A(t) - S(t)) + (A(t) - S(t))^2. \tag{7.8}$$

This inequality is samplewise and holds for every rate-selection policy. For $r(t) = r^*$, Assumption 7.1 makes the current arrival and service independent of $Q(t)$. Hence

$$\begin{aligned}\mathbb{E}[A(t) - S(t) \mid Q(t)] &= \lambda - r^*\mathbb{P}(C(t) \geq r^*) \\ &= \lambda - \mu(r^*) \leq -\varepsilon, \qquad (A(t) - S(t))^2 \leq 1.\end{aligned}$$

The equalities use (7.1) and (7.3). The inequalities use (7.4) and $A(t), S(t) \in [0, 1]$. Thus

$$\begin{aligned}\mathbb{E}[Q(t+1)^2 - Q(t)^2 \mid Q(t)] &\overset{(A)}{\leq} 2Q(t)\mathbb{E}[A(t) - S(t) \mid Q(t)] + 1 \\ &\overset{(B)}{\leq} 1 - 2\varepsilon Q(t). \end{aligned} \tag{7.9}$$

Here (A) uses (7.8) and $(A(t) - S(t))^2 \leq 1$. Step (B) uses the conditional mean above. Taking expectations, summing, and rearranging gives

$$\begin{aligned}2\varepsilon \sum_{t=1}^{T} \mathbb{E}[Q(t)] &\leq T - \sum_{t=1}^{T} \mathbb{E}[Q(t+1)^2 - Q(t)^2] \\ &= T + \mathbb{E}[Q(1)^2] - \mathbb{E}[Q(T+1)^2] \leq T.\end{aligned}$$

Here $Q(1) = 0$. Division by $2\varepsilon T$ gives (7.6). □

7.3 Discretizing the rate space

The following one-sided bound controls the mean-service loss under downward rounding and yields a grid with positive service slack.

Lemma 7.4 (Mean-service bound for lowering a rate)**.** *For* $0 \leq y \leq x \leq 1$,

$$\mu(y) \geq \mu(x) - (x - y). \tag{7.10}$$

Proof. By (7.3),

$$\begin{aligned}\mu(y) = y\mathbb{P}(C(1) \geq y) &\overset{\text{(A)}}{\geq} y\mathbb{P}(C(1) \geq x) \\ &= \mu(x) - (x-y)\mathbb{P}(C(1) \geq x) \geq \mu(x) - (x-y).\end{aligned}$$

Step (A) uses the inclusion $\{C(1) \geq x\} \subseteq \{C(1) \geq y\}$. □

Remark 7.5. *There is no analogous bound for upward rounding. For example, if* $C(t) \equiv 1/2$*, then* $\mu(1/2) = 1/2$*, whereas* $\mu(r) = 0$ *for every* $r > 1/2$*. Thus, an arbitrarily small increase above rate* $1/2$ *causes a mean-service loss of* $1/2$*.*

For an integer $K \geq 1$, use K equally spaced positive rates:

$$r_k \coloneqq \frac{k}{K}, \qquad \mu_k \coloneqq \mu(r_k), \qquad k \in [K]. \tag{7.11}$$

Let

$$k^* \in \arg\max_{k \in [K]} \mu_k, \qquad \mu_K^* \coloneqq \mu_{k^*}, \qquad \sigma \coloneqq \mu_K^* - \lambda. \tag{7.12}$$

The actual grid slack σ is used only in the analysis.

Lemma 7.6 (A grid retaining positive slack)**.** *If* $K \geq 1/\varepsilon$*, then*

$$\mu_K^* \geq \mu(r^*) - \frac{1}{K}, \qquad \sigma \geq \varepsilon - \frac{1}{K}.$$

In particular, choosing

$$K \coloneqq \left\lceil \frac{3}{\varepsilon} \right\rceil \tag{7.13}$$

ensures $\sigma \geq 2\varepsilon/3 > 0$*.*

Proof. By (7.3), (7.4), $\lambda \geq 0$, and $K \geq 1/\varepsilon$,

$$r^* \geq \mu(r^*) \geq \lambda + \varepsilon \geq \varepsilon \geq \frac{1}{K}.$$

Thus

$$r^{\downarrow} := \frac{\lfloor Kr^* \rfloor}{K}, \qquad \frac{1}{K} \leq r^{\downarrow} \leq r^* \leq 1, \tag{7.14}$$

$$0 \leq r^* - r^{\downarrow} < \frac{1}{K}. \tag{7.15}$$

Since $\lfloor Kr^* \rfloor \in \{1, \ldots, K\}$, $r^{\downarrow}$ is a positive grid rate. Its mean service is at most μ^*_K, so

$$\mu^*_K \geq \mu(r^{\downarrow}) \overset{(A)}{\geq} \mu(r^*) - (r^* - r^{\downarrow}) \geq \mu(r^*) - \frac{1}{K}, \tag{7.16}$$

$$\sigma = \mu^*_K - \lambda \geq \mu(r^*) - \lambda - \frac{1}{K} \geq \varepsilon - \frac{1}{K}. \tag{7.17}$$

Step (A) applies (7.10). The remaining bounds use (7.15) and (7.4). For the choice (7.13),

$$K = \left\lceil \frac{3}{\varepsilon} \right\rceil \geq \frac{3}{\varepsilon} \geq \frac{1}{\varepsilon}, \qquad \frac{1}{K} \leq \frac{\varepsilon}{3}, \tag{7.18}$$

$$\sigma \geq \varepsilon - \frac{1}{K} \geq \frac{2\varepsilon}{3} > 0. \tag{7.19}$$

□

7.4 Rate selection on a fixed grid

Use the same grid of $K = \lceil 3/\varepsilon \rceil$ rates from Section 7.3 for every horizon. Selecting a grid rate is now a finite-armed bandit problem: each rate is an arm, and its offered service is the observed reward. We apply UCB1 [95] to these arms.

Algorithm 4 UCB1 rate adaptation with a known lower bound on service slack

Require: A known $\varepsilon > 0$ satisfying Assumption 7.2
Set $K \leftarrow \lceil 3/\varepsilon \rceil$ and $r_k \leftarrow k/K$ for $k = 1, \dots, K$
Initialize $N_k(0) \leftarrow 0$ for every k
for $t = 1, 2, \dots$ **do**
    if $t \le K$ **then**
        $J(t) \leftarrow t$
    else
        Choose $J(t) \in \arg\max_{k \in [K]} U_k(t-1)$
        Resolve ties using a fixed rule
    end if
    Set $r(t) \leftarrow r_{J(t)}$, observe $B(t)$, and compute $S(t) \leftarrow r(t)B(t)$
    Update the statistics in (7.20)
end for

Let $J(t)$ be the index of the rate selected in slot t. For each rate, the controller records its selection count and average offered service, with $S(t) = r_{J(t)}B(t)$. After slot t, these statistics are

$$\begin{aligned} N_k(t) &\coloneqq \sum_{\tau=1}^{t} \mathbf{1}_{\{J(\tau)=k\}}, \\ \widehat{\mu}_k(t) &\coloneqq \frac{\sum_{\tau=1}^{t} S(\tau)\mathbf{1}_{\{J(\tau)=k\}}}{N_k(t)}, \qquad N_k(t) > 0. \end{aligned} \tag{7.20}$$

Define the confidence radius

$$b(n,u) \coloneqq \sqrt{\frac{2\log u}{n}}, \qquad b_k(t) \coloneqq b(N_k(t), t), \tag{7.21}$$

where $n \ge 1$, $u \ge 1$, and the second expression requires $N_k(t) > 0$. The controller first samples every grid rate once. It then adds the confidence radius to each estimated mean service to form the UCB index

$$U_k(t) \coloneqq \widehat{\mu}_k(t) + b_k(t). \tag{7.22}$$

After initialization, it selects a rate in slot t with the largest index evaluated at $t-1$.

7.5 Main guarantees

The first result bounds the queue under UCB1 at every horizon. The next two results distinguish the unavoidable cost for any causal policy from the larger worst-case cost of this particular algorithm.

Theorem 7.7 (Uniform bound on time-average expected queue length)**.** *Under Assumptions 7.1 and 7.2, let $0 < \varepsilon \leq 1$ be the known slack lower bound and run Algorithm 4 from $Q(1) = 0$. For every integer $T \geq 1$,*

$$\overline{Q}_T \leq \frac{170(1 + \log(1/\varepsilon))}{\varepsilon^2}. \tag{7.23}$$

In particular, the worst-case time-average expected queue length is $O(\varepsilon^{-2} \log(1/\varepsilon))$ as $\varepsilon \downarrow 0$.

The following result is restated in this chapter's notation from [91, Theorem 2].

Theorem 7.8 (A lower bound for every causal policy)**.** *For every $0 < \varepsilon \leq 1/144$ and every causal, possibly randomized rate-selection policy given ε, there is an environment satisfying the model assumptions with $\mu(r^*) - \lambda = \varepsilon$ and an integer horizon $T \geq 1$ such that, starting from $Q(1) = 0$,*

$$\overline{Q}_T \geq \frac{6 \times 10^{-7}}{\varepsilon^2}. \tag{7.24}$$

The arrivals in this environment are i.i.d. Bernoulli$(1/2)$ random variables, independent of the i.i.d. channel capacities.

The proof is given in [91, Appendix J, p. 41 of the PDF] and is omitted here.

Theorem 7.9 (Necessity of the logarithmic factor for UCB1)**.** *There is a numerical constant $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon \le \varepsilon_0$, there is an environment satisfying the model assumptions with $\mu(r^*) - \lambda = \varepsilon$ in which Algorithm 4, started from $Q(1) = 0$, satisfies*

$$\overline{Q}_{T_\varepsilon} \ge \frac{\log(1/\varepsilon)}{32\varepsilon^2} \tag{7.25}$$

at a horizon $T_\varepsilon = \Theta(\varepsilon^{-3}\log(1/\varepsilon))$. Thus, for UCB1, the upper bound in Theorem 7.7 and this lower bound have the same dependence on ε.

The upper bound is proved in Section 7.6. The UCB1 lower bound is proved in Section 7.7.1 using deterministic arrivals and capacities.

7.6 Upper-bound analysis

Recall that K is the number of grid rates and $\sigma = \mu_K^* - \lambda > 0$ is the service slack of the best grid rate. The grid slack satisfies $\sigma \ge 2\varepsilon/3$ by Lemma 7.6. For the analysis, define the mean-service gap of each grid rate and of the selected rate by

$$\Delta_k := \mu_K^* - \mu_k, \quad k \in [K], \qquad \Delta(t) := \Delta_{J(t)}. \tag{7.26}$$

The controller does not need these gaps.

Let $\mathcal{F}_{t-1}$ contain the observations through slot $t-1$ and the controller randomization used to select the rate in slot t. This information determines both the UCB1 decision $J(t)$ and the current

queue length $Q(t)$. It excludes the current arrival $A(t)$ and capacity $C(t)$. Those two random variables are independent of this history, so the selected gap $\Delta(t)$ gives the decomposition

$$\mathbb{E}[A(t) - S(t) \mid \mathcal{F}_{t-1}] = \lambda - \mu_{J(t)} = -\sigma + \Delta(t). \tag{7.27}$$

Following [105], Lemmas 7.11 and 7.13 derive an auxiliary-queue comparison and a queue-weighted drift bound from this identity. For fixed K and σ, the comparison bound grows logarithmically with the horizon, while the learning term in the drift bound tends to zero. Section 7.6.4 therefore uses the comparison bound for shorter horizons and the drift bound for longer horizons to obtain a bound independent of T. This split is used only in the proof. The controller follows the same UCB1 rule throughout.

7.6.1 Accuracy of service estimates under adaptive sampling

We use the standard concentration argument from the analysis of UCB1 [95]. For each rate, we apply Hoeffding's inequality [117] at a fixed sample count, then take a union over all possible counts to account for adaptive sampling, as in [118, Lemma 3.5].

Here, u denotes the number of completed slots and n the number of samples of a given rate. The decision in slot t uses $u = t - 1$.

We use the standard representation of a stochastic bandit by independent sample sequences [98, Section 4.6]. For each grid rate r_k, generate independent service samples $X_{k,1}, X_{k,2}, \ldots$. Each sample has the distribution of $r_k \mathbf{1}_{\{C(1) \geq r_k\}}$. On the nth selection of that rate, reveal $X_{k,n}$. The sequences are independent across rates and independent of the arrivals.

This construction does not change the distribution of what the controller observes: given the past and the selected rate, the next service has exactly the same distribution as in the original model. The observations, decisions, and queue therefore have the same joint distribution as before.

Set $\widehat{\mu}_{k,n} \coloneqq n^{-1} \sum_{j=1}^{n} X_{k,j}$. In this construction,

$$\widehat{\mu}_k(t) = \widehat{\mu}_{k,N_k(t)}, \qquad N_k(t) > 0.$$

Since $X_{k,j} \in [0,1]$, Hoeffding's inequality with the radius in (7.21) gives, for each fixed sample count $n \geq 1$ and integer $u > 1$,

$$\mathbb{P}(|\widehat{\mu}_{k,n} - \mu_k| \geq b(n,u)) \leq 2\exp\Big(-2nb(n,u)^2\Big) = 2\exp\left(-2n\frac{2\log u}{n}\right) = 2u^{-4}. \tag{7.28}$$

For each integer $u > 1$, define

$$\mathcal{E}_u \coloneqq \bigcap_{k=1}^{K} \bigcap_{n=1}^{u} \left\{|\widehat{\mu}_{k,n} - \mu_k| < b(n,u)\right\}. \tag{7.29}$$

A union bound in (7.28) gives

$$\mathbb{P}(\mathcal{E}_u^c) \leq \sum_{k=1}^{K} \sum_{n=1}^{u} 2u^{-4} = \frac{2K}{u^3}. \tag{7.30}$$

On $\mathcal{E}_u$, each rate with $N_k(u) > 0$ has $1 \leq N_k(u) \leq u$, so

$$|\widehat{\mu}_k(u) - \mu_k| < b(N_k(u), u) = b_k(u). \tag{7.31}$$

The inequality selects $n = N_k(u)$ in (7.29). Adding the bonus $b_k(u)$ to the resulting bounds on $\widehat{\mu}_k(u)$ and using (7.22) yields

$$\mu_k \leq U_k(u) \leq \mu_k + 2b_k(u). \tag{7.32}$$

The next lemma is the classical UCB1 sampling bound of Auer, Cesa-Bianchi, and Fischer [95, proof of Theorem 1], expressed in the notation of grid rates.

Lemma 7.10 (Sampling a suboptimal rate)**.** *For every grid rate with $\Delta_k > 0$ and every integer $T \geq 1$,*

$$\mathbb{E}[N_k(T)] \leq \frac{8 \log T}{\Delta_k^2} + 1 + \frac{\pi^2}{3}. \tag{7.33}$$

Proof. During initialization, each rate is sampled at most once, so $N_k(T) \leq 1$ when $T \leq K$. For $T > K$, set

$$m \coloneqq \left\lceil \frac{8 \log T}{\Delta_k^2} \right\rceil. \tag{7.34}$$

For a decision at $K < t \leq T$, put $u \coloneqq t - 1$. If $N_k(u) \geq m$, then (7.21) gives

$$b_k(u) = \sqrt{\frac{2 \log u}{N_k(u)}} \overset{\text{(A)}}{\leq} \sqrt{\frac{2 \log T}{m}} \overset{\text{(B)}}{\leq} \sqrt{\frac{2 \log T}{8 \log T / \Delta_k^2}} = \frac{\Delta_k}{2}. \tag{7.35}$$

Here, (A) uses $u < T$ and $N_k(u) \geq m$. Step (B) uses the ceiling lower bound in (7.34). For $K \leq u < T$, define

$$\mathcal{H}_k(u) \coloneqq \{\widehat{\mu}_k(u) - \mu_k \geq b_k(u)\} \cup \{\mu_K^* - \widehat{\mu}_{k^*}(u) \geq b_{k^*}(u)\}. \tag{7.36}$$

By (7.22) and (7.26), if $N_k(u) \geq m$ and $\mathcal{H}_k(u)$ does not occur, the two indices satisfy

$$\begin{aligned} U_k(u) &= \widehat{\mu}_k(u) + b_k(u) \overset{\text{(A)}}{<} \mu_k + 2b_k(u) \overset{\text{(B)}}{\leq} \mu_k + \Delta_k = \mu_K^*, \\ U_{k^*}(u) &= \widehat{\mu}_{k^*}(u) + b_{k^*}(u) \overset{\text{(C)}}{>} \mu_K^*. \end{aligned} \tag{7.37}$$

Steps (A) and (C) use the absence of the respective errors. Step (B) uses (7.35). Thus the optimal rate has a strictly larger index, and r_k cannot be selected. The probability of either error satisfies

$$\begin{aligned}\mathbb{P}(\mathcal{H}_k(u)) &\overset{(A)}{\le} \sum_{n=1}^{u} \mathbb{P}(\widehat{\mu}_{k,n} - \mu_k \ge b(n,u)) + \sum_{n=1}^{u} \mathbb{P}(\mu_K^* - \widehat{\mu}_{k^*,n} \ge b(n,u)) \\ &\overset{(B)}{\le} 2\sum_{n=1}^{u} \exp(-2nb(n,u)^2) = 2u^{-3}. \end{aligned} \tag{7.38}$$

Here (A) takes a union over the possible sample counts $n = 1, \dots, u$. Step (B) uses the one-sided Hoeffding bounds underlying (7.28) with the radius (7.21). On every sample path,

$$\begin{aligned}\{J(u+1) = k,\ N_k(u) \ge m\} \subseteq \mathcal{H}_k(u), \qquad K \le u < T, \\ N_k(T) \le m + \sum_{u=K}^{T-1} \mathbf{1}_{\{\mathcal{H}_k(u)\}}.\end{aligned} \tag{7.39}$$

The inclusion follows from (7.37) and the maximizing rule in Algorithm 4. The count follows from (7.20): at most m selections occur before the count reaches m, and the inclusion applies to every later selection. Consequently,

$$\mathbb{E}[N_k(T)] \overset{(A)}{\le} m + 2\sum_{u=K}^{T-1} u^{-3} \overset{(B)}{\le} \frac{8\log T}{\Delta_k^2} + 1 + 2\sum_{u=1}^{\infty} u^{-2} = \frac{8\log T}{\Delta_k^2} + 1 + \frac{\pi^2}{3}.$$

Step (A) takes expectations in (7.39) and applies (7.38). Step (B) uses $\lceil z \rceil \le z+1$ in (7.34), $u^{-3} \le u^{-2}$ for $u \ge 1$, and enlarges the summation range. The last equality uses $\sum_{u=1}^{\infty} u^{-2} = \pi^2/6$. □

7.6.2 Auxiliary-queue comparison bound

Fix $\delta \in (0, \sigma)$. Following [105], we compare the actual queue with an auxiliary queue whose conditional mean net input is $-(\sigma - \delta)$.

Lemma 7.11 (Auxiliary-queue comparison bound)**.** *For every integer $T \geq 2$ and every $\delta \in (0, \sigma)$,*

$$\overline{Q}_T \leq \frac{1}{4(\sigma - \delta)} + \frac{\sigma - \delta}{2} + \frac{2K \log T}{\delta}. \tag{7.40}$$

It follows that

$$\overline{Q}_T \leq \frac{1}{2} + \frac{1}{\sigma}\left(\frac{1}{2} + \sqrt{2K \log T}\right)^2. \tag{7.41}$$

Proof. Define

$$\zeta(t) \coloneqq (A(t) - \lambda) - (S(t) - \mu_{J(t)}). \tag{7.42}$$

Given the history $\mathcal{F}_{t-1}$, the selected rate is fixed. The arrival and offered service are then independent, and each lies in $[0, 1]$. For any such random variable X,

$$\begin{aligned} \mathrm{Var}(X \mid \mathcal{F}_{t-1}) &= \mathbb{E}[(X - 1/2)^2 \mid \mathcal{F}_{t-1}] - (\mathbb{E}[X \mid \mathcal{F}_{t-1}] - 1/2)^2 \\ &\leq \mathbb{E}[(X - 1/2)^2 \mid \mathcal{F}_{t-1}] \leq \frac{1}{4}. \end{aligned} \tag{7.43}$$

We have

$$\begin{aligned} \mathbb{E}[\zeta(t) \mid \mathcal{F}_{t-1}] &= 0, \\ \mathbb{E}[\zeta(t)^2 \mid \mathcal{F}_{t-1}] &= \mathrm{Var}(A(t) \mid \mathcal{F}_{t-1}) + \mathrm{Var}(S(t) \mid \mathcal{F}_{t-1}) \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}. \end{aligned} \tag{7.44}$$

The mean identity follows from (7.42) and (7.27). Conditional independence gives the sum of variances, and (7.43) bounds each variance by $1/4$. The net input can be written as

$$A(t) - S(t) \overset{\text{(A)}}{=} \zeta(t) + \lambda - \mu_{J(t)} \overset{\text{(B)}}{=} \zeta(t) - \sigma + \Delta(t). \tag{7.45}$$

Here (A) uses (7.42). Step (B) uses (7.12) and (7.26). Set

$$\begin{aligned} Q(t+1) &= [Q(t) + \zeta(t) - \sigma + \Delta(t)]_+, \\ \widetilde{Q}(t+1) &\coloneqq [\widetilde{Q}(t) + \zeta(t) - (\sigma - \delta)]_+, \qquad \widetilde{Q}(1) \coloneqq 0. \end{aligned} \tag{7.46}$$

Put $\kappa \coloneqq \sigma - \delta > 0$. Since $\mathcal{F}_{t-1}$ determines $\widetilde{Q}(t)$,

$$\begin{aligned} &\mathbb{E}[\widetilde{Q}(t+1)^2 - \widetilde{Q}(t)^2 \mid \mathcal{F}_{t-1}] \\ &\quad \overset{(A)}{\leq} 2\widetilde{Q}(t)\mathbb{E}[\zeta(t) - \kappa \mid \mathcal{F}_{t-1}] + \mathbb{E}[(\zeta(t) - \kappa)^2 \mid \mathcal{F}_{t-1}] \\ &\quad = 2\widetilde{Q}(t)(\mathbb{E}[\zeta(t) \mid \mathcal{F}_{t-1}] - \kappa) + \mathbb{E}[\zeta(t)^2 \mid \mathcal{F}_{t-1}] \\ &\qquad - 2\kappa\mathbb{E}[\zeta(t) \mid \mathcal{F}_{t-1}] + \kappa^2 \qquad (7.47) \\ &\quad \overset{(B)}{=} \mathbb{E}[\zeta(t)^2 \mid \mathcal{F}_{t-1}] + \kappa^2 - 2\kappa\widetilde{Q}(t) \\ &\quad \overset{(C)}{\leq} \frac{1}{2} + \kappa^2 - 2\kappa\widetilde{Q}(t). \qquad (7.48) \end{aligned}$$

Step (A) applies (7.7) to (7.46). After the expansion in (7.47), (B) and (C) use (7.44). Taking expectations and summing gives

$$\begin{aligned} 2\kappa \sum_{t=1}^{T} \mathbb{E}[\widetilde{Q}(t)] &\leq T\left(\frac{1}{2} + \kappa^2\right) - \sum_{t=1}^{T} \mathbb{E}[\widetilde{Q}(t+1)^2 - \widetilde{Q}(t)^2] \\ &= T\left(\frac{1}{2} + \kappa^2\right) + \mathbb{E}[\widetilde{Q}(1)^2] - \mathbb{E}[\widetilde{Q}(T+1)^2] \\ &\leq T\left(\frac{1}{2} + \kappa^2\right). \end{aligned} \tag{7.49}$$

Here $\widetilde{Q}(1) = 0$. Division by $2\kappa T$ gives

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\widetilde{Q}(t)] \leq \frac{1/2 + \kappa^2}{2\kappa} = \frac{1}{4\kappa} + \frac{\kappa}{2} = \frac{1}{4(\sigma - \delta)} + \frac{\sigma - \delta}{2}. \tag{7.50}$$

Define

$$L_\delta(t) \coloneqq \sum_{\tau=1}^{t} [\Delta(\tau) - \delta]_+ = \sum_{k:\Delta_k > \delta} (\Delta_k - \delta) N_k(t), \qquad L_\delta(0) \coloneqq 0. \tag{7.51}$$

We prove $Q(t) \le \widetilde{Q}(t) + L_\delta(t-1)$ by induction. At $t = 1$, all three quantities are zero. Assuming the inequality holds at slot t, the coupled updates give

$$\begin{aligned} Q(t+1) &\overset{(A)}{\le} [\widetilde{Q}(t) + \zeta(t) - (\sigma - \delta) + L_\delta(t-1) + \Delta(t) - \delta]_+ \\ &\le [\widetilde{Q}(t) + \zeta(t) - (\sigma - \delta)]_+ + L_\delta(t-1) + [\Delta(t) - \delta]_+ \\ &\overset{(B)}{=} \widetilde{Q}(t+1) + L_\delta(t). \end{aligned}$$

Step (A) uses the actual queue update in (7.46) and the induction hypothesis. Step (B) applies the auxiliary update in (7.46) and the definition (7.51). Thus the induction proves, for every $t \ge 1$,

$$Q(t) \le \widetilde{Q}(t) + \sum_{\tau=1}^{t-1} [\Delta(\tau) - \delta]_+ = \widetilde{Q}(t) + L_\delta(t-1). \tag{7.52}$$

Taking expectations in (7.52), averaging over $t = 1, \dots, T$, and applying (7.50) gives

$$\overline{Q}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[Q(t)] \le \frac{1}{4(\sigma - \delta)} + \frac{\sigma - \delta}{2} + \frac{1}{T} \sum_{\tau=0}^{T-1} \mathbb{E}[L_\delta(\tau)]. \tag{7.53}$$

So far, we have used no property specific to UCB1. The comparison applies to any rule that selects a grid rate using only the history before the current arrival and capacity are drawn. We now use the UCB1 sampling bound to control L_δ.

For $\Delta > \delta > 0$,

$$\frac{1}{4\delta} - \frac{\Delta - \delta}{\Delta^2} = \frac{(\Delta - 2\delta)^2}{4\delta\Delta^2} \ge 0. \tag{7.54}$$

We also have

$$\sum_{\tau=1}^{T-1} \log \tau \le \int_1^T \log x \, \mathrm{d}x = T \log T - T + 1. \tag{7.55}$$

For $0 < \Delta_k \le 1$, the sampling bound (7.33) and $1 + \pi^2/3 < 8$ imply

$$\mathbb{E}[N_k(\tau)] \le \frac{8 \log \tau}{\Delta_k^2} + 1 + \frac{\pi^2}{3} \le \frac{8(1 + \log \tau)}{\Delta_k^2}.$$

The upper bound $\Delta_k \le 1$ follows from $0 \le \mu_k \le \mu_K^* \le 1$. For each grid rate with $\Delta_k > \delta$,

$$\begin{aligned} (\Delta_k - \delta) \sum_{\tau=1}^{T-1} \mathbb{E}[N_k(\tau)] &\overset{\text{(A)}}{\le} 8 \frac{\Delta_k - \delta}{\Delta_k^2} \sum_{\tau=1}^{T-1} (1 + \log \tau) \\ &\overset{\text{(B)}}{\le} \frac{2}{\delta} \sum_{\tau=1}^{T-1} (1 + \log \tau) \overset{\text{(C)}}{\le} \frac{2}{\delta} T \log T. \end{aligned} \tag{7.56}$$

Here (A) sums the preceding sampling estimate over $\tau = 1, \dots, T-1$. Step (B) uses (7.54). Step (C) uses (7.55). Combining these estimates gives

$$\begin{aligned} \overline{Q}_T &\overset{\text{(A)}}{\le} \frac{1}{4(\sigma - \delta)} + \frac{\sigma - \delta}{2} + \frac{1}{T} \sum_{k:\Delta_k > \delta} (\Delta_k - \delta) \sum_{\tau=1}^{T-1} \mathbb{E}[N_k(\tau)] \\ &\overset{\text{(B)}}{\le} \frac{1}{4(\sigma - \delta)} + \frac{\sigma - \delta}{2} + \frac{2K \log T}{\delta}. \end{aligned}$$

Step (A) is (7.53) after expanding L_δ by (7.51). Step (B) applies (7.56) to at most K rates. This proves (7.40). Since $\sigma - \delta < \sigma = \mu_K^* - \lambda \le 1$, its middle term is at most $1/2$. To obtain (7.41), set $R \coloneqq \sqrt{2K \log T} > 0$ for this calculation and choose

$$\delta \coloneqq \frac{2\sigma R}{1 + 2R} \in (0, \sigma).$$

Then $\sigma - \delta = \sigma/(1+2R)$ and $2K\log T = R^2$, so

$$\begin{aligned}\overline{Q}_T &\overset{\text{(A)}}{\le} \frac{1}{2} + \frac{1}{4(\sigma-\delta)} + \frac{R^2}{\delta} = \frac{1}{2} + \frac{1+2R}{4\sigma} + \frac{R(1+2R)}{2\sigma} \\ &= \frac{1}{2} + \frac{(1+2R)^2}{4\sigma} = \frac{1}{2} + \frac{1}{\sigma}\left(\frac{1}{2} + \sqrt{2K\log T}\right)^2. \end{aligned} \tag{7.57}$$

Here (A) uses (7.40) and the preceding bound $(\sigma-\delta)/2 \le 1/2$. □

7.6.3 Queue-weighted drift bound

To bound $\mathbb{E}[Q(t)\Delta(t)]$, define

$$h(t) \coloneqq \begin{cases} \Delta(t), & 1 \le t \le K, \\ 2b_{J(t)}(t-1), & t > K. \end{cases} \tag{7.58}$$

We also use

$$\Gamma(T) \coloneqq \frac{1}{3} + 8(1+\log T)\log T. \tag{7.59}$$

Let

$$Z_T \coloneqq \sum_{t=1}^{T} Q(t).$$

Then $\mathbb{E}[Z_T] = T\overline{Q}_T$ by (7.5). The queue-length bound refines an inequality of Huang, Golubchik, and Huang [103], using the largest backlog as in [118, Lemma 3.4].

Lemma 7.12 (Two deterministic bounds)**.** *For every integer $T \ge 1$, on every sample path,*

$$\sum_{t=1}^{T} h(t)^2 \le K\Gamma(T), \qquad \sum_{t=1}^{T} Q(t)^2 \le \sqrt{2}\left(\sum_{t=1}^{T} Q(t)\right)^{3/2}. \tag{7.60}$$

Proof. During initialization, each rate is sampled once. Since $\Delta_k = \mu_K^* - \mu_k$ and $\mu_k = \mu(k/K)$ by (7.26) and (7.11), respectively,

$$\begin{aligned} \Delta_k &= \mu(k^*/K) - \mu(k/K) \overset{\text{(A)}}{\leq} \frac{k^* - k}{K}, \quad k \leq k^*, \\ \Delta_k &= \mu_K^* - \mu_k \overset{\text{(B)}}{\leq} \frac{k^*}{K}, \qquad\qquad\qquad k > k^*. \end{aligned} \tag{7.61}$$

Step (A) applies (7.10) with $x = k^*/K$ and $y = k/K$. Step (B) uses $\mu_k \geq 0$ and $\mu_K^* = \mu(k^*/K) = (k^*/K)\mathbb{P}(C(1) \geq k^*/K) \leq k^*/K$ from the service definition (7.3). Thus

$$\begin{aligned} \sum_{k=1}^{K} \Delta_k^2 &\overset{\text{(A)}}{\leq} \frac{1}{K^2} \sum_{j=0}^{k^*-1} j^2 + (K - k^*)\frac{(k^*)^2}{K^2} \\ &= \frac{k^*(k^* - 1)(2k^* - 1)}{6K^2} + (K - k^*)\frac{(k^*)^2}{K^2} \\ &\overset{\text{(B)}}{\leq} \frac{(k^*)^3}{3K^2} + (K - k^*)\frac{(k^*)^2}{K^2} \\ &= \frac{(k^*)^2}{K} - \frac{2(k^*)^3}{3K^2} = \frac{K}{3} - \frac{(K - k^*)^2(K + 2k^*)}{3K^2} \leq \frac{K}{3}. \end{aligned} \tag{7.62}$$

Here (A) squares the nonnegative bounds in (7.61) and reindexes the first sum by $j = k^* - k$. Step (B) uses $(k^* - 1)(2k^* - 1) \leq 2(k^*)^2$, since $k^* \geq 1$. The first equality uses $\sum_{j=0}^{m-1} j^2 = m(m-1)(2m-1)/6$. Equation (7.62) also bounds the contribution if the horizon ends before initialization is complete. If $T > K$, each further selection of rate k increases its sample count by one. Its counts just before those selections are therefore exactly $1, \ldots, N_k(T) - 1$. Using $h(t)^2 = 8\log(t-1)/N_{J(t)}(t-1)$, from (7.58) and (7.21), gives

$$\begin{aligned} \sum_{t=K+1}^{T} h(t)^2 &\overset{\text{(A)}}{\leq} 8\log T \sum_{k=1}^{K} \sum_{n=1}^{N_k(T)-1} \frac{1}{n} \\ &\overset{\text{(B)}}{\leq} 8K\log T \sum_{n=1}^{T-1} \frac{1}{n} \overset{\text{(C)}}{\leq} 8K(1 + \log T)\log T. \end{aligned} \tag{7.63}$$

In (A), use $\log(t-1) \le \log T$. Step (B) extends each nonnegative sum using $N_k(T) \le T$. Step (C) uses the harmonic-sum bound $\sum_{n=1}^{T-1} n^{-1} \le 1 + \int_1^{T-1} x^{-1}\,\mathrm{d}x \le 1 + \log T$ for $T \ge 2$. Combining (7.62) and (7.63) with (7.59) proves the first bound in (7.60). Write $Q_T^{\max} := \max_{1\le t\le T} Q(t)$, and choose a slot $t_{\max}$ at which the maximum is attained. The queue recursion (7.2) gives $Q(t+1) = [Q(t) + A(t) - S(t)]_+ \le Q(t) + 1$, since $0 \le A(t) \le 1$ and $S(t) \ge 0$. Iterating this inequality from $Q(1) = 0$ gives

$$0 \le Q(t) \le t - 1. \tag{7.64}$$

Applying (7.64) at $t_{\max}$ and iterating the one-slot bound backward, respectively, gives

$$Q_T^{\max} \le t_{\max} - 1, \qquad Q(t_{\max} - j) \ge Q_T^{\max} - j \quad (0 \le j < \lceil Q_T^{\max} \rceil). \tag{7.65}$$

Hence

$$Z_T \overset{\text{(A)}}{\ge} \sum_{j=0}^{\lceil Q_T^{\max}\rceil - 1} (Q_T^{\max} - j) \overset{\text{(B)}}{\ge} \int_0^{Q_T^{\max}} (Q_T^{\max} - x)\,\mathrm{d}x = \frac{(Q_T^{\max})^2}{2}. \tag{7.66}$$

In (A), use (7.65) and the nonnegativity of the remaining queue lengths. All displayed slots exist because $Q_T^{\max} \le t_{\max} - 1$. For (B), the decreasing nonnegative function $Q_T^{\max} - x$ on $[0, Q_T^{\max}]$ is at most $Q_T^{\max} - j$ on each interval $[j, \min\{j+1, Q_T^{\max}\}]$, whose length is at most one. Consequently,

$$\sum_{t=1}^{T} Q(t)^2 \le Q_T^{\max} Z_T \overset{\text{(A)}}{\le} \sqrt{2}\, Z_T^{3/2}.$$

Step (A) uses $Q_T^{\max} \le \sqrt{2Z_T}$ from (7.66). □

Lemma 7.13 (Queue-weighted drift bound)**.** *For every integer $T \ge 1$,*

$$\sigma\overline{Q}_T \le \frac{1}{2} + \frac{4}{T} + \left(\frac{2K^2\Gamma(T)^2}{T\sigma^3}\right)^{1/4} (\sigma\overline{Q}_T)^{3/4}. \tag{7.67}$$

Consequently,

$$\overline{Q}_T \le \frac{2}{\sigma} + \frac{16}{T\sigma} + \frac{2K^2\Gamma(T)^2}{T\sigma^4}. \tag{7.68}$$

Proof. Since $Q(t)$ is determined by the history $\mathcal{F}_{t-1}$ and $|A(t) - S(t)| \le 1$, taking conditional expectations in (7.8) and using (7.27) gives

$$\begin{aligned} \frac{1}{2}\mathbb{E}[Q(t+1)^2 - Q(t)^2 \mid \mathcal{F}_{t-1}] &\le \frac{1}{2} + Q(t)(-\sigma + \Delta(t)), \\ \frac{1}{2}\mathbb{E}[Q(t+1)^2 - Q(t)^2] &\le \frac{1}{2} - \sigma\mathbb{E}[Q(t)] + \mathbb{E}[Q(t)\Delta(t)]. \end{aligned} \tag{7.69}$$

For $t \le K$, the gap term equals $\mathbb{E}[Q(t)h(t)]$ by (7.58). For $t > K$, we put $u \coloneqq t-1$ and recall the event $\mathcal{E}_u$ from (7.29). The following comparison is the standard UCB optimism argument [98, Section 7.1], applied here to mean service. On $\mathcal{E}_u$,

$$\begin{aligned} \mu_K^* &\overset{\text{(A)}}{\le} U_{k^*}(u) \overset{\text{(B)}}{\le} U_{J(t)}(u) \overset{\text{(C)}}{\le} \mu_{J(t)} + 2b_{J(t)}(u), \\ \Delta(t) &= \mu_K^* - \mu_{J(t)} \le 2b_{J(t)}(u) = h(t). \end{aligned} \tag{7.70}$$

Here (A) and (C) use (7.32). Step (B) holds because UCB1 selects a maximizing index. The second line uses the definitions in (7.26) and (7.58). We therefore obtain

$$\mathbb{E}[Q(t)\Delta(t)] \overset{\text{(A)}}{\le} \mathbb{E}[Q(t)h(t)\mathbf{1}_{\{\mathcal{E}_u\}}] + u\mathbb{P}(\mathcal{E}_u^c) \overset{\text{(B)}}{\le} \mathbb{E}[Q(t)h(t)] + 2Ku^{-2}. \tag{7.71}$$

In (A), split the expectation over $\mathcal{E}_u$ and its complement, then use (7.70) on the first event and $Q(t)\Delta(t) \le u$ on the second, using $\Delta(t) \le 1$ and $Q(t) \le t-1 = u$ from (7.64). Step (B) applies $\mathbb{P}(\mathcal{E}_u^c) \le 2Ku^{-3}$ from (7.30). The total contribution of the failure events is bounded by

$$2K\sum_{u=K}^{\infty} u^{-2} \overset{\text{(A)}}{\le} 2K\left(\frac{1}{K^2} + \int_K^{\infty} x^{-2}\,\mathrm{d}x\right) = 2 + \frac{2}{K} \le 4. \tag{7.72}$$

Step (A) keeps the $u = K$ term and bounds the remaining sum by the integral of the decreasing function x^{-2}. Summing (7.69) over time and applying (7.71)–(7.72) gives

$$\begin{aligned}\sigma\mathbb{E}\left[\sum_{t=1}^{T} Q(t)\right] &\le \frac{T}{2} + 4 + \mathbb{E}\left[\sum_{t=1}^{T} Q(t)h(t)\right] + \frac{1}{2}\mathbb{E}[Q(1)^2] - \frac{1}{2}\mathbb{E}[Q(T+1)^2] \\ &\overset{\text{(A)}}{\le} \frac{T}{2} + 4 + \mathbb{E}\left[\sum_{t=1}^{T} Q(t)h(t)\right]. \end{aligned} \tag{7.73}$$

Step (A) uses $Q(1) = 0$. For $T \le K$, there are no terms from $\mathcal{E}_u^c$.

Set

$$y_T \coloneqq \sigma\overline{Q}_T = \frac{\sigma\mathbb{E}[Z_T]}{T}, \qquad a_T \coloneqq \frac{2K^2\Gamma(T)^2}{T\sigma^3}. \tag{7.74}$$

On every sample path,

$$\begin{aligned}\sum_{t=1}^{T} Q(t)h(t) &\overset{\text{(A)}}{\le} \left(\sum_{t=1}^{T} Q(t)^2\right)^{1/2}\left(\sum_{t=1}^{T} h(t)^2\right)^{1/2} \\ &\overset{\text{(B)}}{\le} 2^{1/4}\sqrt{K\Gamma(T)}\, Z_T^{3/4}. \end{aligned} \tag{7.75}$$

Here (A) is Cauchy–Schwarz (2.3); (B) uses (7.60). Thus

$$\begin{aligned} y_T &\overset{\text{(A)}}{\le} \frac{1}{2} + \frac{4}{T} + \frac{1}{T}\mathbb{E}\left[\sum_{t=1}^{T} Q(t)h(t)\right] \\ &\overset{\text{(B)}}{\le} \frac{1}{2} + \frac{4}{T} + \frac{2^{1/4}\sqrt{K\Gamma(T)}}{T}\mathbb{E}[Z_T^{3/4}] \\ &\overset{\text{(C)}}{\le} \frac{1}{2} + \frac{4}{T} + \frac{2^{1/4}\sqrt{K\Gamma(T)}}{T}(\mathbb{E}[Z_T])^{3/4} \\ &= \frac{1}{2} + \frac{4}{T} + \frac{2^{1/4}\sqrt{K\Gamma(T)}}{T}\left(\frac{Ty_T}{\sigma}\right)^{3/4} \\ &= \frac{1}{2} + \frac{4}{T} + \left(\frac{2K^2\Gamma(T)^2}{T\sigma^3}\right)^{1/4} y_T^{3/4} = \frac{1}{2} + \frac{4}{T} + a_T^{1/4} y_T^{3/4} \\ &\overset{\text{(D)}}{\le} \frac{1}{2} + \frac{4}{T} + \frac{(a_T^{1/4})^4}{4} + \frac{3}{4}(y_T^{3/4})^{4/3} = \frac{1}{2} + \frac{4}{T} + \frac{1}{4}a_T + \frac{3}{4}y_T. \end{aligned}$$

Step (A) divides (7.73) by T; (B) uses (7.75); (C) is Jensen's inequality (2.9) with $\theta = 3/4$. The equalities use (7.74) and give (7.67). Step (D) is Young's inequality (2.4) with $(p, q) = (4, 4/3)$ and $(a, b) = (a_T^{1/4}, y_T^{3/4})$. Rearranging and dividing by $\sigma > 0$ gives

$$\begin{aligned} \frac{1}{4} y_T &\le \frac{1}{2} + \frac{4}{T} + \frac{1}{4} a_T, \qquad y_T \le 2 + \frac{16}{T} + a_T, \\ \overline{Q}_T &= \frac{y_T}{\sigma} \le \frac{2}{\sigma} + \frac{16}{T\sigma} + \frac{2K^2 \Gamma(T)^2}{T\sigma^4}. \end{aligned} \tag{7.76}$$

This is (7.68). □

7.6.4 Combining the two bounds

Proof of Theorem 7.7. For $T = 1$, $\overline{Q}_1 = Q(1) = 0$. For $T \ge 2$, set $\ell_\varepsilon := \log(1/\varepsilon) \ge 0$ and choose

$$T_{\mathrm{cut}} := e^{29/2} \varepsilon^{-6}. \tag{7.77}$$

Apply (7.57) for $2 \le T \le T_{\mathrm{cut}}$ and (7.76) for $T \ge T_{\mathrm{cut}}$. The numerical estimates verified in Section 7.6.5 give, for $T \ge 2$,

$$\varepsilon^2 \overline{Q}_T \overset{\text{(A)}}{\le} \max\{169.903 + 64\ell_\varepsilon, 167.959\} \le 170 + 64\ell_\varepsilon \le 170(1 + \ell_\varepsilon).$$

Here (A) uses (7.84) and (7.89) on their respective ranges, together with (7.90) and (7.91). Together with the case $T = 1$, this proves (7.23). □

7.6.5 Verification of the numerical constant

We verify the two numerical estimates used in the proof of Theorem 7.7. Set $\ell_\varepsilon := \log(1/\varepsilon) \ge 0$ and $c := 29/2$. The cutoff in (7.77) is therefore $T_{\mathrm{cut}} = e^c \varepsilon^{-6}$.

To account for rounding the grid size up to an integer, put $\kappa := K\varepsilon$. Since $K = \lceil 3/\varepsilon \rceil$, $3 \le \kappa \le 4$. Consequently,

$$\frac{1}{\sigma} \overset{\text{(A)}}{\le} \frac{1}{\varepsilon}\frac{\kappa}{\kappa-1} \overset{\text{(B)}}{\le} \frac{3}{2\varepsilon}, \tag{7.78}$$

$$\frac{K}{\sigma} \overset{\text{(A)}}{\le} \frac{1}{\varepsilon^2}\frac{\kappa^2}{\kappa-1} \overset{\text{(B)}}{\le} \frac{16}{3\varepsilon^2}, \tag{7.79}$$

$$\frac{\sqrt{2K}}{\sigma} \overset{\text{(A)}}{\le} \frac{1}{\varepsilon^{3/2}}\frac{\sqrt{2\kappa^3}}{\kappa-1} \overset{\text{(B)}}{\le} \sqrt{\frac{128}{9}}\,\varepsilon^{-3/2}, \tag{7.80}$$

$$\frac{K^2}{\sigma^4} \overset{\text{(A)}}{\le} \frac{1}{\varepsilon^6}\frac{\kappa^6}{(\kappa-1)^4} \overset{\text{(B)}}{\le} \frac{4096}{81\varepsilon^6}. \tag{7.81}$$

In these four bounds, (A) substitutes $K = \kappa/\varepsilon$ and $\sigma \ge \varepsilon(\kappa-1)/\kappa$, the latter being (7.17). For (B), $\kappa/(\kappa-1) = 1 + 1/(\kappa-1)$ is at most $3/2$ because $\kappa \ge 3$. The last three coefficients are largest at $\kappa = 4$. To check this, note that on $[3,4]$,

$$\frac{\mathrm{d}}{\mathrm{d}\kappa}\frac{\kappa^2}{\kappa-1} = \frac{\kappa(\kappa-2)}{(\kappa-1)^2} \ge 0, \qquad \frac{\mathrm{d}}{\mathrm{d}\kappa}\frac{\kappa^3}{(\kappa-1)^2} = \frac{\kappa^2(\kappa-3)}{(\kappa-1)^3} \ge 0.$$

The third coefficient is the square root of $2\kappa^3/(\kappa-1)^2$. The fourth is $\kappa^6/(\kappa-1)^4 = (\kappa^3/(\kappa-1)^2)^2$.

First consider the shorter horizons, $2 \le T \le T_{\mathrm{cut}}$. The cutoff gives

$$\begin{aligned} \log T &\overset{\text{(A)}}{\le} c + 6\ell_\varepsilon, \\ \varepsilon \log T \le e^{-\ell_\varepsilon}(c + 6\ell_\varepsilon) &\overset{\text{(B)}}{\le} c. \end{aligned} \tag{7.82}$$

Here (A) takes logarithms of $T \le T_{\mathrm{cut}} = e^c\varepsilon^{-6}$ from (7.77). Step (B) evaluates the decreasing function at $\ell_\varepsilon = 0$, since $c = 29/2 > 6$ gives, for $\ell_\varepsilon \ge 0$,

$$\frac{\mathrm{d}}{\mathrm{d}\ell_\varepsilon}\left[e^{-\ell_\varepsilon}(c + 6\ell_\varepsilon)\right] = e^{-\ell_\varepsilon}(6 - c - 6\ell_\varepsilon) < 0.$$

The constant terms in the comparison bound satisfy

$$\frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{4\sigma} \overset{\text{(A)}}{\leq} \frac{\varepsilon^2}{2} + \frac{3\varepsilon}{8} \overset{\text{(B)}}{\leq} \frac{1}{2} + \frac{3}{8} = \frac{7}{8}, \tag{7.83}$$

where (A) uses (7.78) and (B) uses $0 < \varepsilon \leq 1$. We obtain

$$\begin{aligned} \varepsilon^2 \overline{Q}_T &\overset{\text{(A)}}{\leq} \frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{4\sigma} + \frac{\varepsilon^2\sqrt{2K}}{\sigma}\sqrt{\log T} + \frac{2\varepsilon^2 K}{\sigma}\log T \\ &\overset{\text{(B)}}{\leq} \frac{7}{8} + \sqrt{\frac{128}{9}\varepsilon\log T} + \frac{32}{3}\log T \\ &\overset{\text{(C)}}{\leq} \frac{7}{8} + \sqrt{\frac{128c}{9}} + \frac{32c}{3} + 64\ell_\varepsilon. \end{aligned} \tag{7.84}$$

Step (A) uses (7.57). Step (B) uses (7.83), (7.80), and (7.79) for the three terms, respectively. Step (C) applies both bounds in (7.82).

Now consider the longer horizons, $T \geq T_{\text{cut}}$, and use (7.76). We will show that its learning term decreases throughout this range. With $z \coloneqq \log T$, set

$$G(z) \coloneqq \Gamma(e^z) = \frac{1}{3} + 8z + 8z^2.$$

The factor $\Gamma(T)^2/T$ then becomes $e^{-z}G(z)^2$. For $z > 0$,

$$2G(z) - zG'(z) = \frac{2}{3} + 8z > 0, \qquad \frac{G'(z)}{G(z)} \leq \frac{2}{z}, \tag{7.85}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\log\Big(e^{-z}G(z)^2\Big) = -1 + 2\frac{G'(z)}{G(z)} \leq -1 + \frac{4}{z}. \tag{7.86}$$

Thus $\Gamma(T)^2/T$ is nonincreasing for $T \geq e^4$. Since $T_{\mathrm{cut}} \geq e^c > e^4$, substituting $T_{\mathrm{cut}} = e^c\varepsilon^{-6} = e^{c+6\ell_\varepsilon}$ and $\Gamma(T_{\mathrm{cut}}) = G(c+6\ell_\varepsilon)$ gives

$$\begin{aligned}\overline{Q}_T &\overset{\text{(A)}}{\leq} \frac{3}{\varepsilon} + \frac{24}{T\varepsilon} + \frac{8192}{81\varepsilon^6}\frac{\Gamma(T)^2}{T} \\ &\overset{\text{(B)}}{\leq} \frac{3}{\varepsilon} + \frac{24}{T_{\mathrm{cut}}\varepsilon} + \frac{8192}{81\varepsilon^6}\frac{\Gamma(T_{\mathrm{cut}})^2}{T_{\mathrm{cut}}} \\ &= \frac{3}{\varepsilon} + 24e^{-c}\varepsilon^5 + \frac{8192}{81}e^{-c}G(c+6\ell_\varepsilon)^2. \end{aligned} \tag{7.87}$$

Here (A) applies (7.78) and (7.81) to (7.76). Step (B) uses $T \geq T_{\mathrm{cut}}$ for $1/T$ and (7.86) for the learning term. After multiplying the queue bound by $\varepsilon^2 = e^{-2\ell_\varepsilon}$, the remaining expression involving G is also decreasing in ℓ_ε. Indeed,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\ell_\varepsilon}\log\left(e^{-2\ell_\varepsilon}G(c+6\ell_\varepsilon)^2\right) &= -2 + 12\frac{G'(c+6\ell_\varepsilon)}{G(c+6\ell_\varepsilon)} \\ &\overset{\text{(A)}}{\leq} -2 + \frac{24}{c+6\ell_\varepsilon} \overset{\text{(B)}}{<} 0. \end{aligned} \tag{7.88}$$

Step (A) is (7.85) at $z = c + 6\ell_\varepsilon$. Step (B) uses $c = 29/2 > 12$ and $\ell_\varepsilon \geq 0$. Multiplying (7.87) by ε^2 therefore gives

$$\begin{aligned}\varepsilon^2\overline{Q}_T &\leq 3\varepsilon + 24e^{-c}\varepsilon^7 + \frac{8192}{81}e^{-c}e^{-2\ell_\varepsilon}G(c+6\ell_\varepsilon)^2 \\ &\overset{\text{(A)}}{\leq} 3 + e^{-c}\left(24 + \frac{8192}{81}G(c)^2\right). \end{aligned} \tag{7.89}$$

Step (A) uses $\varepsilon \leq 1$ for the first two terms and (7.88) to bound the last term by its value at $\ell_\varepsilon = 0$. For the fixed choice $c = 29/2$, both numerical coefficients are smaller than 170:

$$\frac{7}{8} + \sqrt{\frac{128c}{9}} + \frac{32c}{3} = \frac{7}{8} + \frac{8\sqrt{29}}{3} + \frac{464}{3} < 169.903, \tag{7.90}$$

$$G(c) = \frac{1}{3} + 8\frac{29}{2} + 8\left(\frac{29}{2}\right)^2 = \frac{5395}{3},$$

$$3 + e^{-c}\left(24 + \frac{8192}{81}G(c)^2\right) = 3 + e^{-29/2}\left(24 + \frac{8192}{81}\left(\frac{5395}{3}\right)^2\right) < 167.959. \tag{7.91}$$

These are direct evaluations of the constants in (7.84) and (7.89), with the displayed decimals rounded upward.

Remark 7.14 (Long-run queue length)**.** *For each fixed environment and supplied slack lower bound ε, the long-run time-average expected queue length satisfies*

$$\limsup_{T\to\infty} \overline{Q}_T \le \frac{3}{4\varepsilon}. \tag{7.92}$$

With K and σ fixed, the definitions in (7.74) and $\Gamma(T) = O((\log T)^2)$ give $a_T \to 0$. The explicit estimate (7.68) shows that y_T stays bounded, so the learning term in (7.67) vanishes. Taking the limit superior in (7.67) and using $y_T = \sigma\overline{Q}_T$ and $\sigma \ge 2\varepsilon/3$ from (7.19) proves the bound. This long-run bound has the same $O(1/\varepsilon)$ order as the fixed-rate benchmark bound (7.6).

7.7 Lower-bound analysis

Theorem 7.8 gives the lower bound for every causal policy. Its proof is in the original paper. Here we prove the stronger bound for the prescribed UCB1 policy by constructing a deterministic environment in which exploration limits the service supplied to the queue.

7.7.1 A matching lower bound for UCB1

The deterministic construction and index comparison have a two-arm precedent in Audibert, Munos, and Szepesvári [116, Appendix A.2, Proposition 1 and Lemma 4]. The proof below applies this approach to the prescribed rate grid and queueing objective.

Proof of Theorem 7.9. Take deterministic arrivals and capacities

$$A(t) \equiv \varepsilon, \qquad C(t) \equiv 2\varepsilon, \qquad 0 < \varepsilon \le \frac{1}{12}. \tag{7.93}$$

Here $\lambda = \varepsilon$ and $\mu(r) = r\mathbf{1}_{\{r \le 2\varepsilon\}}$, so $\mu(r^*) = 2\varepsilon$ and $\mu(r^*) - \lambda = \varepsilon$. For the prescribed grid $r_k = k/K$, the choice $K = \lceil 3/\varepsilon \rceil$ gives $6 \le 2\varepsilon K < 7$. Thus exactly the first six grid rates provide positive service. Define

$$\begin{aligned} &\mathcal{K}_+ \coloneqq \{1, \dots, 6\}, \qquad \mathcal{K}_0 \coloneqq \{7, \dots, K\}, \\ &K_0 \coloneqq |\mathcal{K}_0| = K - 6, \\ &\mu_k = r_k \mathbf{1}_{\{k \in \mathcal{K}_+\}}, \qquad 0 \le \mu_k \le 2\varepsilon \mathbf{1}_{\{k \in \mathcal{K}_+\}}. \end{aligned} \tag{7.94}$$

Set

$$T_\varepsilon \coloneqq \left\lfloor \frac{K_0 \log K_0}{8\varepsilon^2} \right\rfloor, \qquad T \coloneqq T_\varepsilon. \tag{7.95}$$

The grid choice (7.13) and (7.94) give $K_0 = 3/\varepsilon + O(1)$. Hence (7.95) gives

$$T = \Theta\Big(\varepsilon^{-3} \log(1/\varepsilon)\Big) \qquad (\varepsilon \downarrow 0). \tag{7.96}$$

Choose $\varepsilon_0 \in (0, 1/12]$ so that, for every $0 < \varepsilon \le \varepsilon_0$,

$$\begin{gathered} \frac{2}{\varepsilon} \le K_0 \le \frac{3}{\varepsilon}, \qquad K < T, \qquad \frac{K_0 \log K_0}{8\varepsilon^2} \ge 2, \\ \frac{48}{K_0} + \frac{25}{2T} \le \frac{1}{4}. \end{gathered} \tag{7.97}$$

The bounds on K_0 follow from $K_0 = \lceil 3/\varepsilon \rceil - 6$ for $\varepsilon \le 1/12$. The remaining conditions hold by (7.96) and $K_0 = 3/\varepsilon + O(1)$. Assume $0 < \varepsilon \le \varepsilon_0$ for the remainder of the proof.

We first bound how often rates in $\mathcal{K}_+$ are selected by comparing their indices with the index of a rate in $\mathcal{K}_0$.

Each observation of arm k equals μ_k. With the fixed tie-breaking rule in Algorithm 4, all selections, counts, and queue lengths are therefore deterministic. After initialization, the statistics and indices in (7.20) and (7.22) specialize to

$$\begin{aligned} \widehat{\mu}_k(u) = \mu_k, \qquad U_k(u) = \mu_k + b_k(u), \\ J(u+1) \in \arg\max_{k\in[K]} U_k(u), \qquad u \geq K. \end{aligned} \tag{7.98}$$

Fix $K \leq u < T$ and choose $j \in \arg\min_{\ell\in\mathcal{K}_0} N_\ell(u)$. Then

$$N_j(u) \leq \frac{1}{K_0}\sum_{\ell\in\mathcal{K}_0} N_\ell(u) \leq \frac{u}{K_0}, \tag{7.99}$$

$$b_j(u)^2 \overset{(A)}{\geq} \frac{2K_0\log u}{u} \overset{(B)}{\geq} \frac{2K_0\log K_0}{T} \overset{(C)}{\geq} 16\varepsilon^2. \tag{7.100}$$

Step (A) uses (7.99) and (7.21); (B) uses $K_0 < K \leq u < T$; (C) uses (7.95). If $J(u+1) = k \in \mathcal{K}_+$, then

$$b_j(u) = U_j(u) \overset{(A)}{\leq} U_k(u) = \mu_k + b_k(u) \overset{(B)}{\leq} 2\varepsilon + b_k(u) \overset{(C)}{\leq} \tfrac{1}{2}b_j(u) + b_k(u). \tag{7.101}$$

Step (A) uses (7.98); (B) uses $\mu_k \leq 2\varepsilon$ from (7.94); (C) uses (7.100). By (7.21),

$$\begin{aligned} b_k(u) \geq \tfrac{1}{2}b_j(u), \qquad \frac{2\log u}{N_k(u)} = b_k(u)^2 \geq \frac{b_j(u)^2}{4} = \frac{2\log u}{4N_j(u)}, \\ N_k(u) \leq 4N_j(u) \overset{(A)}{\leq} \frac{4u}{K_0}. \end{aligned} \tag{7.102}$$

Here $\log u > 0$ and $N_k(u), N_j(u) > 0$ since $u \geq K \geq 3$; (A) uses (7.99). For each $k \in \mathcal{K}_+$, apply (7.102) just before its last selection in slots $K+1, \ldots, T$ to obtain

$$N_k(T) \leq \frac{4T}{K_0} + 1, \qquad k \in \mathcal{K}_+. \tag{7.103}$$

If there is no such selection, $N_k(T) = 1$. Define

$$N_+(t) \coloneqq \sum_{k \in \mathcal{K}_+} N_k(t), \qquad N_+(0) = 0.$$

By (7.103),

$$N_+(T) \le \frac{24T}{K_0} + 6. \tag{7.104}$$

The service and queue equations (7.1) and (7.2) specialize to

$$\begin{aligned} S(t) &= r_{J(t)} \mathbf{1}_{\{J(t) \in \mathcal{K}_+\}} \le 2\varepsilon \mathbf{1}_{\{J(t) \in \mathcal{K}_+\}}, \\ Q(t+1) &= [Q(t) + \varepsilon - S(t)]_+. \end{aligned} \tag{7.105}$$

Using the service bound (7.94) and iterating (7.105) from $Q(1) = 0$ gives

$$Q(t) \ge \varepsilon(t-1) - \sum_{\tau=1}^{t-1} S(\tau) \ge \varepsilon(t-1) - 2\varepsilon N_+(t-1). \tag{7.106}$$

Since the queue and counts are deterministic here, $\overline{Q}_T = T^{-1} \sum_{\tau=1}^{T} Q(\tau)$. Therefore

$$\begin{aligned} \overline{Q}_T &\overset{\text{(A)}}{\ge} \frac{\varepsilon(T-1)}{2} - \frac{2\varepsilon}{T} \sum_{t=1}^{T} N_+(t-1) \\ &\overset{\text{(B)}}{\ge} \frac{\varepsilon(T-1)}{2} - 2\varepsilon N_+(T) \\ &\overset{\text{(C)}}{\ge} \frac{\varepsilon(T-1)}{2} - 2\varepsilon \left(\frac{24T}{K_0} + 6 \right) \\ &= \varepsilon T \left(\frac{1}{2} - \frac{48}{K_0} - \frac{25}{2T} \right) \overset{\text{(D)}}{\ge} \frac{\varepsilon T}{4}. \end{aligned} \tag{7.107}$$

Step (A) sums (7.106); (B) uses $N_+(t-1) \le N_+(T)$; (C) uses (7.104); (D) uses (7.97). Finally,

$$\overline{Q}_T \ge \frac{\varepsilon}{4} \frac{K_0 \log K_0}{16\varepsilon^2} = \frac{K_0 \log K_0}{64\varepsilon} \ge \frac{\log(1/\varepsilon)}{32\varepsilon^2}. \tag{7.108}$$

The first inequality uses (7.107) and $\lfloor z \rfloor \geq z/2$ for $z \geq 2$, with $z = K_0 \log K_0/(8\varepsilon^2) \geq 2$ by (7.97). The second uses $K_0 \geq 2/\varepsilon$. This proves the stated lower bound for every $0 < \varepsilon \leq \varepsilon_0$. □

7.8 Discussion and conclusion

The controller needs only a known lower bound ε on service slack to choose a finite rate grid and run UCB1. Starting from an empty queue, it achieves the time-average expected queue bound $O((1 + \log(1/\varepsilon))/\varepsilon^2)$ in Theorem 7.7. This bound holds for every horizon.

Theorem 7.8, proved in [91], shows that, for every sufficiently small ε, any causal policy that knows ε but not the channel distribution faces a feasible channel for which the time-average expected queue is $\Omega(\varepsilon^{-2})$ at some horizon. For the prescribed UCB1 policy, Theorem 7.9 gives the stronger bound $\Omega(\varepsilon^{-2} \log(1/\varepsilon))$, even with deterministic arrivals and capacities.

For each fixed environment and supplied ε, the long-run bound (7.92) is $O(\varepsilon^{-1})$. The larger uniform bound accounts for the transient queue buildup during learning. The UCB1 lower bound exhibits such buildup at T_ε. It remains unresolved whether another causal policy can remove the logarithmic factor from the uniform bound.

Chapter 8

Conclusions and Future Directions

This thesis studies constrained decisions with costly domain projections, states that depend on past actions, or limited feedback. The methods use linear optimization over the original domain, separate updates for energy usage and allocation proportions, simplex-constrained storage policies, and a finite grid of transmission rates to obtain tractable updates.

8.1 Main results

Chapter 3 develops a method for nonsmooth convex optimization over compact convex domains. A multiplier couples domain points produced by linear optimization to auxiliary query points. With parameters chosen for a query budget T, the method attains an objective gap of $O(T^{-1/2})$ using at most T calls to each oracle: an exact ordinary linear oracle over the original domain and a subgradient oracle on the auxiliary set. Projections are required only on the auxiliary set. The returned point belongs to the original domain.

Chapter 4 applies the algorithm of Asgari and Neely [17] to the positive parts of the constraints. Keeping the functional inequalities outside the linear oracle preserves the tractability of the oracle on the original domain. With parameters chosen for a query budget T, it gives an $O(T^{-1/2})$ upper bound on the objective gap at the returned domain point. The constraint-violation norm has the

same rate if a Lagrange multiplier exists. The algorithm need not know the multiplier's value, although its norm enters the violation bound.

Chapter 5 studies allocation among n subsystems with stochastic energy arrivals in $[0,1]$ and total usage at most one per slot. Separating total usage from allocation proportions allows battery feedback to regulate expenditure while entropic mirror descent learns proportions on a fixed simplex. The benchmark is the best nonnegative fixed allocation whose total usage does not exceed the mean energy arrival. Under the stated arrival and information assumptions, tuning for a planned horizon of T slots gives expected cumulative regret $O(\sqrt{T(1+\log n)})$ with sufficient battery capacity $B_{\max} = 3\sqrt{T} = O(\sqrt{T})$. The controller uses only the energy available in each slot without knowing the mean arrival.

Chapter 6 studies scalar storage control with arrivals in $[0,1]$ and full cost feedback. The entropic SDAC controller achieves $O(\sqrt{T\log(H+1)})$ regret against policies with memory H. With the stated horizon-dependent memory and step size, it achieves $O(\sqrt{T\log\log T})$ regret against the fixed infinite-memory SDAC class.

For retention $\alpha \in (0,1)$, positive cost Lipschitz bounds L_x, L_u, and an integer $H \geq 1$, Theorem 6.10 gives a lower bound against the fixed memory-H SDAC class. Under the horizon condition (6.50), for every causal feasible controller $\mathcal{A}$, including randomized controllers, there is a deterministic sequence of arrivals and costs, fixed before the interaction, such that

$$\mathbb{E}_{\mathcal{A}}[\mathrm{Reg}_T(\mathcal{M}_H)] \geq \frac{G_\alpha}{8}\sqrt{T}, \qquad G_\alpha \coloneqq \alpha\left(\frac{L_x}{1-\alpha} + L_u\right),$$

where the expectation is over the controller's randomization. For fixed H, α, L_x, L_u, this establishes optimal $\sqrt{T}$ dependence on the horizon T. Gaps remain in the dependence on the retention coefficient α and memory length H.

Chapter 7 studies rate adaptation using success or failure acknowledgments to learn mean offered service across transmission rates. It assumes a known lower bound $\varepsilon > 0$ on the difference between the best achievable mean offered service and the mean arrival rate. This supplied bound determines a finite rate grid containing a stabilizing rate. Starting from an empty queue, the UCB1 controller bounds the time-average expected queue length by $O((1 + \log(1/\varepsilon))/\varepsilon^2)$ uniformly over all horizons.

Theorem 7.8, restated without proof from the original joint paper [91], gives a lower bound of $\Omega(\varepsilon^{-2})$ for every causal learning policy on some admissible instance and horizon. For this UCB1 controller, Theorem 7.9 gives a matching $\Omega(\varepsilon^{-2}\log(1/\varepsilon))$ lower bound at a horizon $T_\varepsilon = \Theta(\varepsilon^{-3}\log(1/\varepsilon))$, even with deterministic arrivals and capacities. For each fixed environment and supplied bound ε, (7.92) gives the separate long-run guarantee

$$\limsup_{T\to\infty} \overline{Q}_T \le \frac{3}{4\varepsilon}.$$

8.2 Models and performance criteria

Table 1.1 summarizes the models, feedback, comparators, and feasibility assumptions. The energy-allocation benchmark satisfies a mean budget and may exceed the available resource on individual slots. The SDAC benchmarks are feasible policies evaluated on their own trajectories.

8.3 Open problems

Strongly convex objectives. An open question is whether the techniques of Chapter 3 can exploit strong convexity to achieve a rate of convergence of the objective gap faster than $O(T^{-1/2})$

for a query budget T. The aim is to retain an ordinary linear optimization oracle over the original domain and projections only onto an auxiliary set, without requiring the objective to be smooth.

Constraint dependence and violation lower bounds. For Chapter 4, can the dependence on the number m of functional constraints be improved when the individual constraint subgradient bounds are held fixed? This dependence enters the violation guarantee through the aggregate bound G and the multiplier norm. A related problem is to establish lower bounds for the constraint-violation norm V_T under the same oracle model and requirement on objective accuracy, and thereby determine the best possible dependence on T and m.

Feasible policy benchmarks for energy allocation. Can the regret guarantees against fixed allocations in Chapter 5 be extended to feasible policies that spend up to a fixed amount each slot? Such a policy fixes a maximum expenditure $u \in [0, 1]$ and allocation proportions, spends u whenever sufficient energy is available, and otherwise spends all available energy. Under the chapter's arrival timing, its total usage is $U_t^\pi := \min\{u, B_t^\pi + A_t\}$, where B_t^π is its own battery level. Each policy is evaluated with the same arrivals, capacity, and initial energy as the controller. The benchmark would select the best fixed threshold and allocation proportions in hindsight.

Memory and retention dependence in storage control. For fixed $L_x, L_u > 0$ and horizons satisfying (6.50), as $\alpha \uparrow 1$ the storage lower bound above is $\Omega((1-\alpha)^{-1}\sqrt{T})$. The upper bound in Theorem 6.5 is $O((1-\alpha)^{-3/2}\sqrt{T\log(H+1)})$ in this regime, by Remark 6.6. For fixed-memory policies, the unresolved factors are $\sqrt{\log(H+1)}$ in memory dependence and $(1-\alpha)^{-1/2}$ in retention dependence. For infinite-memory policies, it is also unknown whether the regret guarantee must contain the factor $\sqrt{\log\log T}$.

Uniform queue bounds. This thesis leaves open whether a causal learning policy can achieve $\overline{Q}_T = O(\varepsilon^{-2})$ uniformly over all horizons under the assumptions of Chapter 7. Theorem 7.8 establishes a lower bound of this order. Theorem 7.9 shows that removing the logarithm requires a different controller, rather than a sharper analysis of the current UCB1 policy alone.

Feedback along the executed trajectory. Another open problem is to extend the scalar storage regret bounds to costs observed only along the executed trajectory. The controller in Chapter 6 uses the revealed cost function to obtain a subgradient at the current fixed-policy state and action. An estimator based on feedback observed only along the executed trajectory must account for the dependence of the executed state on past actions.

Horizon-independent storage control. Can the storage regret rates be attained without knowing the horizon in advance? Online projected subgradient descent with $\eta_t := t^{-1/2}$ achieves $O(\sqrt{T})$ regret without knowing T in advance on a compact convex domain with uniformly bounded subgradients [2, Theorem 1]. For the scalar storage controller, the memory length in (6.47) also depends on the horizon. A restart scheme must account for the current storage state and the continuing influence of past arrivals.

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